

Notion My 14 step plan (tier 1.5)

▼ (Phase 1 Planning & Setup)

▼ Step 1 - Lock the thesis goal, scope, and research questions

chapter 1 intro

chapter 2 thoery

▼ chapter 1 intro

▼ chapter 1 deliverables

- background and motivation
- problem statement
- aim
- research questions
- scope and limitations
- usage of AI tools section
- division of work section
- chapter outline

▼ chapter 1 feeds from

docs/ongoing/project-overview.md
docs/ongoing/research-questions.md
docs/ongoing/yacoub-scope.md
docs/ongoing/obid-minimum-integration.md
docs/plans/implementation-plan.md
docs/decisions.md

▼ chapter 2 thoery

▼ chapter 2 delivers

- initial theory boundaries
- what theory is relevant
- what theory is out of scope

▼ chapter 2 feeds from

docs/n8n-version.md
docs/architecture/README.md
literature notes
n8n notes
edge/IoT notes
API/validation notes

▼ Goal

Freeze the thesis direction so it does not keep changing.

This step exists to stop the project from drifting back into the old broader Yacoub/Obid plan.

▼ Work to complete

Finalize the core planning documents:

- `docs/plans/implementation-plan.md`
- `docs/ongoing/project-overview.md`
- `docs/ongoing/yacoub-scope.md`
- `docs/ongoing/obid-minimum-integration.md`
- `docs/ongoing/research-questions.md`
- `docs/ongoing/report-outline.md`
- `docs/n8n-version.md`
- `docs/decisions.md`

▼ Research questions

The locked research questions should stay close to this:

1. Can a self-hosted n8n-based IoT system run in a resource-constrained environment with acceptable latency?
2. What RAM, CPU, and thermal/resource effects are observed during operation?
3. Can a minimal workflow/safety layer safely mediate one physical action?
4. How does a deterministic baseline compare to an agent-enhanced workflow on the same task?

▼ Scope lock

The project is limited to:

- one scenario
- one sensor/event path
- one action path
- one middleware bridge
- one deterministic baseline
- one minimal agent-enhanced path
- one validation/approval path
- one evaluation loop

▼ Evidence produced

- locked scopelocked scope
- research questions
- decision log
- report outline
- n8n version pin
- scope documents
- narrowed implementation plan

▼ Step 2 - Create the repo foundation and reproducible environment
Chapter 3 Methodology

Chapter 4 Choice of approach

▼ Chapter 3 — Methodology feeds from:

```
docs/plans/implementation-plan.md
docs/decisions.md
README.md
AGENTS.md
Git commit history
AI-tool workflow notes
```

▼ Chapter 3 deloiv

- project/work method
- tool workflow
- use of Git/GitHub
- use of AI tools in development
- iterative development method

▼ Chapter 4 — Choice of approach feeds from:

```
docs/decisions.md
docs/architecture/README.md
docs/ongoing/integration-contract.md
```

▼ Chapter 4 dev

- why this architecture was chosen
- why scope was narrowed
- why Antigravity/Codex are used as support tools
- why implementation is built as one vertical slice

▼ Goal

Make the project usable, version-controlled, and ready for implementation.

▼ Work to complete

▼ 1

- GitHub repo is live
- branch/commit discipline is active
- `README.md` exists
- `AGENTS.md` exists
- Antigravity rules/workflows/skill exist
- Codex threads are set up
- repo folders are created
- basic docs are populated

▼ 2

Work to complete

Set up:

```
README.md
AGENTS.md
.agents/rules/thesis-boundaries.md
.agents/workflows/audit-implementation.md
.agents/skills/thesis-review/SKILL.md
docs/architecture/README.md
docs/ongoing/integration-contract.md
GitHub repo
Codex audit-review thread
Codex repair-fix thread
Antigravity workspace
```

▼ Keep minimal

Do not start with:

- MCP
- n8n core modification
- advanced multi-agent logic
- multiple devices
- full Raspberry Pi deployment
- broad benchmarking

▼ Evidence produced

▼ 1

- GitHub repository
- clean commit history
- repo structure
- README
- AGENTS rules
- planning docs
 - Antigravity rules
 - Codex thread setup
 - architecture overview
 - initial contract document

▼ 2

Evidence produced

- GitHub repository
- clean commit history
- README
- AGENTS file
- Antigravity rules
- Codex thread setup

- architecture overview
- initial contract document

▼ (Phase 2: Building the Foundation (No AI yet))

▼ Step 3 - Build the local n8n baseline via docker (`infrastructure`)

(chapter 4 Choice of approach)

(chapter 5 Implementation)

▼ AI

▼ cx step runner

▼ q

You are the step-runner thread for the new-yacoub thesis repository.

Your job is to execute the thesis implementation through the 14-step process, one step at a time.

Use these files as source of truth:

- [AGENTS.md](#)
- [README.md](#)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/ongoing/integration-contract.md
- docs/evaluation/evaluation-protocol.md
- docs/n8n-version.md

Scope:

- Yacoub complete in narrowed form
- Obid minimum integration only
- no broad multi-agent expansion
- no MCP in phase 0
- no n8n core modification
- no multiple devices/models
- no heavy Prometheus/Grafana stack

Working rule:

Only work on the current step. Do not jump ahead.

Current step:

Step 3 — local n8n baseline.

Task:

Create the minimal local self-hosted n8n baseline.

Create or update:

- infrastructure/docker/docker-compose.yml
- infrastructure/docker/.env.example
- infrastructure/docker/README.md

- docs/n8n-version.md if a Docker image tag/version is chosen
- README.md only if a short run pointer is needed

Requirements:

1. Use Docker Compose.
2. Keep it simple and reproducible.
3. Use a pinned explicit n8n Docker image tag if you can verify it from the environment or existing docs. If not, use a clearly documented temporary tag and mark docs/n8n-version.md as needing final verification.
4. Include local startup and shutdown instructions.
5. Include notes for future Raspberry Pi validation, but do not implement Pi deployment yet.
6. Do not add PostgreSQL unless necessary for the first baseline.
7. Do not add Prometheus/Grafana.
8. Do not add MCP.
9. Do not implement middleware or workflows yet.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to test it locally.

▼ a

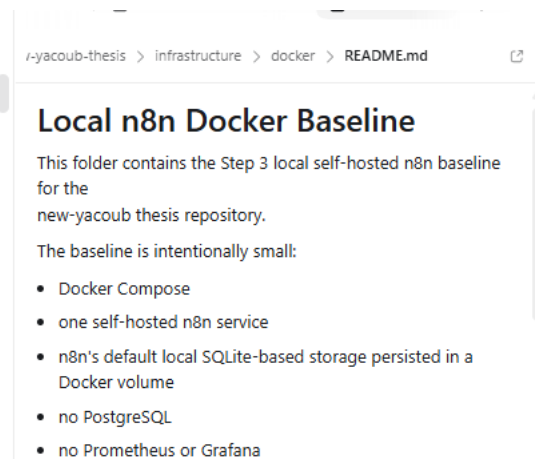
▼ files

6 files changed +313 -13	Undo ↶
infrastructure/docker/.env.example +22 -0 •	🔗 ▼
infrastructure/docker/README.md +90 -0 •	▼
infrastructure/docker/docker-compose.yml +22 -0 •	▼
docs/n8n-version.md +121 -0 •	▼
README.md +24 -8	▼
docs/decisions.md +34 -5	▼

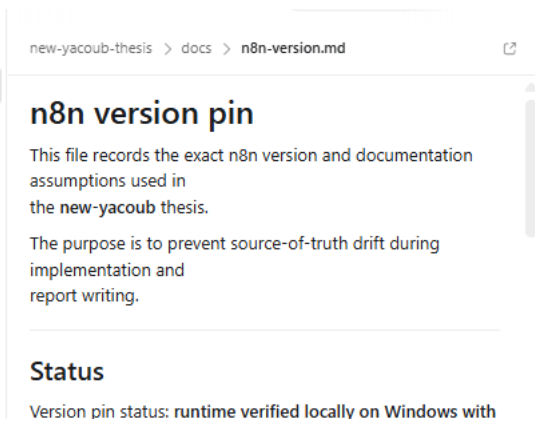
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exi:
Run
No

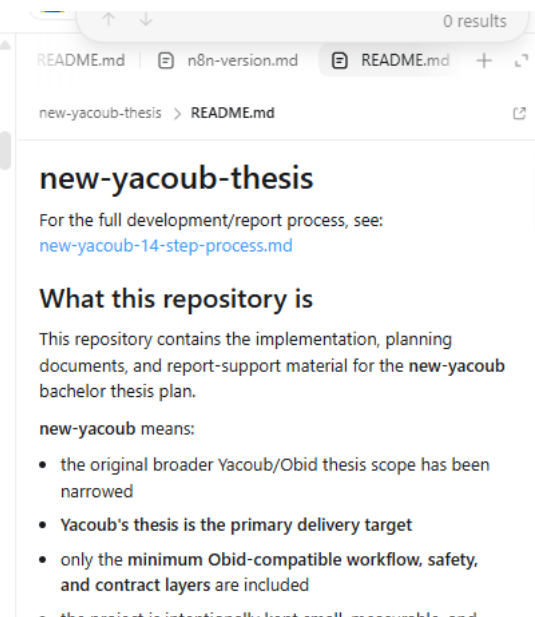
▼ 1



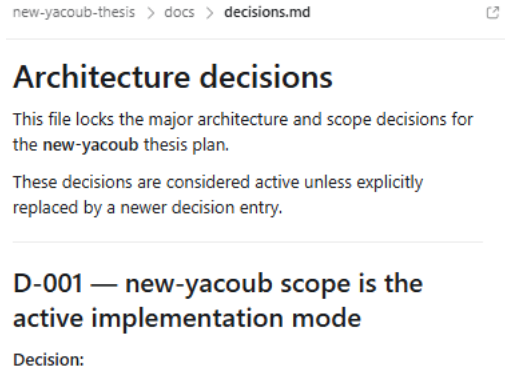
▼ 2



▼ 3



▼ 4



Step 3 is done, kept narrow.

Changed:

- Added `docker-compose.yml` with one pinned `n8nio/n8n:1.123.37` service, local port `5678`, and persisted Docker volume storage.
- Added `.env.example` for local config.
- Added `infrastructure/docker/README.md` with startup, shutdown, reset, basic checks, and future Raspberry Pi notes.
- Updated `docs/n8n-version.md` to pin `n8nio/n8n:1.123.37`, with runtime verification still pending.
- Updated `docs/decisions.md` with the local Docker Compose/default storage decision.
- Updated `README.md` with a short local n8n run pointer.

The n8n tag was verified from Docker Hub tag metadata: [n8nio/n8n tags](#). I could not run the container here because `docker` is not installed in this environment.

To test locally on a Docker-enabled machine:

```
cd "C:\\Users\\Jake_\\OneDrive - Mittuniversitetet\\EDU\\4. RED\\1. kaniddat\\new-yacoub-thesis\\infrastructure\\docker"
Copy-Item .env.example .env
docker compose up -d
docker compose ps
docker compose logs -f n8n
```

Then open `http://localhost:5678`.

To stop:

```
docker compose down
```

To reset local n8n state:

```
docker compose down -v
```

▼ cx step runenr 2

▼ q

The only thing that is still missing is runtime proof:

docker compose up -d succeeds
n8n opens at <http://localhost:5678>
screenshot or terminal log saved somewhere
docs/n8n-version.md can be marked as runtime verified

Do this now

In PowerShell:

```
cd "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"  
Copy-Item .env.example .env  
docker compose up -d  
docker compose ps  
docker compose logs n8n
```

Then open:

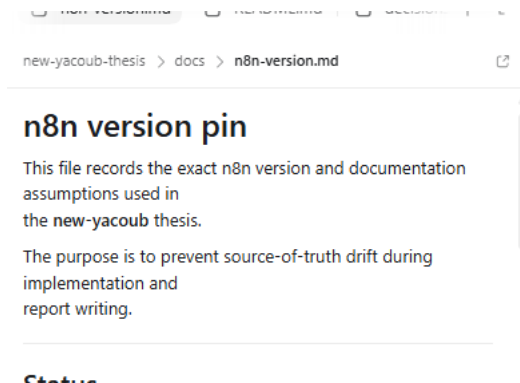
<http://localhost:5678>

If it works, take one screenshot of n8n running.

Or let me take a screenshot if you cant

▼ a

▼ files



I tried to complete the runtime proof, but this environment is blocked because Docker is not installed or not available in PowerShell.

What happened:

- `Copy-Item .env.example .env` succeeded.
- `docker compose up -d` failed: `docker` command not recognized.
- `docker compose ps` failed for the same reason.
- `docker compose logs n8n` failed for the same reason.
- `localhost:5678` TCP check failed.
- Docker Desktop was not found under `C:\Program Files\Docker`.
- WSL is not installed.
- I opened `http://localhost:5678` via Windows fallback, but n8n is not running.

Saved proof log here:

<infrastructure/docker/evidence/n8n-runtime-proof-2026-04-25.txt>

I also updated:

[docs/n8n-version.md](#)

It is now marked as **runtime verification blocked**, not verified. That is the honest state. Once Docker Desktop is installed/running, the same command sequence should be rerun and then the doc can be marked runtime verified.

▼ og follow up

▼ q

Step-runner did **good work** for Step 3.

But important distinction:

| Step 3 is **implemented**, but not yet **verified**.

Codex created the files, documented the setup, pinned an n8n Docker image, and kept scope narrow. The Docker README clearly says it is one n8n service, persisted volume storage, no PostgreSQL, no Prometheus/Grafana, no MCP, and no middleware/workflows yet. The root README also now points to the local n8n baseline and gives a quick start command.

Step 3 checklist

What I think about the Step 3 files

This is good:

- one simple n8n service
- pinned image tag
- persistent Docker volume
- no PostgreSQL yet
- no middleware yet
- no workflow yet
- no MCP
- no Prometheus/Grafana
- Raspberry Pi only as future notes

That is exactly the Step 3 scope.

The only thing that is still missing is **runtime proof**:

- `docker compose up -d` succeeds
- n8n opens at `http://localhost:5678`
- screenshot or terminal log saved somewhere
- `docs/n8n-version.md` can be marked as runtime verified

Do this now

In PowerShell:

```
cd "C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"
Copy-Item .env.example .env
docker compose up -d
```

```
docker compose ps
docker compose logs n8n
```

Then open:

```
http://localhost:5678
```

If it works, take one screenshot of n8n running.

Then stop it:

```
docker compose down
```

Do **not** run `docker compose down -v` unless you intentionally want to delete the local n8n data.

After the local test works

Update `docs/n8n-version.md` from:

```
runtime verification pending
```

to something like:

```
runtime verified locally on Windows with Docker Desktop on YYYY-MM-DD
```

Then ask `audit-review`:

```
Audit the Step 3 local n8n baseline changes only.

Check:
- Docker Compose is minimal and reproducible
- README instructions are clear
- docs/n8n-version.md is correct after local runtime verification
- no middleware/workflow/safety/MCP/PostgreSQL/Prometheus/Grafana scope was added
- Step 3 evidence is enough to move to Step 4: middleware skeleton

Do not suggest broader architecture unless something blocks Step 4.
```

Commit after audit passes

```
git status
git add .
git commit -m "feat(infra): add local n8n docker baseline"
git push
```

Verdict

Step-runner successfully created the **Step 3 baseline files**.

You are now at:

| Step 3 implementation complete, local verification pending.

Once you run Docker and Codex audit passes, Step 3 is officially done and you can move to **Step 4: middleware skeleton**.

▼ a

Step-runner did **good work** for Step 3.

But important distinction:

| Step 3 is **implemented**, but not yet **verified**.

Codex created the files, documented the setup, pinned an n8n Docker image, and kept scope narrow. The Docker README clearly says it is one n8n service, persisted volume storage, no PostgreSQL, no Prometheus/Grafana, no MCP, and no middleware/workflows yet. The root README also now points to the local n8n baseline and gives a quick start command.

Step 3 checklist

What I think about the Step 3 files

This is good:

- one simple n8n service
- pinned image tag
- persistent Docker volume
- no PostgreSQL yet
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- no MCP
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That is exactly the Step 3 scope.

The only thing that is still missing is **runtime proof**:

- `docker compose up -d` succeeds
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- screenshot or terminal log saved somewhere
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Do this now

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cd "C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"
Copy-Item .env.example .env
docker compose up -d
docker compose ps
docker compose logs n8n
```

Then open:

```
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```

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Update `docs/n8n-version.md` from:

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```

Then ask `audit-review`:

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Audit the Step 3 local n8n baseline changes only.

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- Docker Compose is minimal and reproducible
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Do not suggest broader architecture unless something blocks Step 4.
```

Commit after audit passes

```
git status
git add .
git commit -m "feat(infra): add local n8n docker baseline"
git push
```

Verdict

Step-runner successfully created the **Step 3 baseline files**.

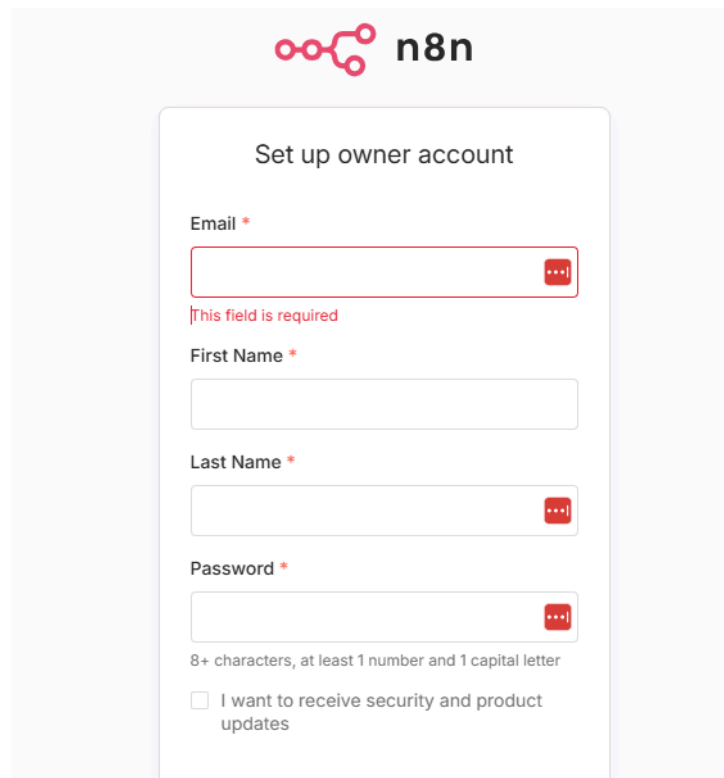
You are now at:

| Step 3 implementation complete, local verification pending.

Once you run Docker and Codex audit passes, Step 3 is officially done and you can move to **Step 4: middleware skeleton**.

▼ terminal log

▼ screenshot (<http://localhost:5678/>) of local host n8n running after terminal docke



The screenshot shows the n8n logo at the top, followed by the title 'Set up owner account'. The form contains four input fields: 'Email *', 'First Name *', 'Last Name *', and 'Password *'. The 'Email' field has a red border and a red error message 'This field is required' below it. The 'Password' field has a red border and a red error message '8+ characters, at least 1 number and 1 capital letter' below it. At the bottom, there is a checkbox labeled 'I want to receive security and product updates'.

▼ report scribe

▼ q

PROMPT - REPORT WRITER AGENT

report-scribe

You are the report-scribe thread for the new-yacoub thesis repository.

Your job is not to write the final thesis yet.

Your job is to maintain concise report notes after each implementation step.

Use:

- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/evaluation/evaluation-protocol.md
- docs/architecture/README.md

For each completed step, create or update:

docs/report-notes/step-XX-name.md

Each report note must include:

- what was built
- why it matters for the thesis
- evidence produced

- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Do not invent results.

Do not write final conclusions before measurements exist.

Do not expand scope.

▼ a

▼ Step 01 - Lock Scope

What was built

- The active thesis direction was narrowed to the new-yacoub plan.
- The repository now prioritizes Yacoub's self-hosted n8n IoT system while keeping only the minimum Obid-compatible workflow, safety, and contract layer.
- Core planning documents define the scope, research questions, report outline, architecture assumptions, and implementation order.

Why it matters for the thesis

This step prevents the project from drifting back into the broader historical Yacoub/Obid dual-thesis scope. It gives the report a defensible aim, research-question set, limitation boundary, and implementation target before technical work expands.

Evidence produced

- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/ongoing/project-overview.md
- docs/ongoing/yacoub-scope.md
- docs/ongoing/obid-minimum-integration.md
- docs/ongoing/research-questions.md
- docs/ongoing/report-outline.md
- docs/decisions.md
- docs/architecture/README.md
- docs/evaluation/evaluation-protocol.md

Report chapters it feeds

- Chapter 1 - Introduction: aim, problem statement, scope, limitations, research questions, AI-tool usage boundaries.
- Chapter 2 - Theory: initial theory boundaries and excluded areas.
- Chapter 3 - Methodology: evidence-first development method.
- Chapter 4 - Choice of approach: why the thesis uses one narrow vertical slice.

Limitations or assumptions

- The scope is a planning lock, not a measured result.

- No latency, resource, safety, or baseline-vs-agent conclusions can be drawn from this step.
- Historical broader plan files still exist under docs/plans/ and must be treated as reference-only when they conflict with the new-yacoub plan.

Screenshots/logs still needed

- None required for the scope lock itself.
- Later report writing may need a short evidence table showing which planning files define the active scope.

▼ Step 02 - Repo Foundation

What was built

- The repository foundation was created with the main thesis folders, README, AGENTS guidance, architecture notes, planning documents, evaluation protocol, and a Git remote.
- The repo structure separates high-priority Yacoub implementation areas from minimum-integration-only folders.
- Local Git history exists and the repository is connected to <https://github.com/Rumple12/new-yacoub-thesis.git>.

Why it matters for the thesis

This step makes the project reproducible and reviewable. It gives the report a concrete development method: small scoped changes, version control, explicit AI-agent boundaries, and documentation that links implementation artifacts to thesis chapters.

Evidence produced

- README.md
- AGENTS.md
- docs/architecture/README.md
- docs/ongoing/integration-contract.md
- docs/evaluation/evaluation-protocol.md
- Top-level folders: infrastructure/, middleware/, evaluation/, shared_interfaces/, cognitive_logic/, safety_layer/, d
- Git remote: origin <https://github.com/Rumple12/new-yacoub-thesis.git>
- Recent Git history includes commits such as feat(infra): add local n8n docker baseline

Report chapters it feeds

- Chapter 3 - Methodology: version control, iterative development, AI-tool workflow, evidence-driven report writing.
- Chapter 4 - Choice of approach: one vertical slice, narrow architecture, repo organization.
- Appendix A - Source Code: repository structure and source-code reference.

Limitations or assumptions

- The local repository is available and connected to a GitHub remote, but this note does not prove repository visibility, branch protection, or external reviewer access.

- .agents/ directories exist, but no concrete Antigravity rule, workflow, or skill files were found during this report-scribe pass.
- No implementation behavior is proven by the scaffold alone.

Screenshots/logs still needed

- Optional GitHub repository screenshot for the methodology chapter.
- Optional commit-history excerpt for the development-method section.
- If Antigravity workspace rules/workflows/skills are required as evidence, add the concrete files or record why the directories alone are sufficient.

▼ Step 03 - Local n8n Baseline

What was built

- A local self-hosted n8n runtime baseline was created under infrastructure/docker/.
- The baseline uses Docker Compose with the pinned image n8nio/n8n:1.123.37.
- n8n uses its default local SQLite-backed storage persisted in a Docker volume.
- The baseline intentionally excludes PostgreSQL, Prometheus/Grafana, MCP, middleware, workflows, and Raspberry Pi deployment.

Why it matters for the thesis

This step proves the first Yacoub runtime layer can be started locally before adding middleware, contracts, workflows, safety logic, and hardware complexity. It supports the PC-first development decision and creates a reproducible workflow-engine baseline for later measurement.

Evidence produced

- infrastructure/docker/docker-compose.yml
- infrastructure/docker/.env.example
- infrastructure/docker/README.md
- docs/n8n-version.md
- docs/decisions.md decision D-011
- infrastructure/docker/evidence/n8n-runtime-proof-2026-04-25.txt
- infrastructure/docker/evidence/n8n-localhost-5678-success-2026-04-25.png

Report chapters it feeds

- Chapter 4 - Choice of approach: local self-hosted n8n, PC-first development, simple Docker baseline.
- Chapter 5 - Implementation: infrastructure setup and runtime reproducibility.
- Chapter 6 - Results: later runtime evidence can reference the baseline, but no performance result is produced yet.

Limitations or assumptions

- The runtime was verified locally on Windows with Docker Desktop on 2026-04-25.
- The proof shows n8n reaches the owner setup screen, not that workflows or middleware integration exist.

- No latency, RAM, CPU, or thermal measurement has been produced by this step.
- Raspberry Pi validation remains a later Tier 1.5 step and is not part of this baseline.

Screenshots/logs still needed

- The existing screenshot proves the local n8n owner setup screen.
- Later steps still need workflow screenshots, middleware request/response logs, Docker/resource logs, and measurement CSVs.

▼ ~~step 3 theory~~

```

step 3

creating a tiny mini computer inside of my pc
dedicated entirely to running n8n
-
( why did we create this PC )
* docker is just a isolated testbench for n8n separate from our windows installion ... Also helps us testbench our setup befo
re moving to raspberry pi
-
-
( why did we do this for yacoub-thesis)
This gave us ( Isolation and Reproducibility.)
* n8n installation and runtime is not tangled with other up with other software on the Windows PC
* AND when its time to move project to Raspberry Pi
no need to reinstall things manually
* We just copy the docker-compose recipe over, run docker compose up -d, and it will behave exactly the same way it does
on your PC right now.

* docker allows us to run n8n simply
* (blueprint for container) n8n version 1.123.37
  * It contains the exact OS, code, and tools n8n needs to run perfectly.
* (container itself) - physical running instance of n8n blueprint
* (docker compose up -d) - heavy lifting happens
* (http://localhost:5678) - your browser walked across that port bridge (5678) into the Docker container and asked n8n for a
webpage.
  * page shows ( Setup Owner Account )

wsl
-
docker needs linux
updated to ensure the linux enviroment was up to date for docker to safely run

-

```

▼ step 3 rundown

```

(goal)
before running n8n engine on hardware like raspberry pi
get it running locally on my pc via docker
(what this proves)
This proves that the thesis has a reproducible local runtime.

(create files)
-
the files we created/updated directly line up with the "Evidence Produced" requirement for your thesis report.
In academic terms, this shows how your implementation artifacts (the files) generate the proof (the evidence) that you comple
ted the step.

```

```

-
(1. The Code Maps to the Evidence)
(Docker Compose file)
* infrastructure/docker/(docker-compose.yml)
* infrastructure/docker/(.env.example)
* proves you didn't just install software randomly on your PC. You created a codified, portable infrastructure. You can link to this file in your report's appendix.
-
(Pinned n8n version/image tag)
docs/n8n-version.md -pinned n8n version/image tag
* Proves scientific rigor. If someone reads your thesis in 2028, they know exactly which version of the software you tested on (n8nio/n8n:1.123.37), ensuring your experiment is valid.
-
(Basic run instructions)
* infrastructure/docker/(README.md)
* Proves the environment is reproducible. A core requirement of academic engineering is that someone else can follow your instructions and get the exact same system running.
-
-
-
(2. Your Actions Map to the Evidence)
-
The files are only half the story. The actions you took in the terminal and browser provide the rest of the required evidence:
-
(Local n8n startup logs)
* Ran (docker compose logs n8n) and copied the terminal output.
* Hard proof that the container booted successfully, connected to its volume, and ran its internal database migrations without crashing.
-
(Screenshots of running n8n)
* Opened http://localhost:5678 in Chrome and took a picture.
* Visual proof for the methodology/results section of your thesis report. A picture proves the baseline UI is functional and awaiting workflows.
-
summary
Every single file modification and terminal command from Step 3 wasn't just "coding"—it was directly generating the required artifacts you will copy/paste or reference when you sit down to write Chapter 4 (Choice of Approach) and Chapter 5 (Implementation) of your final report!

```

▼ step 3 rundown x AI assistance (final)

```

(step 3) - Build the local n8n baseline via docker

(AI implementing step 3 rundown)
-
Before running the n8n workflow engine on physical edge hardware (like a Raspberry Pi), we must get it running locally on a PC via Docker.
* What this proves: This proves that the thesis has a reproducible, isolated local runtime that can be used as a controlled environment for testing.

(AI code generation)
-
(AI Task)
step-runner cx built minimal baseline, made infrastructure for
-> (docker-compose.yml, env.example, README.md)
Updated architectural documentation
-> (docs/n8n-version.md, docs/decisions.md)
* It kept the scope strictly narrow (no PostgreSQL, no middleware yet).
-
(Human Task) (Execution & Verification)
The human researcher took over to execute the physical commands provided by the AI (docker compose up -d) on their local Windows machine. The human verified the n8n UI in the browser and captured the required screenshot evidence.
-
(AI Task) (Report Generation)
After human verification, a secondary AI agent (report-scribe) automatically generated the academic report notes for Step 3, cataloging the limitations and linking the generated artifacts to specific thesis chapters.

```

```

(code maps to evidence)
-
The files the AI created directly line up with the "Evidence Produced" requirement for the thesis report. In academic terms,
this shows how the implementation artifacts generate the proof that the step was completed.
-
(Evidence: Docker Compose file)
* infrastructure/docker/docker-compose.yml
* infrastructure/docker/.env.example
* Why it matters: Proves you didn't just install software randomly on your PC. You created a codified, portable infrastructure.
  You can link to this file in your report's appendix.
-
(Evidence: Pinned n8n version/image tag)
* docs/n8n-version.md
* Why it matters: Proves scientific rigor. If someone reads your thesis in 2028, they know exactly which version of the software
  you tested on (n8nio/n8n:1.123.37), ensuring your experiment remains valid.
-
(Evidence: Basic run instructions)
* infrastructure/docker/README.md
* Why it matters: Proves the environment is reproducible. A core requirement of academic engineering is that someone else can
  follow your instructions and get the exact same system running.

(Actions Map to the Evidence(HARD PROOF))
-
The code files are only half the story. The manual actions taken by the human in the terminal and browser provide the rest of
the required evidence:
-
(Evidence: Local n8n startup logs)
* Action: Ran docker compose logs n8n and copied the terminal output.
* Why it matters: Hard proof that the container booted successfully, connected to its volume, and ran its internal database migrations
  without crashing.
-
(Evidence: Screenshots of running n8n)
* Action: Opened http://localhost:5678 in Chrome and took a picture.
* Why it matters: Visual proof for the methodology/results section of the thesis report. A picture proves the baseline UI is
  functional and awaiting workflows.

(📌 Summary)
-
Every single file modification generated by the AI and every terminal command executed by the human wasn't just "coding"—it was
directly generating the required academic artifacts. This combined workflow provides the exact references needed to write
Chapter 4 (Choice of Approach) and Chapter 5 (Implementation) of the final thesis report.

```

- -
-
-

▼ chapter 4

Chapter 4 deliverables

- local self-hosted n8n vs broader/cloud-heavy alternatives
- why local baseline comes before Raspberry Pi

Chapter 4 — Choice of approach feeds from:

```

docs/decisions.md
docs/n8n-version.md
infrastructure/docker/

```

▼ chapter 5

Chapter 5 — Implementation feeds from:

```
infrastructure/docker/docker-compose.yml
infrastructure/docker/.env.example
README.md
```

Chapter 5 delv

- infrastructure setup
- self-hosted n8n runtime
- reproducibility instructions

Goal

~~Get the workflow engine running locally before involving hardware complexity.~~

~~Get the local workflow engine running before adding middleware, hardware, or agent logic.~~

Work to complete

▼ 4

Create the local infrastructure baseline:

- `infrastructure/docker/docker-compose.yml`
- `infrastructure/docker/.env.example`
- basic local n8n startup instructions
- note the n8n version/image used in `docs/n8n-version.md`

▼ 2

Create:

```
infrastructure/docker/docker-compose.yml
infrastructure/docker/.env.example
infrastructure/docker/README.md
```

Update:

```
docs/n8n-version.md
README.md
docs/decisions.md
```

▼ ~~What this proves~~

~~This proves that the thesis has a reproducible local runtime.~~

▼ ~~Evidence produced~~

▼ 4

- Docker files
- local n8n startup logs

- n8n version pin
- startup logs/screenshots - screenshots of running n8n
- startup logs/screenshots - screenshots of running n8n
- README run instructions - basic run instructions

▼ 2

- ~~Docker Compose file~~
- ~~local n8n startup logs~~
- ~~screenshots of running n8n~~
- ~~pinned n8n version/image tag~~
- ~~basic run instructions~~

▼ **Step 4** - Build the middleware ... mock testing fan on/off (middleware)
 (chapter 5 implementation)
 (chapter 3 methodology)

▼ step 4 rundown

▼ files (AI)

▼ cx step runner

▼ q

Current step:

Step 4 — Build the middleware skeleton.

Use these files as source of truth:

- AGENTS.md
- README.md
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/ongoing/integration-contract.md
- docs/evaluation/evaluation-protocol.md

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Obid minimum integration only
- PC-first development with simulated sensor data
- Raspberry Pi comes later as Tier 1.5 validation
- no MCP
- no n8n core modification
- no workflow implementation yet

- no safety-layer implementation yet
- no shared schema implementation yet unless only a tiny placeholder is needed for alignment
- no multiple devices/models
- no heavy observability stack

Goal:

Build the main Yacoub middleware bridge between sensor/action logic and n8n.

Create:

- middleware/api/app.py
- middleware/api/routes.py
- middleware/gpio/mock_sensor.py
- middleware/webhooks/n8n_sender.py
- middleware/tests/manual-test-notes.md

Initial endpoints:

- POST /sensor-event
- POST /fan/on
- POST /fan/off
- GET /status

Requirements:

1. Keep the middleware minimal and runnable locally.
2. Use mock sensor data only at this stage.
3. The middleware should not require real GPIO yet.
4. The fan endpoints may be simulated; do not require real hardware.
5. Include clear request/response examples in the manual test notes.
6. Include basic local run instructions for the middleware.
7. Include manual test commands using curl or PowerShell Invoke-RestMethod.
8. Do not connect to a real n8n workflow yet unless only a configurable placeholder URL is needed.
9. If an n8n webhook URL is needed, keep it configurable and document that it is pending Step 6 workflow work.
10. Do not implement Step 5 shared schemas yet.
11. Do not implement Step 6 deterministic workflow yet.
12. Do not implement Step 7 agent workflow yet.
13. Do not implement Step 8 safety layer yet.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to run the middleware locally.
- tell me how to test each endpoint.
- tell me what evidence should be saved for Chapter 3 and Chapter 5.

▼ a

▼ what it should do

After Codex finishes, your checklist is:

```
middleware files created
middleware starts locally
/status works
/sensor-event accepts mock data
/fan/on works as simulated action
/fan/off works as simulated action
manual-test-notes.md contains commands/results
audit-review passes
report-scribe creates step-04 report note
commit pushed
```

▼ files

5 files changed +451 -0	Undo ↶
middleware/api/app.py +28 -0 •	▼
middleware/api/routes.py +104 -0 •	▼
middleware/gpio/mock_sensor.py +86 -0 •	▼
middleware/tests/manual-test-notes.md +192 -0 •	▼
middleware/webhooks/n8n_sender.py +41 -0 •	▼

Step 4 middleware skeleton is done.

Changed:

- Added app.py as the local server entry point.
- Added routes.py with:
 - `GET /status`
 - `POST /sensor-event`
 - `POST /fan/on`
 - `POST /fan/off`
- Added mock_sensor.py for mock temperature events and simulated fan state.
- Added n8n_sender.py with optional `N8N_WEBHOOK_URL` ; it skips forwarding until Step 6.
- Added manual-test-notes.md with run instructions, PowerShell/curl commands, and example responses.

Run locally from repo root:

```
python -m middleware.api.app
```

Default URL: `http://127.0.0.1:8000`

Test endpoints:

```
Invoke-WebRequest -Method Get -Uri "<http://127.0.0.1:8000/status>"
Invoke-WebRequest -Method Post -Uri "<http://127.0.0.1:8000/fan/on>"
Invoke-WebRequest -Method Post -Uri "<http://127.0.0.1:8000/fan/off>"
```

Sensor event:


```
$body = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
Invoke-RestMethod -Method Post -Uri "<http://127.0.0.1:8000/sensor-event>" -Co
ntentType "application/json" -Body $body
```

Verified:

- `python -m compileall middleware`
- local smoke test for all endpoints on port `8010`
- malformed `/sensor-event` returns HTTP `400`

Evidence to save:

- Chapter 3: startup command, manual test commands, note that PC-first mock data was used.
- Chapter 5: `/status`, `/sensor-event`, `/fan/on`, `/fan/off`, and malformed-event responses.

▼ cx smoke test

▼ q

Create a small evidence file for Step 4 based on the local smoke test results.

Create:

`middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt`

Include:

- command used to start middleware
- tested endpoints
- valid `/sensor-event` result
- `/fan/on` result
- `/fan/off` result
- `/status` result
- malformed `/sensor-event` returned HTTP 400
- note that n8n forwarding is intentionally skipped unless `N8N_WEBHOOK_URL` is configured

Do not change middleware behavior.

Do not start Step 5.

terminal 1

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> cd
```

```
"C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -m
middleware.api.app
```

new-yacoub middleware running at <http://127.0.0.1:8000>

Press Ctrl+C to stop.

```
127.0.0.1 - "GET /status HTTP/1.1" 200 -
```

```
127.0.0.1 - "POST /fan/on HTTP/1.1" 200 -
```

```
127.0.0.1 - "POST /fan/off HTTP/1.1" 200 -
```

127.0.0.1 - "POST /sensor-event HTTP/1.1" 202 -

127.0.0.1 - "POST /sensor-event HTTP/1.1" 400 -

terminal 2

Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Invoke-
RestMethod -Method Get -Uri "<http://127.0.0.1:8000/status>"

status message state

ok middleware running @{fan=off; last_sensor_event=; hardware=simulated}

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Invoke-
RestMethod -Method Post -Uri "<http://127.0.0.1:8000/fan/on>"

status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Invoke-
RestMethod -Method Post -Uri "<http://127.0.0.1:8000/fan/off>"

status : ok
action : fan_off
fan : off
simulated : True
reason : manual_api_call

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> \$body = @{

```
sensor_id = "temp_sensor_1"  
timestamp = "2026-04-25T20:00:00Z"  
type = "temperature"  
value = 31.4  
unit = "C"  
} | ConvertTo-Json
```

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Invoke-
RestMethod -Method Post -Uri "<http://127.0.0.1:8000/sensor-event>" -ContentType "application/json" -
Body \$body

status message event

accepted sensor event received by middleware @{sensor_id=temp_sensor_1; timestamp=2026-04-
25T20:00:00Z; value=31,4; ...

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> \$badBody
= @{

```
sensor_id = "temp_sensor_1"  
} | ConvertTo-Json  
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>  
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>  
$badBody = @{  
sensor_id = "temp_sensor_1"
```

```

} | ConvertTo-Json
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Invoke-
RestMethod -Method Post -Uri "http://127.0.0.1:8000/sensor-event" -ContentType "application/json" -
Body $badBody
Invoke-RestMethod : {
  "error": "missing_required_field",
  "missing": [
    "timestamp",
    "type",
    "value",
    "unit"
  ],
  "message": "Step 4 uses a minimal inline check; JSON schemas come in Step 5."
}
At line:1 char:1

```

- Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/sensor-eve ...
- + CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebExc eption + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>

▼ a

▼ file - # Step 4 middleware smoke test evidence - 2026-04-25

Step 4 middleware smoke test evidence - 2026-04-25

Scope:

- Step 4 middleware skeleton only.
- Mock sensor data only.
- Simulated fan action only.
- No real GPIO.
- No Step 5 shared schemas.
- No Step 6 n8n workflow.
- No Step 7 agent workflow.
- No Step 8 safety layer.

Command used to start middleware

Working directory:

```text

C:\Users\Jake\_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis

...

Command:

```powershell

python -m middleware.api.app

...

Observed startup output:

```text

new-yacoub middleware running at http://127.0.0.1:8000

Press Ctrl+C to stop.

...

### ## Tested endpoints

- `GET /status`
- `POST /fan/on`
- `POST /fan/off`
- `POST /sensor-event` with a valid temperature event
- `POST /sensor-event` with a malformed event

Observed server log:

```text

127.0.0.1 - "GET /status HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/on HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/off HTTP/1.1" 200 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 202 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 400 -

...

GET /status result

Command:

```powershell

Invoke-RestMethod -Method Get -Uri "http://127.0.0.1:8000/status"

...

Observed result:

```text

status message state

- -----

ok middleware running @{fan=off; last_sensor_event=; hardware=simulated}

...

Result summary:

- HTTP status: `200`
- Middleware status: `ok`
- Hardware mode: `simulated`
- Fan state before action tests: `off`

POST /fan/on result

Command:

```powershell

Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/fan/on"

...

Observed result:

```text

```
status : ok
action : fan_on
fan     : on
simulated : True
reason  : manual_api_call
...
```

Result summary:

- HTTP status: `200`
- Simulated fan state changed to `on`

POST /fan/off result

Created the Step 4 evidence file:

[middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt](#)

It includes the middleware startup command, tested endpoints, observed results for `/status`, `/fan/on`, `/fan/off`, valid `/sensor-event`, malformed `/sensor-event` returning HTTP `400`, and the note that n8n forwarding is intentionally skipped unless `N8N_WEBHOOK_URL` is configured.

No middleware behavior changed, and I did not start Step 5.

▼ report scribe

▼ q

Create the report note for Step 4.

Create:

`docs/report-notes/step-04-middleware-skeleton.md`

Use only evidence already present in the repo and the Step 4 middleware files.

Include:

- what was built
- why it matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent results
- do not claim real GPIO exists
- do not claim n8n workflow integration exists yet
- document that mock sensor data and simulated fan state are used at this stage

▼ Step 04 - Middleware Skeleton

What was built

- A minimal Python HTTP middleware skeleton was added under `middleware/`.
- The middleware can be started with `python -m middleware.api.app`.

- Initial endpoints are present:
 - GET /status
 - POST /sensor-event
 - POST /fan/on
 - POST /fan/off
- Mock temperature-event handling is implemented for PC-first development.
- Fan state is simulated in memory through fan_on, fan_off, and get_state.
- A configurable n8n webhook sender placeholder exists, but forwarding is skipped when N8N_WEBHOOK_URL is not configured.

Why it matters for the thesis

This step creates the controlled execution layer between sensor/event input, future n8n workflow logic, and actuator-style actions. It supports the narrowed Yacoub thesis path by proving that the system can begin with mock sensor data and simulated actuator state before adding shared schemas, workflows, safety validation, measurement scripts, or hardware.

It also preserves the safety boundary: workflow logic does not directly control hardware at this stage, and no real GPIO path is implemented.

Evidence produced

- middleware/api/app.py
- middleware/api/routes.py
- middleware/gpio/mock_sensor.py
- middleware/webhooks/n8n_sender.py
- middleware/tests/manual-test-notes.md
- middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt

The smoke-test evidence records:

- middleware startup at http://127.0.0.1:8000
- GET /status returning middleware status and simulated hardware state
- POST /fan/on changing simulated fan state to on
- POST /fan/off changing simulated fan state to off
- POST /sensor-event accepting a valid temperature event
- malformed sensor input returning HTTP 400
- n8n forwarding intentionally skipped because no Step 6 workflow is connected yet

Report chapters it feeds

- Chapter 3 - Methodology: PC-first development, manual smoke testing, mock sensor data before hardware.
- Chapter 5 - Implementation: middleware API structure, sensor-event input path, simulated actuator endpoints, n8n webhook placeholder.
- Chapter 7 - Discussion: limitations of mock execution and why hardware-facing behavior is deferred until validation layers exist.

Limitations or assumptions

- This step uses mock sensor data only.
- Fan behavior is simulated in memory; no real GPIO or physical fan control exists.
- No shared JSON schemas are used yet; Step 4 uses a minimal inline check for required fields.
- No deterministic n8n workflow integration exists yet.
- No agent-enhanced workflow exists yet.
- No safety layer or human approval checkpoint exists yet.
- No latency, RAM, CPU, or thermal/resource measurements are produced by this step.
- Middleware state is process-local and is not persisted across restarts.

Screenshots/logs still needed

- Terminal screenshot or saved log showing middleware startup.
- Request/response logs for `/status`, `/sensor-event`, `/fan/on`, and `/fan/off` in a consistent evidence format.
- Screenshot or exported evidence once the Step 6 n8n workflow actually receives middleware events.
- Later measurement logs for latency, CPU, RAM, and thermal/resource behavior.

```
step 4 - building the middleware
(translator between thermonitor) -> (n8n brain) -> (fan)
-
Build the technical backbone of Yacoub's thesis: the main bridge between sensor/action logic (hardware) and n8n (the brain).
* What this proves: This proves that the system has a secure, controlled execution layer. It demonstrates that the workflow logic (n8n) does not dangerously control hardware directly, but must pass through a strict API bridge.
```

▼ (How AI assisted)

-

▼ (AI task)

(include the code here)

steprunner was instructed to build a minimal python API structure

▼ * (app.py)

```
"""Local middleware entry point for the Step 4 skeleton."""

from __future__ import annotations

import os
from http.server import ThreadingHTTPServer

from middleware.api.routes import MiddlewareRequestHandler

def main() -> None:
    host = os.getenv("MIDDLEWARE_HOST", "127.0.0.1")
    port = int(os.getenv("MIDDLEWARE_PORT", "8000"))
    server = ThreadingHTTPServer((host, port), MiddlewareRequestHandler)

    print(f"new-yacoub middleware running at http://{host}:{port}")
    print("Press Ctrl+C to stop.")

    try:
```

```

        server.serve_forever()
    except KeyboardInterrupt:
        print("\nStopping middleware.")
    finally:
        server.server_close()

if __name__ == "__main__":
    main()

```

▼ * (routes.py)

```

"""HTTP routes for the minimal Step 4 middleware skeleton."""

from __future__ import annotations

import json
from http import HTTPStatus
from http.server import BaseHTTPRequestHandler
from typing import Any
from urllib.parse import urlparse

from middleware.gpio.mock_sensor import (
    fan_off,
    fan_on,
    get_state,
    normalize_sensor_event,
)
from middleware.webhooks.n8n_sender import send_sensor_event

class MiddlewareRequestHandler(BaseHTTPRequestHandler):
    server_version = "NewYacoubMiddleware/0.1"

    def do_GET(self) -> None:
        path = urlparse(self.path).path

        if path == "/status":
            self._send_json(
                HTTPStatus.OK,
                {
                    "status": "ok",
                    "message": "middleware running",
                    "state": get_state(),
                },
            )
            return

        self._send_json(HTTPStatus.NOT_FOUND, {"error": "not_found", "path": path})

    def do_POST(self) -> None:
        path = urlparse(self.path).path

        if path == "/sensor-event":
            self._handle_sensor_event()
            return

        if path == "/fan/on":
            self._send_json(HTTPStatus.OK, fan_on(reason="manual_api_call"))
            return

        if path == "/fan/off":
            self._send_json(HTTPStatus.OK, fan_off(reason="manual_api_call"))
            return

        self._send_json(HTTPStatus.NOT_FOUND, {"error": "not_found", "path": path})

    def log_message(self, format: str, *args: Any) -> None:
        print(f"{self.address_string()} - {format % args}")

    def _handle_sensor_event(self) -> None:
        payload = self._read_json_body()
        if payload is None:
            self._send_json(
                HTTPStatus.BAD_REQUEST,

```



```

        {
            "error": "invalid_json",
            "message": "Request body must be valid JSON.",
        },
    )
    return

    event = normalize_sensor_event(payload)
    if "error" in event:
        self._send_json(HTTPStatus.BAD_REQUEST, event)
        return

    webhook_result = send_sensor_event(event)
    self._send_json(
        HTTPStatus.ACCEPTED,
        {
            "status": "accepted",
            "message": "sensor event received by middleware",
            "event": event,
            "n8n": webhook_result,
        },
    )

def _read_json_body(self) -> dict[str, Any] | None:
    length = int(self.headers.get("Content-Length", "0"))
    raw_body = self.rfile.read(length) if length else b"{}"

    try:
        payload = json.loads(raw_body.decode("utf-8"))
    except json.JSONDecodeError:
        return None

    return payload if isinstance(payload, dict) else None

def _send_json(self, status: HTTPStatus, payload: dict[str, Any]) -> None:
    body = json.dumps(payload, indent=2).encode("utf-8")
    self.send_response(status.value)
    self.send_header("Content-Type", "application/json")
    self.send_header("Content-Length", str(len(body)))
    self.end_headers()
    self.wfile.write(body)

```

▼ * (mock_sensor.py)

```

"""Mock sensor and actuator state for PC-first middleware development."""

from __future__ import annotations

from datetime import UTC, datetime
from typing import Any

_state: dict[str, Any] = {
    "fan": "off",
    "last_sensor_event": None,
}

def utc_now() -> str:
    return datetime.now(UTC).replace(microsecond=0).isoformat().replace("+00:00", "Z")

def mock_temperature_event(value: float = 31.4) -> dict[str, Any]:
    return {
        "sensor_id": "temp_sensor_1",
    }

```

```

        "timestamp": utc_now(),
        "type": "temperature",
        "value": value,
        "unit": "C",
    }

def normalize_sensor_event(payload: dict[str, Any]) -> dict[str, Any]:
    event = mock_temperature_event() if not payload else dict(payload)
    required_fields = ("sensor_id", "timestamp", "type", "value", "unit")
    missing = [field for field in required_fields if field not in event]

    if missing:
        return {
            "error": "missing_required_field",
            "missing": missing,
            "message": "Step 4 uses a minimal inline check; JSON schemas come
in Step 5.",
        }

    if event["type"] != "temperature":
        return {
            "error": "unsupported_sensor_type",
            "message": "Step 4 middleware only accepts temperature events.",
        }

    try:
        event["value"] = float(event["value"])
    except (TypeError, ValueError):
        return {
            "error": "invalid_value",
            "message": "Temperature value must be numeric.",
        }

    _state["last_sensor_event"] = event
    return event

def fan_on(reason: str) -> dict[str, Any]:
    _state["fan"] = "on"
    return {
        "status": "ok",
        "action": "fan_on",
        "fan": _state["fan"],
        "simulated": True,
        "reason": reason,
    }

def fan_off(reason: str) -> dict[str, Any]:
    _state["fan"] = "off"
    return {
        "status": "ok",
        "action": "fan_off",
    }

```

```

        "fan": _state["fan"],
        "simulated": True,
        "reason": reason,
    }

def get_state() -> dict[str, Any]:
    return {
        "fan": _state["fan"],
        "last_sensor_event": _state["last_sensor_event"],
        "hardware": "simulated",
    }

```

▼ * (n8n_sender.py)

```

"""Configurable n8n webhook sender placeholder for later workflow steps."""

from __future__ import annotations

import json
import os
from typing import Any
from urllib.error import URLError
from urllib.request import Request, urlopen

def send_sensor_event(event: dict[str, Any]) -> dict[str, Any]:
    webhook_url = os.getenv("N8N_WEBHOOK_URL", "").strip()

    if not webhook_url:
        return {
            "status": "skipped",
            "reason": "N8N_WEBHOOK_URL is not configured; n8n workflow wiring is pending Step 6.",
        }

    body = json.dumps(event).encode("utf-8")
    request = Request(
        webhook_url,
        data=body,
        headers={"Content-Type": "application/json"},
        method="POST",
    )

    try:
        with urlopen(request, timeout=5) as response:
            response_body = response.read().decode("utf-8", errors="replace")
            return {
                "status": "sent",
                "status_code": response.status,
                "response": response_body,
            }
    except URLError as exc:
        return {
            "status": "error",
            "reason": str(exc),
        }

```

▼ * HTTP endpoints

- **Human Task (Execution & Smoke Testing):** The human researcher took over to verify the code. Opening two separate PowerShell windows, the human booted the Python middleware server in Terminal 1. In Terminal 2, the human executed manual API calls using `Invoke-RestMethod` to test the `/status`, `/fan/on`, `/fan/off`, and `/sensor-event` endpoints.

▼ **AI Task (Evidence Compilation):** The human researcher fed the raw terminal outputs back into the AI. The AI parsed the human's testing logs and formally structured them into an academic artifact (`middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt`), proving the endpoints responded correctly (e.g., returning HTTP `200` for success and `400` for malformed events).

`middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt`

```
# Step 4 middleware smoke test evidence - 2026-04-25
```

```
Scope:
```

- Step 4 middleware skeleton only.
- Mock sensor data only.
- Simulated fan action only.
- No real GPIO.
- No Step 5 shared schemas.
- No Step 6 n8n workflow.
- No Step 7 agent workflow.
- No Step 8 safety layer.

```
## Command used to start middleware
```

```
Working directory:
```

▼ **AI Task (Report Generation):** Finally, the `report-scribe` AI agent analyzed the generated files and evidence to draft the formal report notes (`docs/report-notes/step-04-middleware-skeleton.md`), mapping exactly how this middleware step feeds into Chapter 3 (Methodology) and Chapter 5 (Implementation).

`docs/report-notes/step-04-middleware-skeleton.md`

```
# Middleware Manual Test Notes
```

```
These notes cover the Step 4 middleware skeleton only.
```

```
Scope limits:
```

- mock sensor data only
- simulated fan action only
- no real GPIO
- shared JSON schemas are defined in Step 5 but are not enforced by this Step 4 middleware yet
- no deterministic n8n workflow yet
- no agent workflow yet
- no safety layer yet

```
## Local run
```

```
From the repository root:
```

▼ 📁 1. The Code Maps to the Evidence

The files the AI created directly line up with the thesis requirements.

- **Evidence: Middleware Source Files**

▼ `middleware/api/app.py` & `routes.py` (The Server)

▼ * (app.py)

```
"""Local middleware entry point for the Step 4 skeleton."""

from __future__ import annotations

import os
from http.server import ThreadingHTTPServer

from middleware.api.routes import MiddlewareRequestHandler

def main() -> None:
    host = os.getenv("MIDDLEWARE_HOST", "127.0.0.1")
    port = int(os.getenv("MIDDLEWARE_PORT", "8000"))
    server = ThreadingHTTPServer((host, port), MiddlewareRequestHandler)

    print(f"new-yacoub middleware running at http://{host}:{port}")
    print("Press Ctrl+C to stop.")

    try:
        server.serve_forever()
    except KeyboardInterrupt:
        print("\nStopping middleware.")
    finally:
        server.server_close()

if __name__ == "__main__":
    main()
```

▼ * (routes.py)

```
"""HTTP routes for the minimal Step 4 middleware skeleton."""

from __future__ import annotations

import json
from http import HTTPStatus
from http.server import BaseHTTPRequestHandler
from typing import Any
from urllib.parse import urlparse

from middleware.gpio.mock_sensor import (
    fan_off,
    fan_on,
    get_state,
    normalize_sensor_event,
)
from middleware.webhooks.n8n_sender import send_sensor_event

class MiddlewareRequestHandler(BaseHTTPRequestHandler):
    server_version = "NewYacoubMiddleware/0.1"

    def do_GET(self) -> None:
        path = urlparse(self.path).path

        if path == "/status":
            self._send_json(
                HTTPStatus.OK,
                {
                    "status": "ok",
                    "message": "middleware running",
                    "state": get_state(),
                },
            )
            return

        self._send_json(HTTPStatus.NOT_FOUND, {"error": "not_found", "path": path})
```

```

def do_POST(self) -> None:
    path = urlparse(self.path).path

    if path == "/sensor-event":
        self._handle_sensor_event()
        return

    if path == "/fan/on":
        self._send_json(HTTPStatus.OK, fan_on(reason="manual_api_call"))
        return

    if path == "/fan/off":
        self._send_json(HTTPStatus.OK, fan_off(reason="manual_api_call"))
        return

    self._send_json(HTTPStatus.NOT_FOUND, {"error": "not_found", "path": path})

def log_message(self, format: str, *args: Any) -> None:
    print(f"{self.address_string()} - {format % args}")

def _handle_sensor_event(self) -> None:
    payload = self._read_json_body()
    if payload is None:
        self._send_json(
            HTTPStatus.BAD_REQUEST,
            {
                "error": "invalid_json",
                "message": "Request body must be valid JSON.",
            },
        )
        return

    event = normalize_sensor_event(payload)
    if "error" in event:
        self._send_json(HTTPStatus.BAD_REQUEST, event)
        return

    webhook_result = send_sensor_event(event)
    self._send_json(
        HTTPStatus.ACCEPTED,
        {
            "status": "accepted",
            "message": "sensor event received by middleware",
            "event": event,
            "n8n": webhook_result,
        },
    )

def _read_json_body(self) -> dict[str, Any] | None:
    length = int(self.headers.get("Content-Length", "0"))
    raw_body = self.rfile.read(length) if length else b"{}"

    try:
        payload = json.loads(raw_body.decode("utf-8"))
    except json.JSONDecodeError:
        return None

    return payload if isinstance(payload, dict) else None

def _send_json(self, status: HTTPStatus, payload: dict[str, Any]) -> None:
    body = json.dumps(payload, indent=2).encode("utf-8")
    self.send_response(status.value)
    self.send_header("Content-Type", "application/json")
    self.send_header("Content-Length", str(len(body)))
    self.end_headers()
    self.wfile.write(body)

```

▼ `middleware/gpio/mock_sensor.py` (The Fake Hardware)

▼ * `(mock_sensor.py)`

```

"""Mock sensor and actuator state for PC-first middleware development."""

```

```

from __future__ import annotations

from datetime import UTC, datetime
from typing import Any

_state: dict[str, Any] = {
    "fan": "off",
    "last_sensor_event": None,
}

def utc_now() -> str:
    return datetime.now(UTC).replace(microsecond=0).isoformat().replace("+00:00", "Z")

def mock_temperature_event(value: float = 31.4) -> dict[str, Any]:
    return {
        "sensor_id": "temp_sensor_1",
        "timestamp": utc_now(),
        "type": "temperature",
        "value": value,
        "unit": "C",
    }

def normalize_sensor_event(payload: dict[str, Any]) -> dict[str, Any]:
    event = mock_temperature_event() if not payload else dict(payload)
    required_fields = ("sensor_id", "timestamp", "type", "value", "unit")
    missing = [field for field in required_fields if field not in event]

    if missing:
        return {
            "error": "missing_required_field",
            "missing": missing,
            "message": "Step 4 uses a minimal inline check; JSON schemas come in Step 5.",
        }

    if event["type"] != "temperature":
        return {
            "error": "unsupported_sensor_type",
            "message": "Step 4 middleware only accepts temperature events.",
        }

    try:
        event["value"] = float(event["value"])
    except (TypeError, ValueError):
        return {
            "error": "invalid_value",
            "message": "Temperature value must be numeric.",
        }

```

```

        _state["last_sensor_event"] = event
    return event

def fan_on(reason: str) -> dict[str, Any]:
    _state["fan"] = "on"
    return {
        "status": "ok",
        "action": "fan_on",
        "fan": _state["fan"],
        "simulated": True,
        "reason": reason,
    }

def fan_off(reason: str) -> dict[str, Any]:
    _state["fan"] = "off"
    return {
        "status": "ok",
        "action": "fan_off",
        "fan": _state["fan"],
        "simulated": True,
        "reason": reason,
    }

def get_state() -> dict[str, Any]:
    return {
        "fan": _state["fan"],
        "last_sensor_event": _state["last_sensor_event"],
        "hardware": "simulated",
    }

```

▼ `middleware/webhooks/n8n_sender.py` (The Messenger)

▼ * (n8n_sender.py)

```

"""Configurable n8n webhook sender placeholder for later workflow steps."""

from __future__ import annotations

import json
import os
from typing import Any
from urllib.error import URLError
from urllib.request import Request, urlopen

def send_sensor_event(event: dict[str, Any]) -> dict[str, Any]:
    webhook_url = os.getenv("N8N_WEBHOOK_URL", "").strip()

    if not webhook_url:
        return {
            "status": "skipped",
            "reason": "N8N_WEBHOOK_URL is not configured; n8n workflow wiring is pending Step 6.",
        }

    body = json.dumps(event).encode("utf-8")
    request = Request(

```



```

        webhook_url,
        data=body,
        headers={"Content-Type": "application/json"},
        method="POST",
    )

    try:
        with urlopen(request, timeout=5) as response:
            response_body = response.read().decode("utf-8", errors="replace")
            return {
                "status": "sent",
                "status_code": response.status,
                "response": response_body,
            }
    except URLError as exc:
        return {
            "status": "error",
            "reason": str(exc),
        }

```

- *Why it matters:* Proves the architecture is decoupled. You have a dedicated API layer ready to accept physical hardware connections in later validation tiers (Tier 1.5).

▼ 2. The Actions Map to the Evidence

The human testing process was converted into formal documentation.

- **Evidence: Endpoint Responses & Manual Test Notes**

▼ *Action:* Human ran `python -m middleware.api.app` and pinged the server with PowerShell command

```

PS C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -m middleware.api.app
new-yacoub middleware running at http://127.0.0.1:8000
Press Ctrl+C to stop.
127.0.0.1 - "GET /status HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/on HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/off HTTP/1.1" 200 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 202 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 400 -
127.0.0.1 - "GET /status HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/on HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/off HTTP/1.1" 200 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 202 -

```

▼ manually ping the server

```

PS C:\WINDOWS\system32> wsl --update
Den begärda åtgärden kräver en privilegiehöjning.
Downloading: Windows Subsystem for Linux 2.6.3
Installing: Windows Subsystem for Linux 2.6.3
Windows Subsystem for Linux 2.6.3 has been installed.
Åtgärden har slutförts.
Leta efter uppdateringar.
Den senaste versionen av Windows-undersystem för Linux är redan installerad.
PS C:\WINDOWS\system32> cd "C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"
PS C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker --version
Docker version 29.4.0, build 9d7ad9f
PS C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker compose version

```

```

Docker Compose version v5.1.2
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker compose up -d
[+] up 19/19
✓ Image n8nio/n8n:1.123.37 Pulled
291.8s
✓ Network docker_default Created
0.2s
✓ Volume docker_n8n_data Created
0.0s
✓ Container new-yacoub-n8n Started
2.2s
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker compose ps

```

| NAME | IMAGE | COMMAND | SERVICE | CREATED |
|----------------|--------------------|--------------------------|---------|----------------|
| new-yacoub-n8n | n8nio/n8n:1.123.37 | "tini -- /docker-ent..." | n8n | 26 seconds ago |

```

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker compose logs n8n
new-yacoub-n8n | Permissions 0644 for n8n settings file /home/node/.n8n/config are too wide. This is ignored for now, but in the future n8n will attempt to change the permissions automatically. To automatically enforce correct permissions now set N8N_ENFORCE_SETTINGS_FILE_PERMISSIONS=true (recommended), or turn this check off set N8N_ENFORCE_SETTINGS_FILE_PERMISSIONS=false.
new-yacoub-n8n | (node:7) [DEP0040] DeprecationWarning: The `punycode` module is deprecated. Please use a userland alternative instead.
new-yacoub-n8n | (Use `node --trace-deprecation ...` to show where the warning was created)
new-yacoub-n8n | Initializing n8n process
new-yacoub-n8n | n8n ready on ::, port 5678
new-yacoub-n8n | Migrations in progress, please do NOT stop the process.
new-yacoub-n8n | Starting migration InitialMigration1588102412422
new-yacoub-n8n | Finished migration InitialMigration1588102412422
new-yacoub-n8n | Starting migration WebhookModel1592445003908
new-yacoub-n8n | Finished migration WebhookModel1592445003908
new-yacoub-n8n | Starting migration CreateIndexStoppedAt1594825041918
new-yacoub-n8n | Finished migration CreateIndexStoppedAt1594825041918
new-yacoub-n8n | Starting migration MakeStoppedAtNullable1607431743769
new-yacoub-n8n | Finished migration MakeStoppedAtNullable1607431743769
new-yacoub-n8n | Starting migration AddWebhookId1611071044839
new-yacoub-n8n | Finished migration AddWebhookId1611071044839
new-yacoub-n8n | Starting migration CreateTagEntity1617213344594
new-yacoub-n8n | Finished migration CreateTagEntity1617213344594
new-yacoub-n8n | Starting migration UniqueWorkflowNames1620821879465
new-yacoub-n8n | Finished migration UniqueWorkflowNames1620821879465
new-yacoub-n8n | Starting migration AddWaitColumn1621707690587
new-yacoub-n8n | Finished migration AddWaitColumn1621707690587
new-yacoub-n8n | Starting migration UpdateWorkflowCredentials1630330987096
new-yacoub-n8n | Finished migration UpdateWorkflowCredentials1630330987096
new-yacoub-n8n | Starting migration AddExecutionEntityIndexes1644421939510
new-yacoub-n8n | Finished migration AddExecutionEntityIndexes1644421939510
new-yacoub-n8n | Starting migration CreateUserManagement1646992772331
new-yacoub-n8n | Finished migration CreateUserManagement1646992772331

```

| | | | | |
|----------------|--|----------|-----------|--|
| new-yacoub-n8n | | Starting | migration | LowerCaseUserEmail1648740597343 |
| new-yacoub-n8n | | Finished | migration | LowerCaseUserEmail1648740597343 |
| new-yacoub-n8n | | Starting | migration | CommunityNodes1652254514001 |
| new-yacoub-n8n | | Finished | migration | CommunityNodes1652254514001 |
| new-yacoub-n8n | | Starting | migration | AddUserSettings1652367743993 |
| new-yacoub-n8n | | Finished | migration | AddUserSettings1652367743993 |
| new-yacoub-n8n | | Starting | migration | AddAPIKeyColumn1652905585850 |
| new-yacoub-n8n | | Finished | migration | AddAPIKeyColumn1652905585850 |
| new-yacoub-n8n | | Starting | migration | IntroducePinData1654089251344 |
| new-yacoub-n8n | | Finished | migration | IntroducePinData1654089251344 |
| new-yacoub-n8n | | Starting | migration | AddNodeIds1658930531669 |
| new-yacoub-n8n | | Finished | migration | AddNodeIds1658930531669 |
| new-yacoub-n8n | | Starting | migration | AddJsonKeyPinData1659888469333 |
| new-yacoub-n8n | | Finished | migration | AddJsonKeyPinData1659888469333 |
| new-yacoub-n8n | | Starting | migration | CreateCredentialsUserRole1660062385367 |
| new-yacoub-n8n | | Finished | migration | CreateCredentialsUserRole1660062385367 |
| new-yacoub-n8n | | Starting | migration | CreateWorkflowsEditorRole1663755770892 |
| new-yacoub-n8n | | Finished | migration | CreateWorkflowsEditorRole1663755770892 |
| new-yacoub-n8n | | Starting | migration | WorkflowStatistics1664196174000 |
| new-yacoub-n8n | | Finished | migration | WorkflowStatistics1664196174000 |
| new-yacoub-n8n | | Starting | migration | CreateCredentialUsageTable1665484192211 |
| new-yacoub-n8n | | Finished | migration | CreateCredentialUsageTable1665484192211 |
| new-yacoub-n8n | | Starting | migration | RemoveCredentialUsageTable1665754637024 |
| new-yacoub-n8n | | Finished | migration | RemoveCredentialUsageTable1665754637024 |
| new-yacoub-n8n | | Starting | migration | AddWorkflowVersionIdColumn1669739707124 |
| new-yacoub-n8n | | Finished | migration | AddWorkflowVersionIdColumn1669739707124 |
| new-yacoub-n8n | | Starting | migration | AddTriggerCountColumn1669823906993 |
| new-yacoub-n8n | | Finished | migration | AddTriggerCountColumn1669823906993 |
| new-yacoub-n8n | | Starting | migration | MessageEventBusDestinations1671535397530 |
| new-yacoub-n8n | | Finished | migration | MessageEventBusDestinations1671535397530 |
| new-yacoub-n8n | | Starting | migration | RemoveWorkflowDataLoadedFlag1671726148419 |
| new-yacoub-n8n | | Finished | migration | RemoveWorkflowDataLoadedFlag1671726148419 |
| new-yacoub-n8n | | Starting | migration | DeleteExecutionsWithWorkflows1673268682475 |
| new-yacoub-n8n | | Finished | migration | DeleteExecutionsWithWorkflows1673268682475 |
| new-yacoub-n8n | | Starting | migration | AddStatusToExecutions1674138566000 |
| new-yacoub-n8n | | Finished | migration | AddStatusToExecutions1674138566000 |
| new-yacoub-n8n | | Starting | migration | CreateLdapEntities1674509946020 |
| new-yacoub-n8n | | Finished | migration | CreateLdapEntities1674509946020 |
| new-yacoub-n8n | | Starting | migration | PurgeInvalidWorkflowConnections1675940580449 |
| new-yacoub-n8n | | Finished | migration | PurgeInvalidWorkflowConnections1675940580449 |
| new-yacoub-n8n | | Starting | migration | MigrateExecutionStatus1676996103000 |
| new-yacoub-n8n | | Finished | migration | MigrateExecutionStatus1676996103000 |
| new-yacoub-n8n | | Starting | migration | UpdateRunningExecutionStatus1677237073720 |
| new-yacoub-n8n | | Finished | migration | UpdateRunningExecutionStatus1677237073720 |
| new-yacoub-n8n | | Starting | migration | CreateVariables1677501636752 |
| new-yacoub-n8n | | Finished | migration | CreateVariables1677501636752 |
| new-yacoub-n8n | | Starting | migration | CreateExecutionMetadataTable1679416281777 |
| new-yacoub-n8n | | Finished | migration | CreateExecutionMetadataTable1679416281777 |
| new-yacoub-n8n | | Starting | migration | AddUserActivatedProperty1681134145996 |
| new-yacoub-n8n | | Finished | migration | AddUserActivatedProperty1681134145996 |

| | | |
|----------------|--|---|
| new-yacoub-n8n | Starting migration | RemoveSkipOwnerSetup1681134145997 |
| new-yacoub-n8n | Finished migration | RemoveSkipOwnerSetup1681134145997 |
| new-yacoub-n8n | Starting migration | MigrateIntegerKeysToString1690000000002 |
| new-yacoub-n8n | Finished migration | MigrateIntegerKeysToString1690000000002 |
| new-yacoub-n8n | Starting migration | SeparateExecutionData1690000000010 |
| new-yacoub-n8n | Finished migration | SeparateExecutionData1690000000010 |
| new-yacoub-n8n | Starting migration | FixMissingIndicesFromStringIdMigration1690000000020 |
| new-yacoub-n8n | Finished migration | FixMissingIndicesFromStringIdMigration1690000000020 |
| new-yacoub-n8n | Starting migration | RemoveResetPasswordColumns1690000000030 |
| new-yacoub-n8n | Finished migration | RemoveResetPasswordColumns1690000000030 |
| new-yacoub-n8n | Starting migration | AddMfaColumns1690000000030 |
| new-yacoub-n8n | Finished migration | AddMfaColumns1690000000030 |
| new-yacoub-n8n | Starting migration | CreateWorkflowNameIndex1691088862123 |
| new-yacoub-n8n | Finished migration | CreateWorkflowNameIndex1691088862123 |
| new-yacoub-n8n | Starting migration | CreateWorkflowHistoryTable1692967111175 |
| new-yacoub-n8n | Finished migration | CreateWorkflowHistoryTable1692967111175 |
| new-yacoub-n8n | Starting migration | ExecutionSoftDelete1693491613982 |
| new-yacoub-n8n | Finished migration | ExecutionSoftDelete1693491613982 |
| new-yacoub-n8n | Starting migration | DisallowOrphanExecutions1693554410387 |
| new-yacoub-n8n | Finished migration | DisallowOrphanExecutions1693554410387 |
| new-yacoub-n8n | Starting migration | AddWorkflowMetadata1695128658538 |
| new-yacoub-n8n | Finished migration | AddWorkflowMetadata1695128658538 |
| new-yacoub-n8n | Starting migration | ModifyWorkflowHistoryNodesAndConnections1695829275184 |
| new-yacoub-n8n | Finished migration | ModifyWorkflowHistoryNodesAndConnections1695829275184 |
| new-yacoub-n8n | Starting migration | AddGlobalAdminRole1700571993961 |
| new-yacoub-n8n | Finished migration | AddGlobalAdminRole1700571993961 |
| new-yacoub-n8n | Starting migration | DropRoleMapping1705429061930 |
| new-yacoub-n8n | Finished migration | DropRoleMapping1705429061930 |
| new-yacoub-n8n | Starting migration | RemoveFailedExecutionStatus1711018413374 |
| new-yacoub-n8n | Finished migration | RemoveFailedExecutionStatus1711018413374 |
| new-yacoub-n8n | Starting migration | MoveSshKeysToDatabase1711390882123 |
| new-yacoub-n8n | [MoveSshKeysToDatabase1711390882123] No SSH keys in filesystem, skipping | |
| new-yacoub-n8n | Finished migration | MoveSshKeysToDatabase1711390882123 |
| new-yacoub-n8n | Starting migration | RemoveNodesAccess1712044305787 |
| new-yacoub-n8n | Finished migration | RemoveNodesAccess1712044305787 |
| new-yacoub-n8n | Starting migration | CreateProject1714133768519 |
| new-yacoub-n8n | Finished migration | CreateProject1714133768519 |
| new-yacoub-n8n | Starting migration | MakeExecutionStatusNonNullable1714133768521 |
| new-yacoub-n8n | Finished migration | MakeExecutionStatusNonNullable1714133768521 |
| new-yacoub-n8n | Starting migration | AddActivatedAtUserSetting1717498465931 |
| new-yacoub-n8n | Finished migration | AddActivatedAtUserSetting1717498465931 |
| new-yacoub-n8n | Starting migration | AddConstraintToExecutionMetadata1720101653148 |
| new-yacoub-n8n | Finished migration | AddConstraintToExecutionMetadata1720101653148 |
| new-yacoub-n8n | Starting migration | CreateInvalidAuthTokenTable1723627610222 |
| new-yacoub-n8n | Finished migration | CreateInvalidAuthTokenTable1723627610222 |

| | | | |
|----------------|--|--------------------|---|
| new-yacoub-n8n | | Starting migration | RefactorExecutionIndices1723796243146 |
| new-yacoub-n8n | | Finished migration | RefactorExecutionIndices1723796243146 |
| new-yacoub-n8n | | Starting migration | CreateAnnotationTables1724753530828 |
| new-yacoub-n8n | | Finished migration | CreateAnnotationTables1724753530828 |
| new-yacoub-n8n | | Starting migration | AddApiKeysTable1724951148974 |
| new-yacoub-n8n | | Finished migration | AddApiKeysTable1724951148974 |
| new-yacoub-n8n | | Starting migration | CreateProcessedDataTable1726606152711 |
| new-yacoub-n8n | | Finished migration | CreateProcessedDataTable1726606152711 |
| new-yacoub-n8n | | Starting migration | SeparateExecutionCreationFromStart1727427440136 |
| new-yacoub-n8n | | Finished migration | SeparateExecutionCreationFromStart1727427440136 |
| new-yacoub-n8n | | Starting migration | AddMissingPrimaryKeyOnAnnotationTagMapping1728659839644 |
| new-yacoub-n8n | | Finished migration | AddMissingPrimaryKeyOnAnnotationTagMapping1728659839644 |
| new-yacoub-n8n | | Starting migration | UpdateProcessedDataValueColumnToText1729607673464 |
| new-yacoub-n8n | | Finished migration | UpdateProcessedDataValueColumnToText1729607673464 |
| new-yacoub-n8n | | Starting migration | AddProjectIcons1729607673469 |
| new-yacoub-n8n | | Finished migration | AddProjectIcons1729607673469 |
| new-yacoub-n8n | | Starting migration | CreateTestDefinitionTable1730386903556 |
| new-yacoub-n8n | | Finished migration | CreateTestDefinitionTable1730386903556 |
| new-yacoub-n8n | | Starting migration | AddDescriptionToTestDefinition1731404028106 |
| new-yacoub-n8n | | Finished migration | AddDescriptionToTestDefinition1731404028106 |
| new-yacoub-n8n | | Starting migration | MigrateTestDefinitionKeyToString1731582748663 |
| new-yacoub-n8n | | Finished migration | MigrateTestDefinitionKeyToString1731582748663 |
| new-yacoub-n8n | | Starting migration | CreateTestMetricTable1732271325258 |
| new-yacoub-n8n | | Finished migration | CreateTestMetricTable1732271325258 |
| new-yacoub-n8n | | Starting migration | CreateTestRun1732549866705 |
| new-yacoub-n8n | | Finished migration | CreateTestRun1732549866705 |
| new-yacoub-n8n | | Starting migration | AddMockedNodesColumnToTestDefinition1733133775640 |
| new-yacoub-n8n | | Finished migration | AddMockedNodesColumnToTestDefinition1733133775640 |
| new-yacoub-n8n | | Starting migration | AddManagedColumnToCredentialsTable1734479635324 |
| new-yacoub-n8n | | Finished migration | AddManagedColumnToCredentialsTable1734479635324 |
| new-yacoub-n8n | | Starting migration | AddStatsColumnsToTestRun1736172058779 |
| new-yacoub-n8n | | Finished migration | AddStatsColumnsToTestRun1736172058779 |
| new-yacoub-n8n | | Starting migration | CreateTestCaseExecutionTable1736947513045 |
| new-yacoub-n8n | | Finished migration | CreateTestCaseExecutionTable1736947513045 |
| new-yacoub-n8n | | Starting migration | AddErrorColumnsToTestRuns1737715421462 |
| new-yacoub-n8n | | Finished migration | AddErrorColumnsToTestRuns1737715421462 |
| new-yacoub-n8n | | Starting migration | CreateFolderTable1738709609940 |
| new-yacoub-n8n | | Finished migration | CreateFolderTable1738709609940 |
| new-yacoub-n8n | | Starting migration | CreateAnalyticsTables1739549398681 |
| new-yacoub-n8n | | Finished migration | CreateAnalyticsTables1739549398681 |

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|----------------|--|----------|-----------|---|
| new-yacoub-n8n | | Starting | migration | UpdateParentFolderIdColumn1740445074052 |
| new-yacoub-n8n | | Finished | migration | UpdateParentFolderIdColumn1740445074052 |
| new-yacoub-n8n | | Starting | migration | RenameAnalyticsToInsights1741167584277 |
| new-yacoub-n8n | | Finished | migration | RenameAnalyticsToInsights1741167584277 |
| new-yacoub-n8n | | Starting | migration | AddScopesColumnToApiKeys1742918400000 |
| new-yacoub-n8n | | Finished | migration | AddScopesColumnToApiKeys1742918400000 |
| new-yacoub-n8n | | Starting | migration | ClearEvaluation1745322634000 |
| new-yacoub-n8n | | Finished | migration | ClearEvaluation1745322634000 |
| new-yacoub-n8n | | Starting | migration | AddWorkflowStatisticsRootCount1745587087521 |
| new-yacoub-n8n | | Finished | migration | AddWorkflowStatisticsRootCount1745587087521 |
| new-yacoub-n8n | | Starting | migration | AddWorkflowArchivedColumn1745934666076 |
| new-yacoub-n8n | | Finished | migration | AddWorkflowArchivedColumn1745934666076 |
| new-yacoub-n8n | | Starting | migration | DropRoleTable1745934666077 |
| new-yacoub-n8n | | Finished | migration | DropRoleTable1745934666077 |
| new-yacoub-n8n | | Starting | migration | AddProjectDescriptionColumn1747824239000 |
| new-yacoub-n8n | | Finished | migration | AddProjectDescriptionColumn1747824239000 |
| new-yacoub-n8n | | Starting | migration | AddLastActiveAtColumnToUser1750252139166 |
| new-yacoub-n8n | | Finished | migration | AddLastActiveAtColumnToUser1750252139166 |
| new-yacoub-n8n | | Starting | migration | AddScopeTables1750252139166 |
| new-yacoub-n8n | | Finished | migration | AddScopeTables1750252139166 |
| new-yacoub-n8n | | Starting | migration | AddRolesTables1750252139167 |
| new-yacoub-n8n | | Finished | migration | AddRolesTables1750252139167 |
| new-yacoub-n8n | | Starting | migration | LinkRoleToUserTable1750252139168 |
| new-yacoub-n8n | | Finished | migration | LinkRoleToUserTable1750252139168 |
| new-yacoub-n8n | | Starting | migration | RemoveOldRoleColumn1750252139170 |
| new-yacoub-n8n | | Finished | migration | RemoveOldRoleColumn1750252139170 |
| new-yacoub-n8n | | Starting | migration | AddInputsOutputsToTestCaseExecution1752669793000 |
| new-yacoub-n8n | | Finished | migration | AddInputsOutputsToTestCaseExecution1752669793000 |
| new-yacoub-n8n | | Starting | migration | LinkRoleToProjectRelationTable1753953244168 |
| new-yacoub-n8n | | Finished | migration | LinkRoleToProjectRelationTable1753953244168 |
| new-yacoub-n8n | | Starting | migration | CreateDataStoreTables1754475614601 |
| new-yacoub-n8n | | Finished | migration | CreateDataStoreTables1754475614601 |
| new-yacoub-n8n | | Starting | migration | ReplaceDataStoreTablesWithDataTables1754475614602 |
| new-yacoub-n8n | | Finished | migration | ReplaceDataStoreTablesWithDataTables1754475614602 |
| new-yacoub-n8n | | Starting | migration | AddTimestampsToRoleAndRoleIndexes1756906557570 |
| new-yacoub-n8n | | Finished | migration | AddTimestampsToRoleAndRoleIndexes1756906557570 |
| new-yacoub-n8n | | Starting | migration | AddAudienceColumnToApiKeys1758731786132 |
| new-yacoub-n8n | | Finished | migration | AddAudienceColumnToApiKeys1758731786132 |
| new-yacoub-n8n | | Starting | migration | AddProjectIdToVariableTable1758794506893 |
| new-yacoub-n8n | | Finished | migration | AddProjectIdToVariableTable1758794506893 |
| new-yacoub-n8n | | Starting | migration | ChangeValueTypesForInsights1759399811000 |
| new-yacoub-n8n | | Finished | migration | ChangeValueTypesForInsights1759399811000 |
| new-yacoub-n8n | | Starting | migration | CreateChatHubTables1760019379982 |
| new-yacoub-n8n | | Finished | migration | CreateChatHubTables1760019379982 |

| | | |
|----------------|--------------------|---|
| new-yacoub-n8n | Starting migration | CreateChatHubAgentTable1760020000000 |
| new-yacoub-n8n | Finished migration | CreateChatHubAgentTable1760020000000 |
| new-yacoub-n8n | Starting migration | UniqueRoleNames1760020838000 |
| new-yacoub-n8n | Finished migration | UniqueRoleNames1760020838000 |
| new-yacoub-n8n | Starting migration | Create0AuthEntities1760116750277 |
| new-yacoub-n8n | Finished migration | Create0AuthEntities1760116750277 |
| new-yacoub-n8n | Starting migration | CreateWorkflowDependencyTable1760314000000 |
| new-yacoub-n8n | Finished migration | CreateWorkflowDependencyTable1760314000000 |
| new-yacoub-n8n | Starting migration | DropUnusedChatHubColumns1760965142113 |
| new-yacoub-n8n | Finished migration | DropUnusedChatHubColumns1760965142113 |
| new-yacoub-n8n | Starting migration | AddWorkflowVersionColumn1761047826451 |
| new-yacoub-n8n | Finished migration | AddWorkflowVersionColumn1761047826451 |
| new-yacoub-n8n | Starting migration | ChangeDependencyInfoToJson1761655473000 |
| new-yacoub-n8n | Finished migration | ChangeDependencyInfoToJson1761655473000 |
| new-yacoub-n8n | Starting migration | AddAttachmentsToChatHubMessages1761773155024 |
| new-yacoub-n8n | Finished migration | AddAttachmentsToChatHubMessages1761773155024 |
| new-yacoub-n8n | Starting migration | AddToolsColumnToChatHubTables1761830340990 |
| new-yacoub-n8n | Finished migration | AddToolsColumnToChatHubTables1761830340990 |
| new-yacoub-n8n | Starting migration | AddWorkflowDescriptionColumn1762177736257 |
| new-yacoub-n8n | Finished migration | AddWorkflowDescriptionColumn1762177736257 |
| new-yacoub-n8n | Starting migration | BackfillMissingWorkflowHistoryRecords1762763704614 |
| new-yacoub-n8n | Finished migration | BackfillMissingWorkflowHistoryRecords1762763704614 |
| new-yacoub-n8n | Starting migration | AddIsGlobalColumnToCredentialsTable1762771954619 |
| new-yacoub-n8n | Finished migration | AddIsGlobalColumnToCredentialsTable1762771954619 |
| new-yacoub-n8n | Starting migration | AddWorkflowHistoryAutoSaveFields1762847206508 |
| new-yacoub-n8n | Finished migration | AddWorkflowHistoryAutoSaveFields1762847206508 |
| new-yacoub-n8n | Starting migration | AddActiveVersionIdColumn1763047800000 |
| new-yacoub-n8n | Finished migration | AddActiveVersionIdColumn1763047800000 |
| new-yacoub-n8n | Starting migration | Change0AuthStateColumnToUnboundedVarchar1763572724000 |
| new-yacoub-n8n | Finished migration | Change0AuthStateColumnToUnboundedVarchar1763572724000 |
| new-yacoub-n8n | Starting migration | CreateBinaryDataTable1763716655000 |
| new-yacoub-n8n | Finished migration | CreateBinaryDataTable1763716655000 |
| new-yacoub-n8n | Starting migration | CreateWorkflowPublishHistoryTable1764167920585 |
| new-yacoub-n8n | Finished migration | CreateWorkflowPublishHistoryTable1764167920585 |
| new-yacoub-n8n | Starting migration | BackfillMissingWorkflowHistoryRecords1765448186933 |
| new-yacoub-n8n | Finished migration | BackfillMissingWorkflowHistoryRecords1765448186933 |

```

new-yacoub-n8n |
new-yacoub-n8n | There are deprecations related to your environment variable
s. Please take the recommended actions to update your configuration:
new-yacoub-n8n | - DB_SQLITE_POOL_SIZE -> Running SQLite without a pool of r
ead connections is deprecated. Please set `DB_SQLITE_POOL_SIZE` to a value hig
her than zero. See: https://docs.n8n.io/hosting/configuration/environment-variables/database/#sqlite
new-yacoub-n8n | - N8N_RUNNERS_ENABLED -> Running n8n without task runners i
s deprecated. Task runners will be turned on by default in a future version. P
lease set `N8N_RUNNERS_ENABLED=true` to enable task runners now and avoid pote
ntial issues in the future. Learn more: https://docs.n8n.io/hosting/configuration/task-runners/
new-yacoub-n8n | - N8N_BLOCK_ENV_ACCESS_IN_NODE -> The default value of N8N_
BLOCK_ENV_ACCESS_IN_NODE will be changed from false to true in a future versio
n. If you need to access environment variables from the Code Node or from expr
essions, please set N8N_BLOCK_ENV_ACCESS_IN_NODE=false. Learn more: https://do
cs.n8n.io/hosting/configuration/environment-variables/security/
new-yacoub-n8n | - N8N_GIT_NODE_DISABLE_BARE_REPOS -> Support for bare repos
itories in the Git Node will be removed in a future version due to security co
ncerns. If you are not using bare repositories in the Git Node, please set N8N
_GIT_NODE_DISABLE_BARE_REPOS=true. Learn more: https://docs.n8n.io/hosting/con
figuration/environment-variables/security/
new-yacoub-n8n |
new-yacoub-n8n | [license SDK] Skipping renewal on init: license cert is not
initialized
new-yacoub-n8n | Version: 1.123.37
new-yacoub-n8n |
new-yacoub-n8n | Editor is now accessible via:
new-yacoub-n8n | [http://localhost:5678](http://localhost:5678/)
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yaco
ub-thesis\infrastructure\docker> [http://localhost:5678](http://localhost:5678/)
(http://localhost:5678/) : The term '[http://localhost:5678](http://localhost:5678/)' is not recognized as the name of a cmdlet, function, script fi
le, or operable program. Check the spelling of the name, or if a path was incl
uded, verify that the path is correct and
try again.
At line:1 char:1

```

▼ *Artifact Produced:* [middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt](#)

```
# Step 4 middleware smoke test evidence - 2026-04-25
```

```
Scope:
```

- Step 4 middleware skeleton only.
- Mock sensor data only.
- Simulated fan action only.
- No real GPIO.
- No Step 5 shared schemas.
- No Step 6 n8n workflow.
- No Step 7 agent workflow.
- No Step 8 safety layer.

```
## Command used to start middleware
```

```
Working directory:
```

```
```text
```



```

C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis
```

Command:

```powershell
python -m middleware.api.app
```

Observed startup output:

```text
new-yacoub middleware running at http://127.0.0.1:8000
Press Ctrl+C to stop.
```

## Tested endpoints

- `GET /status`
- `POST /fan/on`
- `POST /fan/off`
- `POST /sensor-event` with a valid temperature event
- `POST /sensor-event` with a malformed event

Observed server log:

```text
127.0.0.1 - "GET /status HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/on HTTP/1.1" 200 -
127.0.0.1 - "POST /fan/off HTTP/1.1" 200 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 202 -
127.0.0.1 - "POST /sensor-event HTTP/1.1" 400 -
```

## GET /status result

Command:

```powershell
Invoke-RestMethod -Method Get -Uri "http://127.0.0.1:8000/status"
```

Observed result:

```text
status message state

ok middleware running @{fan=off; last_sensor_event=; hardware=simulated}
```

Result summary:

- HTTP status: `200`
- Middleware status: `ok`
- Hardware mode: `simulated`
- Fan state before action tests: `off`

## POST /fan/on result

Command:

```powershell
Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/fan/on"
```

Observed result:

```text
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

Result summary:

- HTTP status: `200`

```

```

- Simulated fan state changed to `on`

## POST /fan/off result

Command:

```powershell
Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/fan/off"
```

Observed result:

```text
status : ok
action : fan_off
fan : off
simulated : True
reason : manual_api_call
```

Result summary:

- HTTP status: `200`
- Simulated fan state changed to `off`

## Valid POST /sensor-event result

Command:

```powershell
$body = @{
 sensor_id = "temp_sensor_1"
 timestamp = "2026-04-25T20:00:00Z"
 type = "temperature"
 value = 31.4
 unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/sensor-event" -ContentType "application/json" -Body $body
```

Observed result:

```text
status message event

accepted sensor event received by middleware @{sensor_id=temp_sensor_1; timestamp=2026-04-25T20:00:00Z; value=31,4; ...
```

Result summary:

- HTTP status: `202`
- Middleware accepted the temperature event.
- Event source: `temp_sensor_1`
- Event type: `temperature`
- Event value: `31.4 C`

## n8n forwarding note

n8n forwarding is intentionally skipped unless `N8N_WEBHOOK_URL` is configured.

This is expected for Step 4 because the n8n workflow is not implemented until Step 6. The middleware bridge has a configurable placeholder, but this smoke test did not connect to a real n8n workflow.

## Malformed POST /sensor-event result

Command:

```powershell
$badBody = @{
 sensor_id = "temp_sensor_1"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/sensor-event" -ContentType "application/json" -Body $badBody
```

```

```

Observed result:

```text
Invoke-RestMethod : {
 "error": "missing_required_field",
 "missing": [
 "timestamp",
 "type",
 "value",
 "unit"
],
 "message": "Step 4 uses a minimal inline check; JSON schemas come in Step 5."
}
```

Result summary:

- HTTP status: `400`
- Missing fields were reported.
- The response explicitly notes that Step 4 uses a minimal inline check and JSON schemas come in Step 5.

## Evidence conclusion

The Step 4 middleware skeleton ran locally on PC with simulated sensor and fan state. All required initial endpoints responded, valid sensor input was accepted, simulated fan actions worked, and malformed sensor input returned HTTP `400`.

```

- *Why it matters:* Hard proof for your Results and Methodology chapters. It proves the code doesn't just exist in theory, but was actively smoke-tested and verified to handle both correct data (HTTP 202) and reject bad data (HTTP 400).

▼ .

-
The AI intentionally kept the scope narrow for PC-first development by utilizing mock_sensor data and simulated fan state rather than requiring real Raspberry Pi GPIO pins.

-
-
-

(Human Task) (Execution & Smoke Testing)
(include the terminal/screenshot shit here)

The human researcher took over to verify the code.

Opening two separate PowerShell windows, the human booted the Python middleware server in Terminal 1. In Terminal 2, the human executed manual API calls using Invoke-RestMethod to test the /status, /fan/on, /fan/off, and /sensor-event endpoints.

-
-
-

(AI Task) (Report Generation)
(INCLUDE REPORT SCRIBE MATERIAL)

Finally, the report-scribe AI agent analyzed the generated files and evidence to draft the formal report notes (docs/report-notes/step-04-middleware-skeleton.md), mapping exactly how this middleware step feeds into Chapter 3 (Methodology) and Chapter 5 (Implementation).

(The Code Maps to the Evidence)

-

The files the AI created directly line up with the thesis requirements.

-

(Evidence: Middleware Source Files)

middleware/api/app.py & routes.py (The Server)

middleware/gpio/mock_sensor.py (The Fake Hardware)

middleware/webhooks/n8n_sender.py (The Messenger)

Why it matters: Proves the architecture is decoupled. You have a dedicated API layer ready to accept physical hardware connections in later validation tiers (Tier 1.5).

(The Actions Map to the Evidence)

-

HARD PROOF

The human testing process was converted into formal documentation.

-

(Evidence: Endpoint Responses & Manual Test Notes)

Action: Human ran `python -m middleware.api.app` and pinged the server with PowerShell commands.

Artifact Produced: `middleware/tests/evidence/step-04-middleware-smoke-test-2026-04-25.txt`

Why it matters: Hard proof for your Results and Methodology chapters. It proves the code doesn't just exist in theory, but was actively smoke-tested and verified to handle both correct data (HTTP 202) and reject bad data (HTTP 400).

summary

-

The development of Step 4 showcases a tight human-AI feedback loop: The AI wrote the theoretical bridge, the human physically tested it in the terminal, and the AI transformed those human test results into formatted academic evidence ready to be cited in the final thesis report.

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-

▼ step 4 theory

```
review step 4 results

from isolated brain in a jar
-> to something that can touch the physical world
* that is the middleware skeleton-
-
* n8n cant directly make the (thermomotor) talk to the (fan)
* (it needs a translator)
* The middleware is the translator for N8N
* so that the thermotor can talk to the fan
* takes thermomotor signal "It's 30 degrees!" translates it into a message the workshop understands.
* workshop middleware python script translator translate it to fan language - "turn on"
-

(api/app.py): This is the heart of the Waiter. It's a tiny, lightweight web server running on your machine (on port 8000). Its entire job is to stand there and listen for incoming messages.) This is the heart of the Waiter. It's a tiny, lightweight web server running on your machine (on port 8000). Its entire job is to stand there and listen for incoming messages.
*
(api/routes.py): This is the Waiter's menu of acceptable commands. It creates specific "endpoints" (URLs) that n8n can interact with.
GET /status: A quick way to ask, "Are you awake?"
POST /fan/on: n8n will send a message here to activate the fan.
```

```

POST /fan/off: n8n will send a message here to deactivate the fan.
POST /sensor-event: A place to register that a new temperature reading has happened.
*
(gpio/mock_sensor.py) Because we are developing on your Windows PC right now, you don't have a physical temperature sensor plugged into your USB port. This file acts as a "fake" sensor. It generates pretend temperature data so we can build and test the entire system before we ever touch a real Raspberry Pi.
*
(webhooks/n8n_sender.py) This is the messenger pigeon. When the fake sensor generates a temperature reading, this file takes that data, packages it up nicely, and "pushes" it straight into n8n via a Webhook (which we will build in n8n later).
-
-
-
(why make the middleman translator python script)
this make our life easier when we run this in raspberry pie
  * later we will answer Q1 (how long it takes for a signal to jump from the mock sensor, into n8n, and back to the /fan/on endpoint)
  * Q1 "Can a self-hosted n8n-based IoT system run in a resource-constrained environment with acceptable latency?"
-
n8n (chef) - n8n is brilliant ... it make all the logical decision
(too hot, turn on the fan)
(too cold, turn off fan)
  * But the Chef is stick at the kitchen door
  * the docker container was built in step 3
  * the chef cannot physically do shit
(The hardware, the dining room)
  * this is the eventual hardware/Raspberry Pi
  * temperature sensor
  * cooling fans
  * they don't understand complex logic, they only understand electricity(GPIO-pins)
(Middleware(waiter))
  * the python code we just generated

cx step runner developed the
(exactly what step 4 should be)
-
middleware only
mock sensor only
simulated fan only
no real GPIO
no workflow implementation
no safety layer
no schemas
no MCP
operational n8n placeholder only

verificaiton

review step 4
-
middleware files created
middleware starts locally
/status works
/sensor-event accepts mock data
/fan/on works as simulated action
/fan/off works as simulated action
manual-test-notes.md contains commands/results
audit-review passes
report-scribe creates step-04 report note
commit pushed

```

▼ chapter 3 methodology

Chapter 3 delv

- how manual testing was performed
- how mock sensor data was used before hardware

Chapter 3 — Methodology feeds from:

```
middleware/tests/manual-test-notes.md
docs/decisions.md
```

▼ chapter 5 implementation

Chapter 5 delv

- middleware design
- API bridge
- sensor input path
- actuator endpoint path

Chapter 5 — Implementation feeds from:

```
middleware/
middleware/tests/manual-test-notes.md
docs/architecture/README.md
```

- -
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 -

▼ old

▼ Goal

Build the technical backbone of Yacoub's thesis: the bridge between sensor/action logic and n8n.

- -
 -
 -

Build the main Yacoub bridge between sensor/action logic and n8n.

▼ Work to complete

▼ 1

In `middleware/`, create the first minimal Python API structure:

- `middleware/api/app.py`
- `middleware/api/routes.py`
- `middleware/gpio/mock_sensor.py`
- `middleware/webhooks/n8n_sender.py`

Initial endpoints:

- `POST /sensor-event`
- `POST /fan/on`
- `POST /fan/off`
- `GET /status`

At this stage, mock sensor data is fine. Development starts on PC first; Raspberry Pi comes later as Tier 1.5 validation.

▼ 2

Work to complete

Create:

```
middleware/api/app.py
middleware/api/routes.py
middleware/gpio/mock_sensor.py
middleware/webhooks/n8n_sender.py
middleware/tests/manual-test-notes.md
```

Initial endpoints:

```
POST /sensor-event
POST /fan/on
POST /fan/off
GET /status
```

▼ What this proves

This proves that the system has a controlled execution layer and that workflow logic does not directly control hardware.

▼ Evidence produced

▼ 1

- middleware source files
- example request/response logs
- terminal outputs
- manual test notes

▼ 2

- middleware source files
- terminal logs
- curl/Postman examples
- endpoint responses
- mock sensor examples

▼ Step 5 — Define the shared contract layer (`shared_interfaces`)

(chapter 4 Choice of approach)

(chapter 5 implementation)

(chapter 7 Discussion)

▼ AI

▼ rundown on step 5

Here is the red carpet summary for **Step 5: Define the Shared Contract Layer**.

▼ The Core Concept: The "Language Barrier"

In Step 4, we built the Waiter (the Middleware) and in Step 3 we built the Chef (n8n). But right now, if the Waiter yells a temperature reading into the kitchen, the Chef might not understand it.

If the Middleware sends `{"temp": 25}`, but n8n is programmed to expect `{"temperature_celsius": 25.0}`, the whole system crashes.

Step 5 is about writing a legally binding dictionary—a **Contract**—that both sides must strictly obey.

▼ What Actually Happens in Step 5?

Instead of writing Python or Docker code, in this step, you will write **JSON Schemas**. A JSON Schema is essentially a rulebook for data.

You will create two main rulebooks:

1. **The Sensor Event Schema:** This dictates exactly how a temperature reading must look when it goes *from* the Middleware *to* n8n. (e.g., "It must have a field named 'temperature', and it must be a number").
2. **The Agent Action Schema:** This dictates exactly how a command must look when it comes *back* from n8n to the Middleware (e.g., "It must have an 'action' field that only says 'fan_on' or 'fan_off'").

You will also create a few fake "example" files (like a pretend temperature reading) so you can test things later.

▼ Why does this matter for the Yacoub Thesis?

This step is the glue between your thesis and the minimum required parts of the "Obid" AI workflow layer.

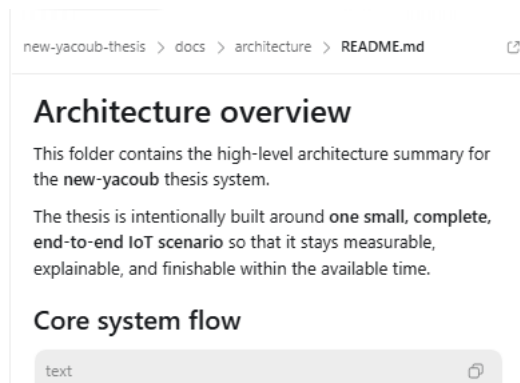
By enforcing a strict contract, you prove that your system is professional and safe. The workflow engine (n8n) isn't allowed to just send random gibberish to your hardware; it is physically forced to send perfectly formatted commands that match the contract. This prevents the system from doing unpredictable or unsafe things in the real world.

Whenever you are ready, we can generate those schema files!

▼ codex `step-runner` developing step 2

▼ files

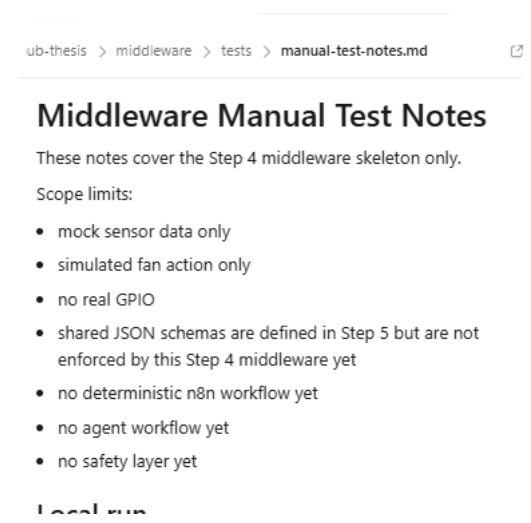
▼ 1



▼ 2



▼ 3



▼ 4



▼ codex verifying

▼ step 5

▼ q

1. Manually inspect the files

Check these quickly:

shared_interfaces/json-schema/sensor-event.schema.json
shared_interfaces/json-schema/agent-action.schema.json
shared_interfaces/examples/sensor-event.example.json
shared_interfaces/examples/fan-on.example.json
shared_interfaces/examples/blocked-action.example.json
docs/ongoing/integration-contract.md
docs/architecture/README.md
middleware/tests/manual-test-notes.md

Make sure:

sensor event fields match Step 4 middleware
fan_on and fan_off are the only allowed actions
fan_1 is the only allowed target
blocked example is clearly described as invalid/blocked, not a normal valid example

▼ a

Manual inspection looks good.

Checks:

- Sensor event schema matches Step 4 middleware fields exactly: sensor_id, timestamp, type, value, unit.
- Step 4 middleware also uses the same required fields in mock_sensor.py.
- Allowed actions in agent-action.schema.json are only fan_on and fan_off.
- Allowed target in agent-action.schema.json is only fan_1.
- Valid examples are clean:
 - sensor-event.example.json matches the sensor schema shape.
 - fan-on.example.json uses fan_on, fan_1, and requires_approval: false.
- Blocked example is clearly not presented as normal valid output:
 - blocked-action.example.json uses open_window and unknown_device.
 - docs/ongoing/integration-contract.md explicitly says it is intentionally **schema-invalid** and **policy-invalid**.
 - The doc also says it is a future Step 8 safety-layer test case, and Step 5 does not implement safety logic.

No changes needed from this review.

Step 5 contract layer is defined.

Changed:

- Added shared_interfaces/json-schema/sensor-event.schema.json
- Added shared_interfaces/json-schema/agent-action.schema.json

- Added shared_interfaces/examples/sensor-event.example.json
- Added shared_interfaces/examples/fan-on.example.json
- Added shared_interfaces/examples/blocked-action.example.json
- Updated docs/ongoing/integration-contract.md with concrete schema/example paths and blocked-example meaning.
- Updated docs/architecture/README.md to point to the Step 5 contract artifacts.
- Updated middleware/tests/manual-test-notes.md to say schemas now exist but are not enforced by middleware yet.

Important detail: blocked-action.example.json is intentionally schema-invalid and policy-invalid because it uses open_window and unknown_device, both outside the minimal Step 5 allowed sets. No safety logic was implemented.

Validated all JSON files under shared_interfaces/ with Python's standard json parser:

```
python - <<'PY'
import json
from pathlib import Path
for path in sorted(Path("shared_interfaces").rglob("*.json")):
    json.load(path.open(encoding="utf-8"))
    print(f"JSON OK: {path}")
PY
```

Evidence to save:

- Chapter 4: schema boundary rationale, allowed actions fan_on/fan_off, allowed target fan_1.
- Chapter 5: the schema files and example payloads as implementation artifacts.
- Chapter 7: the blocked example as a limitation/safety discussion seed, with actual enforcement deferred to Step 8.

▼ step 5 testing 1

Step 5 looks **correctly implemented**, based on the Codex summary.

The important part is this:

| `blocked-action.example.json` is intentionally schema-invalid and policy-invalid.

That is acceptable **because Codex documented it clearly**. Treat it as a **negative test fixture**, not as a valid example that should pass the action schema.

So your Step 5 state is:

```
contract files created
sensor-event schema created
agent-action schema created
valid sensor example created
valid fan-on example created
blocked/policy-invalid example created
integration-contract.md updated
architecture README updated
middleware docs updated without changing middleware behavior
JSON parse check passed
```

Good.

Now do the Step 5 verification workflow.

1. Manually inspect the files

Check these quickly:

```
shared_interfaces/json-schema/sensor-event.schema.json
shared_interfaces/json-schema/agent-action.schema.json
shared_interfaces/examples/sensor-event.example.json
shared_interfaces/examples/fan-on.example.json
shared_interfaces/examples/blocked-action.example.json
docs/ongoing/integration-contract.md
docs/architecture/README.md
middleware/tests/manual-test-notes.md
```

Make sure:

- sensor event fields match Step 4 middleware
- `fan_on` and `fan_off` are the only allowed actions
- `fan_1` is the only allowed target
- blocked example is clearly described as invalid/blocked, not a normal valid example

2. Run JSON parse validation yourself

From repo root:

```
python -c "import json; from pathlib import Path; [json.load(p.open(encoding='utf-8')) or print(f'JSON OK: {p}')] for p in sorted(Path('shared_interfaces').rglob('*.json'))]"
```

Expected:

```
JSON OK: shared_interfaces\examples\blocked-action.example.json
JSON OK: shared_interfaces\examples\fan-on.example.json
JSON OK: shared_interfaces\examples\sensor-event.example.json
JSON OK: shared_interfaces\json-schema\agent-action.schema.json
JSON OK: shared_interfaces\json-schema\sensor-event.schema.json
```

This only proves valid JSON syntax, not full schema validation. That is enough for Step 5 unless you later add a schema validation script.

3. Ask `audit-review`

Use this:

Audit the Step 5 shared contract layer changes only.

Check:

- Step 5 stays inside new-yacoub scope
- sensor-event.schema.json matches the Step 4 middleware input shape:
 - sensor_id
 - timestamp
 - type
 - value
 - unit
- agent-action.schema.json defines the minimal action shape:
 - action_id
 - target
 - reason
 - requires_approval
- allowed actions are limited to fan_on and fan_off
- allowed target is limited to fan_1
- sensor-event.example.json is valid JSON and matches the intended sensor event shape
- fan-on.example.json is valid JSON and represents an allowed action
- blocked-action.example.json is clearly documented as intentionally schema-invalid/policy-invalid and not treated as a valid action example
- docs/ongoing/integration-contract.md references the concrete schema/example files
- docs/architecture/README.md points to the concrete contract artifacts

- middleware/tests/manual-test-notes.md correctly says schemas exist but are not enforced by middleware yet
- no middleware runtime behavior was changed unnecessarily
- no Step 6 n8n workflow was implemented
- no Step 7 agent workflow was implemented
- no Step 8 safety layer was implemented
- no MCP, real GPIO, Raspberry Pi deployment, or measurement harness was added

Do not suggest broader architecture unless something blocks Step 6.

4. Use **repair-fix** only if audit finds real issues

If audit passes, skip repair.

If audit finds something, tell **repair-fix**:

Fix only the Step 5 audit findings.

Constraints:

- do not implement Step 6 workflow
- do not implement Step 7 agent workflow
- do not implement Step 8 safety layer
- do not change middleware runtime behavior unless strictly necessary
- keep blocked-action.example.json clearly marked as a negative/blocked example if retained
- keep changes minimal and testable

After fixing, summarize changed files and how to re-check JSON validity.

5. Ask **report-scribe**

Use this:

Create the report note for Step 5.

Create:

docs/report-notes/step-05-shared-contract.md

Use only evidence already present in the repo and the Step 5 contract files.

Include:

- what was built
- why it matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent results
- do not claim runtime schema enforcement exists yet
- do not claim safety logic exists yet
- document that blocked-action.example.json is a negative example for later safety testing
- document that actual enforcement is deferred to Step 8

6. Commit manually after audit/report note

First check status:

```
git status
```

Make sure you are not committing junk like:

```
__pycache__/  
*.pyc
```

```
*.zip
```

Then commit the intended Step 5 files:

```
git add shared_interfaces/json-schema/sensor-event.schema.json shared_interfaces/js
on-schema/agent-action.schema.json shared_interfaces/examples/sensor-event.example.
json shared_interfaces/examples/fan-on.example.json shared_interfaces/examples/bloc
ked-action.example.json docs/ongoing/integration-contract.md docs/architecture/READ
ME.md middleware/tests/manual-test-notes.md docs/report-notes/step-05-shared-contra
ct.md
git commit -m "feat(contracts): define shared interface schemas"
git push
```

Step 5 completion checklist

Step 5 is complete when:

```
JSON files parse correctly
audit-review passes
report-scribe note exists
no middleware behavior accidentally changed
blocked example is clearly documented as invalid/negative
commit pushed
```

After that, you are cleared for:

Step 6 — Build the deterministic baseline workflow

▼ step 5 testing 2 (fixing powershell command)

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\U. RED\1. kaniddat\new-yacoub-thesis> python -c "import json; from pa
thlib import Path; [json.load(p.open(encoding='utf-8')) or print(f'JSON OK: {p}')] for p in sorted(Path('shared_interfa
ces').rglob('*.json'))]"
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\U. RED\1. kaniddat\new-yacoub-thesis> python -c "import json; from pa
thlib import Path; [print(f'JSON OK: {p}')] for p in sorted(Path('shared_interfaces').rglob('*.json')) if json.load(p.o
pen(encoding='utf-8')) is not None]"
JSON OK: shared_interfaces/examples/blocked-action.example.json
JSON OK: shared_interfaces/examples/fan-on.example.json
JSON OK: shared_interfaces/examples/sensor-event.example.json
JSON OK: shared_interfaces/json-schema/agent-action.schema.json
JSON OK: shared_interfaces/json-schema/sensor-event.schema.json
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\U. RED\1. kaniddat\new-yacoub-thesis> |
```

Yes — your command worked in the sense that **there was no JSON parse error**.

But the command I gave you had a small logic mistake:

```
json.load(...) or print(...)
```

`json.load(...)` returns a dictionary, and a non-empty dictionary is “truthy,” so Python never reaches the `print(...)`.

That is why you got **no output**.

Run this corrected command instead

```
python -c "import json; from pathlib import Path; [print(f'JSON OK: {p}')] for p in
sorted(Path('shared_interfaces').rglob('*.json')) if json.load(p.open(encoding='utf
-8')) is not None]"
```

Expected output:

```
JSON OK: shared_interfaces\examples\blocked-action.example.json
JSON OK: shared_interfaces\examples\fan-on.example.json
JSON OK: shared_interfaces\examples\sensor-event.example.json
JSON OK: shared_interfaces\json-schema\agent-action.schema.json
JSON OK: shared_interfaces\json-schema\sensor-event.schema.json
```

Better PowerShell-friendly version

This one is cleaner and more reliable:

```
python -c "import json; from pathlib import Path; files=sorted(Path('shared_interfaces').rglob('*.json')); [json.load(p.open(encoding='utf-8')) for p in files]; [print(f'JSON OK: {p}') for p in files]"
```

Verdict

Your first run showing **no error** already means the JSON files are probably valid.

Run the corrected command if you want visible proof to paste into your evidence/audit flow.

- -

-

-

▼ **audit-review**

▼ q

Audit the Step 5 shared contract layer changes only.

Check:

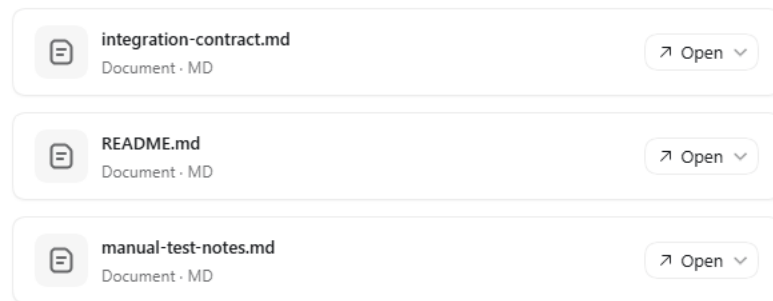
- Step 5 stays inside new-yacoub scope
- sensor-event.schema.json matches the Step 4 middleware input shape:
 - sensor_id
 - timestamp
 - type
 - value
 - unit
- agent-action.schema.json defines the minimal action shape:
 - action_id
 - target
 - reason
 - requires_approval
- allowed actions are limited to fan_on and fan_off
- allowed target is limited to fan_1
- sensor-event.example.json is valid JSON and matches the intended sensor event shape
- fan-on.example.json is valid JSON and represents an allowed action
- blocked-action.example.json is clearly documented as intentionally schema-invalid/policy-invalid and not treated as a valid action example

- docs/ongoing/integration-contract.md references the concrete schema/example files
- docs/architecture/README.md points to the concrete contract artifacts
- middleware/tests/manual-test-notes.md correctly says schemas exist but are not enforced by middleware yet
- no middleware runtime behavior was changed unnecessarily
- no Step 6 n8n workflow was implemented
- no Step 7 agent workflow was implemented
- no Step 8 safety layer was implemented
- no MCP, real GPIO, Raspberry Pi deployment, or measurement harness was added

Do not suggest broader architecture unless something blocks Step 6.

▼ a

▼ files



No Step 5 blocking findings. The shared contract layer is ready to proceed to Step 6.

I checked the current local working tree and the Step 5 changes are confined to the expected contract/doc areas: shared_interfaces, integration-contract.md (line 51), architecture README (line 103), and manual-test-notes.md (line 10). No middleware runtime code changed, and I found no Step 6 workflow, Step 7 agent workflow, Step 8 safety layer, MCP, real GPIO, Raspberry Pi deployment, or measurement harness additions.

The JSON checks passed:

- sensor-event.schema.json (line 8) requires sensor_id, timestamp, type, value, unit.
- agent-action.schema.json (line 8) requires action_id, target, reason, requires_approval.
- Allowed actions are limited to fan_on and fan_off.
- Allowed target is limited to fan_1.
- sensor-event.example.json, fan-on.example.json, and blocked-action.example.json are valid JSON.
- The blocked example is intentionally invalid against the allowed action/target policy, and documents that clearly.

integration-contract.md (line 141)

Residual note: these Step 5 changes are still uncommitted in the working tree. From an audit standpoint, they are good to save and move on to Step 6.

▼ final check

▼ q

quick check

Step 5 is complete when:

JSON files parse correctly

audit-review passes

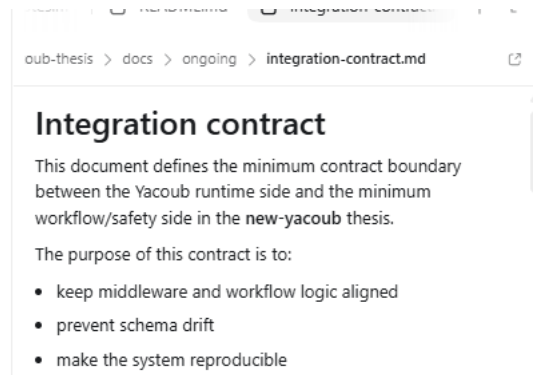
report-scribe note exists

no middleware behavior accidentally changed

blocked example is clearly documented as invalid/negative

commit pushed

▼ a



Step 5 is complete.

I checked each completion condition:

- JSON files parse correctly: passed for both schemas and all three examples.
- audit-review passes: yes, no blocking Step 5 findings from the last audit.
- Report-scribe note exists: yes, docs/report-notes/step-05-shared-contract-layer.md is present.
- No middleware behavior accidentally changed: passed, no diff in middleware/api, middleware/gpio, or middleware/webhooks.
- Blocked example is clearly documented as invalid/negative: yes, states it is intentionally schema-invalid and policy-invalid.
integration-contract.md (line 141)
- Commit pushed: yes, main is clean and aligned with origin/main at 363e118 (step 5 feat(contracts): define shared interface schemas).

- -
 -
 -

▼ report scribe

▼ q

Create the report note for Step 5.

Create:

docs/report-notes/step-05-shared-contract.md

Use only evidence already present in the repo and the Step 5 contract files.

Include:

- what was built
- why it matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent results
- do not claim runtime schema enforcement exists yet
- do not claim safety logic exists yet
- document that blocked-action.example.json is a negative example for later safety testing
- document that actual enforcement is deferred to Step 8

▼ Step 05 - Shared Contract

What was built

- A minimal shared contract layer was added under shared_interfaces/.
- The input-side sensor event schema defines the first scenario's temperature-event payload.
- The output-side action schema defines the minimal fan action contract for workflow-produced actions.
- Example payloads were added for one valid sensor event, one valid fan-on action, and one negative blocked-action case.
- docs/ongoing/integration-contract.md was updated to describe the contract boundary between middleware, shared interfaces, workflow logic, and later safety validation.
- docs/architecture/README.md was updated to list the Step 5 contract artifacts.

Why it matters for the thesis

This step makes the boundary between Yacoub's runtime side and the minimum Obid-compatible workflow/safety side explicit. It supports the thesis by reducing schema drift, making the middleware-to-workflow interface reportable, and preparing a clear place where later validation and approval behavior can be tested.

The contract also keeps the project narrow: it defines one temperature-event path, one fan-action path, and one small set of allowed action and target values.

Evidence produced

- shared_interfaces/json-schema/sensor-event.schema.json
- shared_interfaces/json-schema/agent-action.schema.json
- shared_interfaces/examples/sensor-event.example.json
- shared_interfaces/examples/fan-on.example.json
- shared_interfaces/examples/blocked-action.example.json
- docs/ongoing/integration-contract.md

- docs/architecture/README.md

The contract currently defines:

- sensor event required fields: sensor_id, timestamp, type, value, unit
- supported sensor type: temperature
- supported unit: C
- action required fields: action_id, target, reason, requires_approval
- initially allowed actions: fan_on, fan_off
- initially allowed target: fan_1

blocked-action.example.json is a negative example for later safety testing. It uses open_window and unknown_device, which are outside the Step 5 allowed action and target sets.

Report chapters it feeds

- Chapter 4 - Choice of approach: why a schema-based contract boundary is used instead of informal payloads.
- Chapter 5 - Implementation: shared interface files, sensor-event schema, action schema, example payloads, and contract boundary.
- Chapter 7 - Discussion: limitations of contract-only design before runtime enforcement and safety handling.

Limitations or assumptions

- Step 5 defines schemas and examples only.
- Runtime JSON Schema enforcement is not implemented yet.
- Safety logic is not implemented yet.
- Human approval handling is not implemented yet.
- The Step 4 middleware still uses its minimal inline validation, not the shared JSON schemas.
- blocked-action.example.json is not evidence that unsafe output is blocked at runtime; it is a prepared negative example for later Step 8 safety-layer testing.
- Actual enforcement of malformed, risky, or unsupported actions is deferred to Step 8.
- No n8n workflow is claimed to produce or consume these contract payloads yet.
- No latency, resource, or safety outcome measurements are produced by this step.

Screenshots/logs still needed

- Runtime validation logs once JSON Schema enforcement is connected.
- Safety-layer logs showing the negative blocked-action example being rejected or routed to approval in Step 8.
- n8n workflow evidence showing payloads that follow the shared contract.
- Evaluation records showing allowed and blocked outcomes after enforcement exists.

▼ step 5 rundown

Lock the rigid boundary between Yacoub's runtime side (infrastructure/hardware) and the minimum Obid-compatible workflow/safety side (n8n).

- **What this proves:** This proves that the system communicates through a strictly defined interface instead of relying on informal, undocumented assumptions. It proves the system is professional, decoupled, and safe.

1. **AI Task (Contract Generation):** The AI `step-runner` (Codex) was instructed to build the shared schemas. It generated the JSON rulebooks (`sensor-event.schema.json` and `agent-action.schema.json`) along with valid example payloads. Crucially, the AI purposefully generated a "negative" test case (`blocked-action.example.json`) for future safety testing, heavily documenting it as intentionally invalid so it wouldn't be mistaken for a working file.
2. **Human Task (Syntax Verification & Troubleshooting):** The human researcher ran a Python one-liner in PowerShell to verify the syntax of the generated JSON files. During this step, there was a rapid troubleshooting loop: the initial command provided by the AI yielded no output, the human flagged it, and the AI provided a corrected, PowerShell-friendly script which the human successfully executed to prove the JSON was structurally sound.
3. **AI Task (Peer-Review & Scope Audit):** A secondary AI agent (`audit-review`) was deployed purely to inspect the first AI's work. It rigorously audited the files to ensure the schema strictly adhered to the narrowed scope—confirming that the action schema *only* permitted `fan_on` and `fan_off` targeted at `fan_1`, actively preventing scope-creep.
4. **AI Task (Report Generation):** Finally, the `report-scribe` AI agent drafted the formal report notes (`docs/report-notes/step-05-shared-contract.md`), explicitly documenting the limitation that while the rules exist, *runtime enforcement* is deferred to Step 8.

▼ step 5 rundown

goal
-

Lock the rigid boundary between Yacoub's runtime side (infrastructure/hardware) and the minimum Obid-compatible workflow/safety side (n8n).

- **What this proves:** This proves that the system communicates through a strictly defined interface instead of relying on informal, undocumented assumptions. It proves the system is professional, decoupled, and safe.

▼ 🤖 AI Collaboration Workflow (How it was built)

1. **AI Task (Contract Generation):** The AI `step-runner` (Codex) was instructed to build the shared schemas. It generated the JSON rulebooks (`sensor-event.schema.json` and `agent-action.schema.json`) along with valid example payloads. Crucially, the AI purposefully generated a "negative" test case (`blocked-action.example.json`) for future safety testing, heavily documenting it as intentionally invalid so it wouldn't be mistaken for a working file.
2. **Human Task (Syntax Verification & Troubleshooting):** The human researcher ran a Python one-liner in PowerShell to verify the syntax of the generated JSON files. During this step, there was a rapid troubleshooting loop: the initial command provided by the AI yielded no output, the human flagged it, and the AI provided a corrected, PowerShell-friendly script which the human successfully executed to prove the JSON was structurally sound.
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scope—confirming that the action schema *only* permitted `fan_on` and `fan_off` targeted at `fan_1`, actively preventing scope-creep.

4. **AI Task (Report Generation):** Finally, the `report-scribe` AI agent drafted the formal report notes (`docs/report-notes/step-05-shared-contract.md`), explicitly documenting the limitation that while the rules exist, *runtime enforcement* is deferred to Step 8.

▼ 📁 1. The Code Maps to the Evidence

▼ (Evidence: The JSON Schemas)

▼ `shared_interfaces/json-schema/sensor-event.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://new-yacoub-thesis.local/shared_interfaces/json-schema/sensor-event.schema.json",
  "title": "Sensor Event",
  "description": "Minimal Step 5 sensor-event contract matching the Step 4 middleware input shape.",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "sensor_id",
    "timestamp",
    "type",
    "value",
    "unit"
  ],
  "properties": {
    "sensor_id": {
      "type": "string",
      "minLength": 1,
      "description": "Identifier for the simulated or physical sensor source."
    },
    "timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "ISO 8601 event timestamp."
    },
    "type": {
      "type": "string",
      "const": "temperature",
      "description": "First scenario supports temperature events only."
    },
    "value": {
      "type": "number",
      "description": "Measured temperature value."
    },
    "unit": {
      "type": "string",
      "const": "C",
      "description": "Temperature unit for the first scenario."
    }
  }
}
```

```

    }
  }
}

```

▼ shared_interfaces/json-schema/agent-action.schema.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://new-yacoub-thesis.local/shared_interfaces/json-schema/agent-
action.schema.json",
  "title": "Agent Action",
  "description": "Minimal Step 5 action contract for workflow-produced fan act
ions.",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "action_id",
    "target",
    "reason",
    "requires_approval"
  ],
  "properties": {
    "action_id": {
      "type": "string",
      "enum": [
        "fan_on",
        "fan_off"
      ],
      "description": "Minimal allowed action set for the first scenario."
    },
    "target": {
      "type": "string",
      "enum": [
        "fan_1"
      ],
      "description": "Minimal allowed target set for the first scenario."
    },
    "reason": {
      "type": "string",
      "minLength": 1,
      "description": "Short explanation of why the action was proposed."
    },
    "requires_approval": {
      "type": "boolean",
      "description": "Whether the action must pause for human approval before
execution."
    }
  }
}

```

- *Why it matters:* Proves you established a concrete "language" for the system. This satisfies Chapter 4 (Choice of Approach) by justifying why a schema-based contract boundary is superior to informal payloads.

▼ (Evidence: Example Payloads & The Negative Test)

▼ `shared_interfaces/examples/sensor-event.example.json`

```
{
  "sensor_id": "temp_sensor_1",
  "timestamp": "2026-04-25T20:00:00Z",
  "type": "temperature",
  "value": 31.4,
  "unit": "C"
}
```

▼ `shared_interfaces/examples/fan-on.example.json`

```
{
  "action_id": "fan_on",
  "target": "fan_1",
  "reason": "temperature_above_threshold",
  "requires_approval": false
}
```

▼ `shared_interfaces/examples/blocked-action.example.json`

```
{
  "action_id": "open_window",
  "target": "unknown_device",
  "reason": "hallucinated_action_outside_first_scenario",
  "requires_approval": true
}
```

- *Why it matters:* Proves the schemas work in practice. The `blocked-action` file serves as evidence for Chapter 7 (Discussion) to seed the conversation about system limitations and future safety layers.

▼ Evidence: Contract Documentation

▼ `docs/ongoing/integration-contract.md`

Integration contract

This document defines the minimum contract boundary between the Yacoub runtime side and the minimum workflow/safety side in the ****new-yacoub**** thesis.

The purpose of this contract is to:

- keep middleware and workflow logic aligned
- prevent schema drift
- make the system reproducible
- support testing and evaluation
- ensure unsafe actions cannot bypass validation

Contract boundary

The main contract boundary in this thesis is:

```

```text
Middleware <-> Shared Interfaces <-> Workflow / Safety logic
```

### Input side

Middleware creates or receives sensor events and represents them using a shared event schema.

### Output side

Workflow logic produces a proposed action using a shared action schema.

### Safety rule

No risky or malformed action may reach middleware execution without passing validation and, when needed, approval.

## Initial scenario

The first implementation is intentionally limited to one small end-to-end scenario:

```text
Temperature event
-> Middleware receives or generates the event
-> Middleware pushes the event to n8n
-> Workflow decides whether fan action is needed
-> Validation checks the action
-> Approval is required if the action is risky or malformed
-> Middleware executes /fan/on or /fan/off
```

## Input contract

Sensor events are represented in a shared JSON schema.

Concrete schema file:

- `shared_interfaces/json-schema/sensor-event.schema.json`

### Required fields

- `sensor_id`
- `timestamp`
- `type`
- `value`
- `unit`

### Sensor input example

Concrete example file:

```



```
- `shared_interfaces/examples/sensor-event.example.json`
```

```
```json
{
 "sensor_id": "temp_sensor_1",
 "timestamp": "2026-04-25T20:00:00Z",
 "type": "temperature",
 "value": 31.4,
 "unit": "C"
}
```
```

Input assumptions

- `sensor\_id` uniquely identifies the event source
- `timestamp` uses ISO 8601 format
- `type` identifies the sensor/event kind
- `value` contains the measured reading
- `unit` identifies the measurement unit

Output contract

Workflow decisions are represented in a shared JSON schema for actions.

Concrete schema file:

```
- `shared_interfaces/json-schema/agent-action.schema.json`
```

Required fields

- `action\_id`
- `target`
- `reason`
- `requires\_approval`

Allowed action example

Concrete example file:

```
- `shared_interfaces/examples/fan-on.example.json`
```

```
```json
{
 "action_id": "fan_on",
 "target": "fan_1",
 "reason": "temperature_above_threshold",
 "requires_approval": false
}
```
```

Output assumptions

- `action\_id` identifies the requested action
- `target` identifies the intended device or actuator

- `reason` explains why the action was proposed
- `requires\_approval` indicates whether execution may continue directly or must pause for approval

Validation and approval rules

All workflow-produced actions must be checked before middleware execution.

Approval is required when

- the action is not in the allowed action list
- the target device is unknown
- the output is malformed
- the action includes unexpected fields

Blocked case example

Concrete example file:

- `shared\_interfaces/examples/blocked-action.example.json`

The blocked example is intentionally **schema-invalid** and **policy-invalid** for Step 5 because:

- `open\_window` is outside the minimal allowed action set
- `unknown\_device` is outside the minimal allowed target set

It is included as a future Step 8 safety-layer test case. Step 5 only defines the contract artifact; it does not implement safety logic.

```
```json
{
 "action_id": "open_window",
 "target": "unknown_device",
 "reason": "hallucinated_action_outside_first_scenario",
 "requires_approval": true
}
```
```

Expected handling of blocked/risky output

- malformed or unknown output must not execute directly
- risky output must be stopped or routed to approval
- middleware should only execute actions that passed validation

First allowed action list

The first implementation supports only a very small action set.

Initially allowed

- `fan\_on`
- `fan\_off`

Initially allowed targets

- `fan\_1`

Not allowed in the first implementation

- unknown device actions
- actions outside the defined scenario
- multi-device actions
- free-form or hallucinated commands

Error cases

The first implementation must be able to handle at least these error cases:

1. malformed sensor event
2. missing required input field
3. malformed action output
4. unknown target device
5. unsupported action id
6. unexpected fields in action payload

These error cases should be visible in evaluation and discussion later.

Relationship to implementation

This contract must stay aligned with:

- `shared\_interfaces/json-schema/`
- `shared\_interfaces/examples/`
- `middleware/`
- `cognitive\_logic/`
- `safety\_layer/`
- `docs/architecture/`

If communication behavior changes, this file must be updated.

Step 5 defines these schema and example artifacts only. The Step 4 middleware does not yet enforce JSON Schema validation at runtime; runtime validation and safety behavior remain deferred to later steps.

Scope note

This contract is intentionally minimal.

It is designed only to support:

- one scenario
- one sensor-event path
- one action path
- one validation layer
- one approval checkpoint

It is not intended to define a broad multi-device or multi-agent protocol at this stage.

▼ docs/architecture/README.md

Architecture overview

This folder contains the high-level architecture summary for the **new-yacoub** thesis system.

The thesis is intentionally built around **one small, complete, end-to-end IoT scenario** so that it stays measurable, explainable, and finishable within the available time.

Core system flow

```text

Sensor event

- > Middleware
- > n8n workflow
- > Validation / Approval
- > Action execution
- > Metrics logging

```

Current intended scenario

```text

Temperature event

- > Python middleware receives or generates the event
- > Middleware pushes the event to n8n through webhook flow
- > n8n evaluates whether fan action is needed
- > Safety layer validates the output
- > Approval is required if the action is risky or malformed
- > Middleware executes /fan/on or /fan/off
- > Latency and resource metrics are recorded

```

Main design principle

The system is split into small layers with a clear contract boundary.

This is done to:

- keep the thesis narrow
- reduce scope creep
- make implementation testable
- make the report easier to explain
- prevent workflow logic from directly controlling hardware behavior without checks

Top-level folder responsibilities

`infrastructure/`

Contains runtime and deployment-related material.

Purpose:

- local self-hosted n8n baseline
- Docker configuration
- Raspberry Pi deployment assumptions
- monitoring hooks or runtime notes

`middleware/`

Contains the Python bridge between sensors/actions and n8n.

Purpose:

- receive or generate sensor events
- expose endpoints such as `/status`, `/fan/on`, and `/fan/off`
- push events into n8n through webhook flow
- act as the controlled execution layer for hardware-facing actions

`cognitive\_logic/`

Contains the minimum workflow logic required for integration.

Purpose:

- one workflow
- one prompt set
- one memory choice

In **new-yacoub**, this is minimum integration only. It is not a full cognitive architecture project.

`safety\_layer/`

Contains the minimum safety logic required for the integrated system.

Purpose:

- validate workflow output
- decide whether approval is required
- block malformed or unsafe actions
- preserve one safe case and one blocked case for evaluation

`shared\_interfaces/`

This is the **contract boundary** between middleware and workflow/safety logic.

Purpose:

- define sensor-event schemas

- define action schemas
- store example payloads
- prevent schema drift

This is where Yacoub's runtime side and the minimum Obid-compatible layer meet.

Current Step 5 contract artifacts:

- `shared\_interfaces/json-schema/sensor-event.schema.json`
- `shared\_interfaces/json-schema/agent-action.schema.json`
- `shared\_interfaces/examples/sensor-event.example.json`
- `shared\_interfaces/examples/fan-on.example.json`
- `shared\_interfaces/examples/blocked-action.example.json`

`evaluation/`

Contains the measurement and result material that will feed the thesis report.

Purpose:

- datasets or fixtures
- metrics output
- result files
- baseline-vs-agent comparison evidence

`docs/`

Contains planning, scope, decision, and report-support material.

Purpose:

- lock the narrowed scope
- document architecture decisions
- support report writing
- keep AI tools aligned with thesis boundaries

Contract boundary

The most important architectural boundary in this thesis is:

```text

Middleware <-> Shared Interfaces <-> Workflow / Safety logic

```

Input side

Middleware creates or receives sensor events and represents them using a shared schema.

Concrete input schema:

- `shared\_interfaces/json-schema/sensor-event.schema.json`

Output side

Workflow logic produces a proposed action using a shared schema.

Concrete output schema:

- `shared\_interfaces/json-schema/agent-action.schema.json`

Safety rule

No risky or malformed action may reach middleware execution without passing validation and, when needed, approval.

Current communication model

The first implementation uses a push-based communication flow:

- middleware pushes events to n8n through webhook flow
- n8n does not poll middleware in the first version

Scope note

This architecture is intentionally designed for:

- one complete vertical slice
- one measurable scenario
- one deterministic baseline
- one minimal agent-enhanced path

It is not currently intended to support broad multi-agent expansion, many devices, or deep n8n internals work.

- *Why it matters:* Proves the system architecture is formally documented, fulfilling the academic requirement for traceability.

▼ 2. The Actions Map to the Evidence (HARD PROOF)

▼ (Evidence: JSON Syntax Validation)

▼ terminal

In **Terminal 2**, you ran this specific command: `python -c "import json; from pathlib import Path; [print(f'JSON OK: {p}') for p in sorted(Path('shared_interfaces').rglob('*.json')) if json.load(p.open(encoding='utf-8')) is not None]"`

This is a one-line Python script that loops through the `shared_interfaces` folder, opens every `.json` file, and tries to parse it using `json.load()`. If it succeeds without crashing, it prints `JSON OK: [filename]`. This was the exact action that provided the "Hard proof that the theoretical contract files are syntactically valid."

As for the others:

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacob-thesis> python -c "import json; from pathlib import Path; [json.load(p.open(encoding='utf-8')) or print(f'JSON OK: {p}') for p in sorted(Path('shared_interfaces').rglob('*.json'))]"
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kandidat\new-yacob-thesis> python -c "import json; from pathlib import Path; [print(f'JSON OK: {p}') for p in sorted(Path('shared_interfaces').rglob('*.json')) if json.load(p.open(encoding='utf-8')) is not None]"
JSON OK: shared_interfaces\examples\blocked-action.example.json
JSON OK: shared_interfaces\examples\fan-on.example.json
JSON OK: shared_interfaces\examples\sensor-event.example.json
JSON OK: shared_interfaces\json-schema\agent-action.schema.json
JSON OK: shared_interfaces\json-schema\sensor-event.schema.json
```

- *Action:* Human successfully ran a terminal script to parse all `.json` files in the repository.
- *Why it matters:* Hard proof that the theoretical contract files are syntactically valid and ready to be imported into the n8n engine in the upcoming steps.

- -
-
-

▼ chapter 4

Chapter 4 — Choice of approach feeds from:

```
docs/ongoing/integration-contract.md
shared_interfaces/
docs/decisions.md
```

Chapter 4

- direct action vs validated contract boundary
- why schemas are used

▼ chapter 5

Chapter 5 — Implementation feeds from:

```
shared_interfaces/json-schema/
shared_interfaces/examples/
docs/ongoing/integration-contract.md
```

Chapter 5

- shared interface implementation
- contract boundary

▼ chapter 7

Chapter 7 — Discussion later feeds from:

```
contract limitations
schema drift prevention
blocked/malformed cases
```

Chapter 7

- discussion of safety and system boundaries

▼ Goal

Lock the boundary between Yacoub's runtime side and the minimum Obid-compatible workflow/safety side.

- -
 -
 -

▼ Work to complete

▼ 1

Create:

- `shared_interfaces/json-schema/sensor-event.schema.json`
- `shared_interfaces/json-schema/agent-action.schema.json`
- `shared_interfaces/examples/sensor-event.example.json`
- `shared_interfaces/examples/fan-on.example.json`
- `docs/ongoing/integration-contract.md`

▼ 2

Work to complete

Create:

```
shared_interfaces/json-schema/sensor-event.schema.json
shared_interfaces/json-schema/agent-action.schema.json
shared_interfaces/examples/sensor-event.example.json
shared_interfaces/examples/fan-on.example.json
shared_interfaces/examples/blocked-action.example.json
```

Update:

```
docs/ongoing/integration-contract.md
docs/architecture/README.md
```

▼ What this proves

This proves that the system communicates through a defined interface instead of informal assumptions.

▼ Evidence produced

▼ 1

- schemas
- example payloads
- integration contract
- architecture notes

▼ 2

- JSON schemas
- example payloads

- allowed action examples
- blocked action examples
- contract documentation

▼ (Phase 3: The Experiment (Dumb vs. Smart))

▼ Step 6 — Build the deterministic(non-AI)baseline workflow (`cognitive_logic`)
 (SCREENSHOTS)
 (deterministic IF rule decides fan_on/fan_off)
 chapter 4 Choice of approach
 chapter 5 mplementation
 chapter 6 Results

▼ AI (FINAL)

▼ cx step runner prompt (FINAL SCREENSHOTS)

▼ q

Current step:

Step 6 — Build the deterministic baseline workflow.

Use these files as source of truth:

- [AGENTS.md](#)
- [README.md](#)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/ongoing/integration-contract.md
- docs/evaluation/evaluation-protocol.md
- docs/n8n-version.md
- shared_interfaces/json-schema/sensor-event.schema.json
- shared_interfaces/json-schema/agent-action.schema.json
- shared_interfaces/examples/sensor-event.example.json
- shared_interfaces/examples/fan-on.example.json
- middleware/api/routes.py
- middleware/tests/manual-test-notes.md

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Obid minimum integration only
- Step 6 only
- deterministic baseline only
- no AI agent logic
- no prompt/memory implementation

- no Step 7 agent workflow
- no Step 8 safety layer
- no MCP
- no n8n core modification
- no real GPIO
- no Raspberry Pi deployment
- no broad multi-agent logic

Goal:

Create the deterministic non-AI baseline workflow that the later agent-enhanced workflow will be compared against.

Task:

Create the deterministic workflow artifacts for n8n.

Create:

- cognitive_logic/workflows/deterministic-baseline.json
- cognitive_logic/workflows/deterministic-baseline.md

The workflow should be designed to:

1. receive a temperature event
2. apply a deterministic threshold rule
3. choose fan_on or fan_off
4. call the middleware endpoint
5. log or document the result path

Baseline rule:

- if temperature is greater than or equal to 30.0 C, request fan_on
- otherwise request fan_off

Middleware endpoint assumptions:

- middleware runs locally at <http://127.0.0.1:8000>
- POST /fan/on turns the simulated fan on
- POST /fan/off turns the simulated fan off
- no real hardware is used

Important implementation constraint:

If a valid n8n export JSON can be created safely, create it.

If exact n8n export structure cannot be guaranteed, create a clearly marked draft/export-candidate JSON and explain what must be verified inside the n8n UI before treating it as final.

Documentation requirements for deterministic-baseline.md:

- explain the workflow purpose
- explain the threshold rule
- list expected input shape
- list expected action outputs
- list middleware endpoints called
- explain how to manually recreate/import/test the workflow in n8n

- explain what screenshot/log evidence should be saved
- explain how this feeds Chapter 4, Chapter 5, and Chapter 6
- clearly state that this is non-AI baseline behavior

Do not:

- implement an AI agent node
- create prompts
- create memory files
- implement safety approval logic
- implement Step 7 or Step 8
- change middleware behavior unless absolutely necessary
- change shared schemas unless you find a direct mismatch that blocks Step 6
- add measurement harness scripts yet unless only a tiny manual logging note is needed

Validation:

- Ensure any JSON file created is valid JSON.
- If possible, run Python json parsing on cognitive_logic/workflows/deterministic-baseline.json.
- Do not claim the workflow has been runtime-tested unless it was actually imported/run in n8n.
- If runtime testing is not possible, mark runtime verification as pending.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to inspect/import/test the deterministic workflow in n8n.
- tell me what evidence should be saved for Chapter 4, Chapter 5, and Chapter 6.

▼ a

▼ files

2 files changed +521 -0	Undo ↶
cognitive_logic/workflows/deterministic-baseline.json +246 -0	▼
cognitive_logic/workflows/deterministic-baseline.md +275 -0	▼

Step 6 artifacts are created.

Added:

- deterministic-baseline.json
- deterministic-baseline.md

The JSON is valid and parsed successfully with Python. It is clearly marked as a **draft/export candidate**, because I did not import/run it in n8n yet.

Workflow logic:

- Webhook receives temperature event.
- IF node checks value >= 30.0.

- True branch creates fan_on action and calls POST http://127.0.0.1:8000/fan/on.
- False branch creates fan_off action and calls POST http://127.0.0.1:8000/fan/off.
- No AI agent, prompts, memory, safety approval, GPIO, or measurement harness added.

To inspect/import/test in n8n:

1. Start middleware: python -m middleware.api.app
2. Open n8n: http://localhost:5678
3. Import cognitive_logic/workflows/deterministic-baseline.json
4. Verify node settings in UI, especially IF rule and HTTP URLs.
5. Test with high temp 31.4 and low temp 24.5.
6. Save the final verified export back over the JSON only after import/run succeeds.

Evidence to save:

- Chapter 4: screenshot/note showing this is the non-AI baseline comparison anchor.
- Chapter 5: workflow canvas, IF node config, HTTP Request node config, final verified export.
- Chapter 6: one high-temperature execution log, one low-temperature execution log, and middleware logs showing /fan/on and /fan/off.

Runtime verification remains pending until the workflow is actually imported and run in n8n.

▼ og verify cx step 6 result

Step-runner did a **good Step 6 draft**, but I would **not call Step 6 finished yet**.

It created the right artifacts:

```
cognitive_logic/workflows/deterministic-baseline.json
cognitive_logic/workflows/deterministic-baseline.md
```

The logic is also right:

```
Webhook receives temperature event
IF value >= 30.0
  -> fan_on
else
  -> fan_off
```

And it correctly did **not** add AI, prompts, memory, safety, GPIO, MCP, or measurement harness. Good.

But there are **two likely runtime issues** before you import/test in n8n.

Main issue 1 — 127.0.0.1:8000 may fail from Docker n8n

Your n8n runs inside Docker.

Inside the n8n container, this URL:

```
http://127.0.0.1:8000/fan/on
```

means:

└ "inside the n8n container itself"

It does **not** necessarily mean your Windows host where the middleware is running.

For Docker Desktop on Windows, the safer URL is usually:

```
http://host.docker.internal:8000/fan/on
http://host.docker.internal:8000/fan/off
```

So Step 6 is conceptually correct, but the HTTP Request nodes may fail unless the URL is changed.

Main issue 2 — Webhook body path may be wrong

The workflow checks:

```
$json.value
```

But n8n Webhook nodes often place POST JSON under:

```
$json.body.value
```

So the IF condition may need to be:

```
={{ $json.body.value }}
```

instead of:

```
={{ $json.value }}
```

This must be verified in the n8n execution view.

Verdict

Good as:

```
Step 6 draft/export candidate
```

Not done as:

```
runtime-verified deterministic baseline
```

That matches what Codex said: runtime verification is pending.

What to tell Codex step-runner now

Paste this:

```
Step 6 draft looks structurally correct, but before runtime testing, check and repair two likely n8n/Docker issues.

Potential issue 1:
Because n8n runs inside Docker and middleware runs on the Windows host, HTTP Request nodes should probably call:
- http://host.docker.internal:8000/fan/on
- http://host.docker.internal:8000/fan/off
```

Using `http://127.0.0.1:8000` from inside the n8n container may point to the container itself and fail.

Potential issue 2:

The Webhook node may output POST JSON under `$json.body.value` rather than `$json.value`.

Check whether the IF node should use:

`={{ $json.body.value }}`

instead of:

`={{ $json.value }}`

Task:

Update `deterministic-baseline.json` and `deterministic-baseline.md` only if needed.

Keep scope:

- Step 6 only
- deterministic baseline only
- no AI
- no prompts
- no memory
- no safety layer
- no measurement harness
- no middleware behavior changes

After editing:

- summarize what changed
- tell me exactly how to import/test the workflow in n8n
- tell me which URL and expression should be checked in the n8n UI

Step 6 runtime test checklist

After that repair/check, do this.

1. Start middleware

From repo root:

```
python -m middleware.api.app
```

Keep it running.

2. Start n8n

From Docker folder:

```
cd infrastructure/docker
docker compose up -d
```

Open:

```
http://localhost:5678
```

3. Import workflow

Import:

```
cognitive_logic/workflows/deterministic-baseline.json
```

4. Check these node settings in n8n UI

IF node

Check whether condition uses:

```
={{ $json.body.value }}
```

or whatever the execution view shows is correct.

HTTP Request nodes

Check URLs:

```
http://host.docker.internal:8000/fan/on  
http://host.docker.internal:8000/fan/off
```

5. Run high-temperature test

Use the n8n test webhook URL.

Payload:

```
{  
  "sensor_id": "temp_sensor_1",  
  "timestamp": "2026-04-25T20:00:00Z",  
  "type": "temperature",  
  "value": 31.4,  
  "unit": "C"  
}
```

Expected:

```
IF true branch  
POST /fan/on  
middleware log shows /fan/on
```

6. Run low-temperature test

Payload:

```
{  
  "sensor_id": "temp_sensor_1",  
  "timestamp": "2026-04-25T20:05:00Z",  
  "type": "temperature",  
  "value": 24.5,  
  "unit": "C"  
}
```

Expected:

```
IF false branch  
POST /fan/off  
middleware log shows /fan/off
```

Evidence to save

Create later:

```
cognitive_logic/workflows/evidence/step-06-deterministic-baseline-test-2026-04-XX.md
```

Include:

```
- workflow imported successfully
- IF node screenshot or note
- high temp test result
- low temp test result
- middleware terminal log showing /fan/on and /fan/off
- final verified workflow export saved
```

Bottom line

Step-runner did a good **draft**.

But before audit/commit, fix or verify:

```
127.0.0.1 vs host.docker.internal
$json.value vs $json.body.value
```

After those are handled and the workflow actually runs in n8n, Step 6 is complete.

▼ cx fix

did the step runner fix it?

Updated only the Step 6 workflow files:

deterministic-baseline.json

deterministic-baseline.md

Changed:

IF expression from `={{ $json.value }}` to `={{ $json.body.value }}`

HTTP URLs from <http://127.0.0.1:8000/>... to:

<http://host.docker.internal:8000/fan/on><http://host.docker.internal:8000/fan/off>

Docs now explain that 127.0.0.1:8000 is the Windows-host middleware URL, but Dockerized n8n should call host.docker.internal:8000.

Validated:

deterministic-baseline.json parses as valid JSON.

I did not import/run it in n8n, so runtime verification is still pending.

In n8n UI, check exactly:

IF node expression: `={{ $json.body.value }}`

true branch URL: <http://host.docker.internal:8000/fan/on>

false branch URL: <http://host.docker.internal:8000/fan/off>

Import/test:

Start middleware: `python -m middleware.api.app`

Start/open n8n at <http://localhost:5678>

Import cognitive_logic/workflows/deterministic-baseline.json

Open Webhook node and use its test URL.

Send high temp 31.4: expect /fan/on

Send low temp 24.5: expect /fan/off

I left the unrelated untracked cognitive_logic.zip alone.

▼ og response to cx fix

Yes — that sounds like the **correct fix**.

Step-runner fixed the two important Step 6 issues:

```
$json.value      -> $json.body.value  
127.0.0.1:8000   -> host.docker.internal:8000
```

That is exactly what we wanted for Dockerized n8n talking to middleware running on your Windows host.

Current Step 6 status

You are now at:

```
Step 6 draft fixed  
JSON valid  
runtime verification still pending
```

So it is **good enough to test in n8n now**, but not officially complete yet.

What to do next

1. Start middleware

From repo root:

```
python -m middleware.api.app
```

2. Start n8n

In another terminal:

```
cd "C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub  
-thesis\infrastructure\docker"  
docker compose up -d
```

Open:

```
http://localhost:5678
```

3. Import the workflow

Import:

```
cognitive_logic/workflows/deterministic-baseline.json
```

4. Check these in n8n UI

Make sure the IF node uses:

```
={{ $json.body.value }}
```

Make sure HTTP Request nodes use:

```
http://host.docker.internal:8000/fan/on
http://host.docker.internal:8000/fan/off
```

5. Test high temp

Send this to the n8n test webhook:

```
{
  "sensor_id": "temp_sensor_1",
  "timestamp": "2026-04-25T20:00:00Z",
  "type": "temperature",
  "value": 31.4,
  "unit": "C"
}
```

Expected middleware log:

```
POST /fan/on 200
```

6. Test low temp

```
{
  "sensor_id": "temp_sensor_1",
  "timestamp": "2026-04-25T20:05:00Z",
  "type": "temperature",
  "value": 24.5,
  "unit": "C"
}
```

Expected middleware log:

```
POST /fan/off 200
```

Also clean this before commit

Codex mentioned:

```
untracked cognitive_logic.zip
```

Do **not** commit that.

Remove it:

```
Remove-Item cognitive_logic.zip -Force -ErrorAction SilentlyContinue
```

Also check:

```
git status
```

Make sure no `.zip`, `__pycache__`, or `.pyc` files are staged.

Verdict

Yes, the fix is right.

Now Step 6 needs runtime proof:

```
import workflow
run high-temp test
run low-temp test
save evidence
audit-review
report-scribe
commit
```

Then Step 6 is officially complete.

▼ og setting up the n8n

By **n8n**, I mean the **local n8n web app you already started with Docker**.

Not a separate downloadable app right now.

When I say **“main n8n page”**, I mean open this in your browser:

```
http://localhost:5678
```

That is your local n8n interface.

What is happening

You already have n8n running through Docker from Step 3:

```
cd infrastructure/docker
docker compose up -d
```

That starts n8n in the background.

Then your browser accesses it here:

```
http://localhost:5678
```

That page is what I mean by:

```
n8n main page
```

So the real steps are

1. Make sure n8n Docker is running

In PowerShell:

```
cd "C:\Users\Jake_OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub
-thesis\infrastructure\docker"
docker compose up -d
```

2. Open n8n in browser

Go to:

```
http://localhost:5678
```

That is the n8n UI.

3. Inside that browser page

You should see the n8n interface with workflows.

That is where you import:

```
cognitive_logic/workflows/deterministic-baseline.json
```

Mental model

Think of it like this:

```
Docker runs n8n
Browser opens n8n
n8n UI imports workflow JSON
workflow calls your Python middleware
```

So no, don't download a new n8n desktop software. Use the local browser UI at:

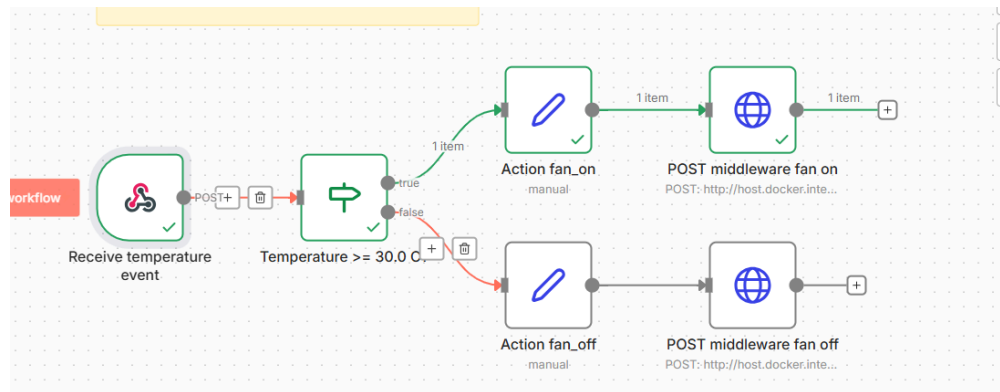
```
http://localhost:5678
```

▼ og running low/high temp test (FINAL)

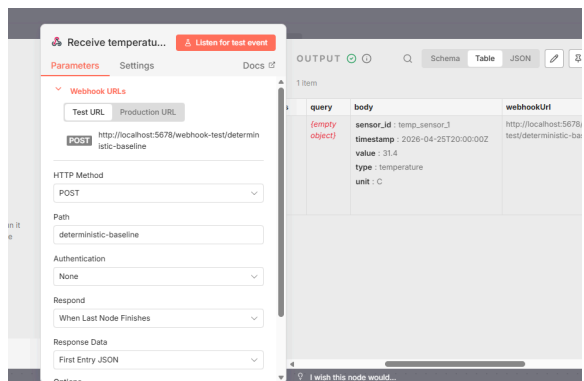
▼ terminal 1 running the high low/temp commands to n8n

▼ high temp

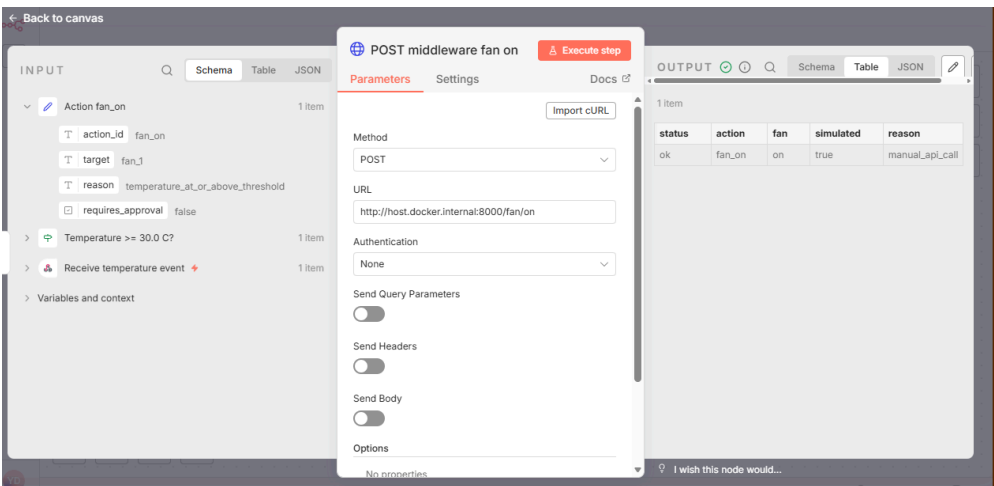
▼ full tree



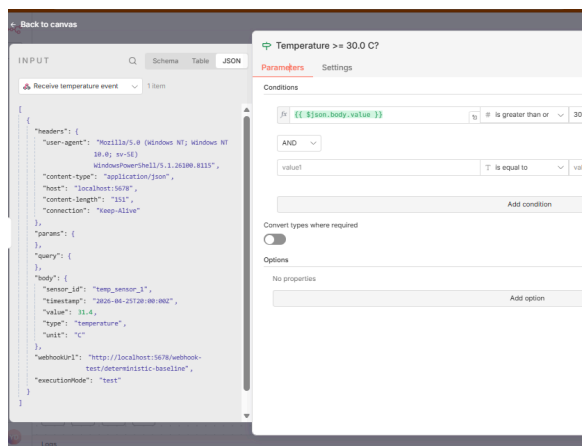
▼ Node confirmation (triangle) - high temp recieve confirm



▼ Node confirmation (globe shape) - POST low temp



▼ (if condition node) (screenshot based on high condition > 30)



▼ prompt

Your test **successfully reached the Webhook node**, but it does **not look like the full workflow ran yet**.

In your screenshot, n8n shows the Webhook node received the body:

```
value: 31.4
```

Good.

But the downstream nodes are not green:

```
Temperature >= 30.0 C?  
Action fan_on  
POST middleware fan on
```

So right now you have only proven:

```
PowerShell -> n8n Webhook works
```

You have **not yet proven**:

```
Webhook -> IF -> fan_on -> middleware
```

Do this instead

Go back to the canvas. Do **not** stay inside the Webhook node panel.

On the canvas, click the big red/orange button:

```
Execute workflow
```

It is at the bottom of your canvas in your screenshot.

After clicking **Execute workflow**, n8n should wait for the webhook event.

Then run the PowerShell command again:

```
$webhookUrl = "http://localhost:5678/webhook-test/deterministic-baseline"  
  
$highBody = @{  
    sensor_id = "temp_sensor_1"  
    timestamp = "2026-04-25T20:00:00Z"  
    type = "temperature"  
    value = 31.4  
    unit = "C"  
} | ConvertTo-Json  
  
Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $highBody
```

What success should look like

After the full workflow executes, you should see green checks on:

```
Receive temperature event
Temperature >= 30.0 C?
Action fan_on
POST middleware fan on
```

Then click the `POST middleware fan on` node and check output. You want to see:

```
status: ok
action: fan_on
fan: on
simulated: true
```

That is the real proof.

Middleware terminal

The middleware terminal is still this one:

```
python -m middleware.api.app
new-yacoub middleware running at http://127.0.0.1:8000
```

If it does not print new request lines, that is annoying but not fatal. The n8n HTTP node output showing `status: ok` and `fan: on` is proof that middleware responded.

Then low temp

After high temp works, click **Execute workflow** again and run:

```
$webhookUrl = "http://localhost:5678/webhook-test/deterministic-baseline"
```

```
$lowBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json
```

```
Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $lowBody
```

Expected green path:

```
Receive temperature event
Temperature >= 30.0 C?
Action fan_off
POST middleware fan off
```

Expected output from `POST middleware fan off`:

```
status: ok
action: fan_off
fan: off
simulated: true
```


Bottom line

Your current screenshot = webhook received data.

Next goal = run via **Execute workflow** so the IF + HTTP Request nodes execute too.

the three should be submitted to terminal separately

```
$webhookUrl = "http://localhost:5678/webhook-test/deterministic-baseline"

$highBody = @{
  sensor_id = "temp_sensor_1"
  timestamp = "2026-04-25T20:00:00Z"
  type = "temperature"
  value = 31.4
  unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $highBody
```

▼ low temp

▼ prompt

Yes — **this high-temperature test worked.**

Your screenshot shows the full successful path:

```
Receive temperature event ✓
Temperature >= 30.0 C? ✓
Action fan_on ✓
POST middleware fan on ✓
```

That means Step 6 high-temp branch is verified:

```
temperature = 31.4
>= 30.0
→ fan_on
→ middleware responded OK
```

Even if the middleware terminal does not visibly print the request line, n8n's output already proved the middleware answered because the `POST middleware fan on` node returned:

```
status: ok
action: fan_on
fan: on
simulated: true
```

That is valid evidence.

Now do the low-temperature test

Go back to the canvas.

Click:

Execute workflow

Then run this in PowerShell:

```
$webhookUrl = "http://localhost:5678/webhook-test/deterministic-baseline"

$lowBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $lowBody
```

Expected green path:

```
Receive temperature event ✓
Temperature >= 30.0 C? ✓
Action fan_off ✓
POST middleware fan off ✓
```

Then click the `POST middleware fan off` node and confirm output says:

```
status: ok
action: fan_off
fan: off
simulated: true
```

Evidence to save now

Take/save screenshots of:

1. high-temp workflow canvas showing fan_on path green
2. POST middleware fan on output showing fan: on
3. low-temp workflow canvas showing fan_off path green
4. POST middleware fan off output showing fan: off
5. IF node condition showing value >= 30.0

After low-temp works, Step 6 runtime verification is basically done.

▼ Node confirmation (triangle) - low temp recieve confirm

Receive temperature event

Webhook URLs

Test URL: `http://localhost:5678/webhook-test/deterministic-baseline`

HTTP Method: POST

Path: `deterministic-baseline`

Authentication: None

Respond: When Last Node Finishes

Response Data: First Entry JSON

OUTPUT

params	query	body	webhookUrl
(empty object)	(empty object)	<pre>{ "sensor_id": "temp_sensor_1", "timestamp": "2026-04-25T20:05:00Z", "value": 24.5, "type": "temperature", "unit": "C" }</pre>	http://localhost:5678/webhook-test/deterministic-baseline

▼ Node confirmation (globe shape) - POST low temp

POST middleware fan off

Method: POST

URL: `http://host.docker.internal:8000/fan/off`

Authentication: None

Send Query Parameters: ☐

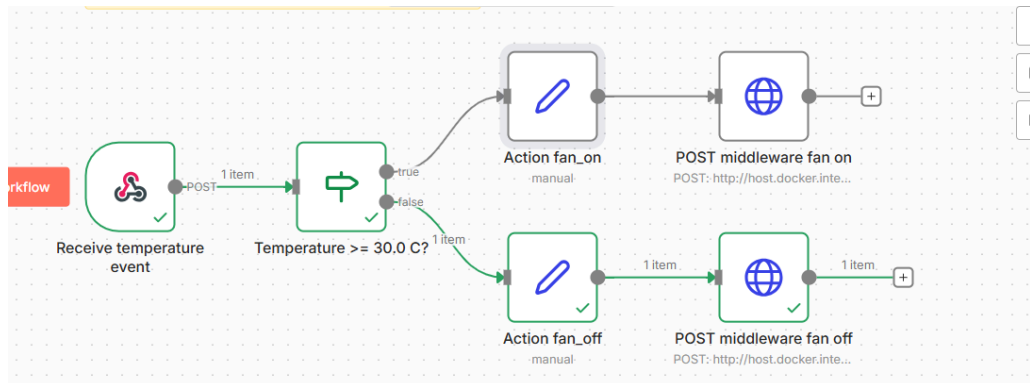
Send Headers: ☐

Send Body: ☐

Options: No properties

OUTPUT

status	action	fan	simulated	reason
ok	fan_off	off	true	manual_api_call



```
$webhookUrl = "http://localhost:5678/webhook-test/deterministic-baseline"
```

```
$lowBody = @{
  sensor_id = "temp_sensor_1"
}
```

```

timestamp = "2026-04-25T20:05:00Z"
type = "temperature"
value = 24.5
unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $lowBody

```

▼ (no terminal log script)

The middleware terminal did not visibly print request lines during this test, but the n8n HTTP Request node returned the middleware JSON response. This confirms that the middleware endpoint was reached successfully.

▼ step 6 full workflows

step 6 workflow

1. Inspect deterministic-baseline.json and [deterministic-baseline.md](#).
2. Run JSON parse check.
3. Import/recreate the workflow in n8n if possible.
4. Run middleware locally.
5. Trigger the workflow with a high-temperature test event.
6. Trigger the workflow with a low-temperature test event.
- x 7. Save screenshot/log evidence.
8. Ask audit-review to audit Step 6.
9. Use repair-fix only if audit finds real problems.
10. Ask report-scribe to create docs/report-notes/step-06-deterministic-baseline.md.
11. Commit manually.

▼ step 6 rundown

Here is the polished integration of your notes and the AI dump for Step 6. This one is highly valuable for your thesis methodology because it highlights a complex, multi-stage troubleshooting loop involving Docker networking, UI execution, and human-in-the-loop testing.

Create the "dumb" baseline—one n8n workflow with strictly deterministic rules and no AI reasoning.

- **What this proves:** This gives the thesis a controlled comparison anchor. Before introducing complex AI agents, you must prove the system can successfully ingest data, apply a basic hardcoded rule (*If Temp >= 30.0°C, turn fan on*), and trigger physical endpoints. This is the foundation for the comparisons in Chapter 6.

▼ AI Collaboration Workflow (How it was built)

This section documents the specific division of labor between the AI assistant and the human researcher.

1. **AI Task (Draft Generation):** The AI `step-runner` was instructed to create the deterministic n8n workflow file (`deterministic-baseline.json`) without any AI nodes. It successfully generated the logic structure (Webhook → IF Node → HTTP Requests), but flagged the file as a "draft" because it lacked the physical ability to test it inside the n8n UI.
2. **AI Task (Architecture Repair):** Before human testing, a secondary AI review caught two critical runtime bugs related to Docker:

- n8n was running inside Docker, so targeting `127.0.0.1` would hit the container's own loopback, not the Windows host middleware. The AI dynamically corrected this to `host.docker.internal`.
 - The webhook JSON path was corrected from `$json.value` to `$json.body.value`. The `step-runner` successfully applied these architectural fixes.
3. **Human Task (Execution & Troubleshooting):** The human researcher imported the JSON into the local n8n interface and began testing via PowerShell. There was a notable human-AI troubleshooting loop here: the human's first test only verified that the webhook received data. The AI stepped in to guide the human to actively click "Execute workflow" on the n8n canvas to prove the *downstream* nodes actually fired.
 4. **Human Task (Evidence Capture):** The human successfully executed the full end-to-end tests. By injecting a high temperature (`31.4°C`) and a low temperature (`24.5°C`) via PowerShell, the human captured visual proof (screenshots) of n8n successfully routing the logic to the `fan_on` and `fan_off` paths, receiving `{"status": "ok"}` responses from the local middleware.

▼ 📁 1. The Code Maps to the Evidence

The files the AI created directly line up with the thesis requirements.

- **Evidence: Exported Workflow & Documentation**
 - `cognitive_logic/workflows/deterministic-baseline.json`
 - `cognitive_logic/workflows/deterministic-baseline.md`
 - *Why it matters:* This feeds directly into **Chapter 4 (Choice of Approach)** and **Chapter 5 (Implementation)**. It proves you codified a baseline that can be universally shared and reproduced by other researchers.

▼ 🖨️ 2. The Actions Map to the Evidence

The physical testing proved the integration contract from Step 5 actually works in practice.

- **Evidence: Baseline Test Logs & Workflow Screenshots**
 - *Action:* Human clicked "Execute workflow" in n8n and fired `Invoke-RestMethod` payloads for both threshold extremes.
 - *Artifact Produced:* Screenshots of the n8n execution canvas showing green checkmarks on the `fan_on` and `fan_off` routes, as well as the JSON response confirming the middleware simulated the hardware action.
 - *Why it matters:* This feeds directly into **Chapter 6 (Results)**. Even though the Python terminal didn't visibly print logs during the test, the visual n8n execution history proving the nodes returned `{"fan": "on"}` and `{"fan": "off"}` serves as undeniable hard evidence that the system successfully bridged n8n and the middleware API.

📝 Summary

Step 6 successfully establishes the control group for the thesis. The AI was utilized to draft and debug complex Docker networking logic, while the human researcher acted as the physical execution layer—navigating the UI, triggering the API, and capturing the visual evidence required for the final academic report.

- -
 -
 -

▼ chapter 4

Chapter 4

- why deterministic baseline is needed
- baseline vs agent-enhanced comparison design

Chapter 4 — Choice of approach feeds from:

```
deterministic-baseline.md  
docs/decisions.md
```

▼ chapter 5

Chapter 5

- deterministic workflow implementation

Chapter 5 — Implementation feeds from:

```
cognitive_logic/workflows/deterministic-baseline.json  
workflow screenshots
```

▼ chapter 6

Chapter 6 — Results feeds from:

```
baseline logs  
baseline latency measurements
```

Chapter 6

- baseline result section

▼ Goal

Create the baseline that the agent-enhanced workflow will be compared against.

▼ Work to complete

▼ 1

Create one n8n workflow with no agent reasoning:

- input: temperature event
- rule: if temperature is above threshold, request `fan_on`
- otherwise no action or `fan_off`
- call the middleware endpoint
- log the result

Store:

- exported workflow JSON
- short explanation of nodes
- workflow screenshot

▼ 2

Work to complete

Create:

```
cognitive_logic/workflows/deterministic-baseline.json  
cognitive_logic/workflows/deterministic-baseline.md
```

The workflow should:

- receive temperature event
- apply threshold rule
- request `fan_on` or `fan_off`
- call middleware endpoint
- log result

▼ What this proves

This gives the thesis a controlled comparison anchor.

▼ Evidence produced

▼ 1

- deterministic workflow export
- workflow screenshot
- baseline test logs
- first latency measurements

▼ 2

- exported workflow JSON
- workflow screenshot
- baseline execution logs
- first basic latency logs

▼ Step 7 — Build the minimum agent-enhanced workflow (`cognitive_logic`)
(minimal AI/LLM step decides fan_on/fan_off using strict structured output)
chapter 4, Choice of approach
chapter 5, mplementation
chapter 7 Discussion

▼ Ai

▼ cx step runner

▼ q

Current step:

Step 7 — Build the minimum agent-enhanced workflow.

Use these files as source of truth:

- [AGENTS.md](#)
- [README.md](#)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/ongoing/integration-contract.md
- docs/evaluation/evaluation-protocol.md
- docs/n8n-version.md
- shared_interfaces/json-schema/sensor-event.schema.json
- shared_interfaces/json-schema/agent-action.schema.json
- shared_interfaces/examples/sensor-event.example.json
- shared_interfaces/examples/fan-on.example.json
- cognitive_logic/workflows/deterministic-baseline.json
- cognitive_logic/workflows/deterministic-baseline.md
- middleware/api/routes.py
- middleware/tests/manual-test-notes.md

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Obid minimum integration only
- Step 7 only
- one minimal agent-enhanced workflow
- one prompt set
- one memory choice
- one decision path
- structured output following the shared contract
- no broad multi-agent architecture
- no model benchmarking
- no multiple LLM providers
- no MCP
- no n8n core modification
- no real GPIO
- no Raspberry Pi deployment
- no Step 8 safety enforcement yet
- no Step 9 measurement harness yet

Goal:

Add only enough workflow intelligence to make the integrated system valid, without turning this into the old full Obid cognitive architecture thesis.

Task:

Create the minimum agent-enhanced workflow artifacts.

Create:

- `cognitive_logic/workflows/agent-minimal.json`
- `cognitive_logic/workflows/agent-minimal.md`
- `cognitive_logic/prompts/system-prompt-v1.md`
- `cognitive_logic/memory/memory-choice-v1.md`

The workflow should be designed to:

1. receive a temperature event
2. pass the relevant event data into one minimal agent/LLM decision step
3. instruct the agent to return structured output matching the agent-action schema
4. map or document the allowed actions:
 - `fan_on`
 - `fan_off`
5. call the same middleware endpoints as the deterministic baseline:
 - <http://host.docker.internal:8000/fan/on>
 - <http://host.docker.internal:8000/fan/off>

Important implementation constraint:

If a valid n8n export JSON can be created safely, create it.

If exact n8n AI node export structure cannot be guaranteed, create a clearly marked draft/export-candidate JSON and explain what must be verified inside the n8n UI before treating it as final.

Prompt requirements:

- `system-prompt-v1.md` must be minimal and strict
- it must tell the agent to output only a structured action object
- it must restrict actions to `fan_on` or `fan_off`
- it must restrict target to `fan_1`
- it must not allow direct hardware instructions outside the contract
- it must mention that safety/approval enforcement is Step 8 and is not implemented yet

Memory requirements:

- `memory-choice-v1.md` should choose one simple memory strategy
- choose either no memory/stateless execution or minimal window buffer memory
- justify why this choice fits the narrowed thesis and edge/resource constraints
- do not compare many memory strategies

Documentation requirements for `agent-minimal.md`:

- explain the workflow purpose
- explain how it differs from deterministic-baseline
- explain expected input shape
- explain expected structured output shape
- explain prompt/memory choice
- explain middleware endpoints

- explain import/test procedure
- explain runtime verification status
- explain what screenshot/log evidence should be saved
- explain how this feeds Chapter 4, Chapter 5, and Chapter 7
- clearly state this is minimum Obid-compatible integration, not a full cognitive architecture implementation

Do not:

- implement Step 8 safety approval logic
- implement a broad ReAct loop
- add multiple memory strategies
- add multiple models/providers
- add MCP
- add measurement harness scripts
- change middleware behavior unless absolutely necessary
- change shared schemas unless there is a direct mismatch that blocks Step 7

Validation:

- Ensure any JSON file created is valid JSON.
- If possible, run Python json parsing on cognitive_logic/workflows/agent-minimal.json.
- Do not claim the workflow has been runtime-tested unless it was actually imported/run in n8n.
- If runtime testing is not possible, mark runtime verification as pending.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to inspect/import/test the minimal agent workflow in n8n.
- tell me what evidence should be saved for Chapter 4, Chapter 5, and Chapter 7.

▼ a

▼ files

4 files changed	+742 -0	Undo ↶
cognitive_logic/memory/memory-choice-v1.md	+35 -0	▼
cognitive_logic/prompts/system-prompt-v1.md	+45 -0	▼
cognitive_logic/workflows/agent-minimal.json	+338 -0	▼
cognitive_logic/workflows/agent-minimal.md	+324 -0	▼

Step 7 artifacts are created and JSON-validated.

Added:

- agent-minimal.json
- agent-minimal.md

- [system-prompt-v1.md](#)
- [memory-choice-v1.md](#)

What it does:

- Receives a temperature event through a Webhook.
- Prepares the event as agent input.
- Uses one minimal LLM decision step as a draft n8n AI node.
- Requires structured output matching `agent-action.schema.json`.
- Routes only `fan_on` and `fan_off`.
- Calls:
 - `http://host.docker.internal:8000/fan/on`
 - `http://host.docker.internal:8000/fan/off`
- Uses stateless/no-memory execution.
- Does not add prompts beyond v1, memory variants, safety logic, MCP, measurement, real GPIO, or middleware changes.

Validation:

- `cognitive_logic/workflows/agent-minimal.json` parses as valid JSON.
- Runtime verification is still pending because it has not been imported/run in n8n.

Import/test in n8n:

1. Start middleware: `python -m middleware.api.app`
2. Open n8n at `http://localhost:5678`
3. Import `cognitive_logic/workflows/agent-minimal.json`
4. Verify/repair the AI node in the UI:
 - exactly one model/provider credential
 - prompt follows `cognitive_logic/prompts/system-prompt-v1.md`
 - no memory node connected
 - output field matches the parse node
5. Test high temp `31.4`: expect structured `fan_on`, then `/fan/on`.
6. Test low temp `24.5`: expect structured `fan_off`, then `/fan/off`.

Evidence to save:

- Chapter 4: why this is minimum Obid-compatible integration, not full cognitive architecture.
- Chapter 5: workflow canvas, AI node config, prompt, no-memory choice, middleware endpoint config.
- Chapter 7: limitations, lack of Step 8 enforcement, structured-output risks, and why safety/approval remains deferred.

▼ setup walkthrough

You're right. We **should not skip directly to audit-review** if you want Step 7 to be fully verified like Step 6.

What I wrote was the **conceptual workflow**, not something you run directly in PowerShell. For Step 7, the real test happens inside **n8n**, same as Step 6, but with one extra complication:

Step 7 needs an actual AI/model node configured with credentials. Step 6 did not.

So yes, we do a full test run **if your n8n AI node imports and you have a model credential available**.

Step 7 status right now

Codex created:

```
agent-minimal.json
agent-minimal.md
system-prompt-v1.md
memory-choice-v1.md
```

That means:

```
static Step 7 artifacts exist ✔
runtime test not done yet ✖
```

So the correct flow is:

1. Try importing agent-minimal.json into n8n
2. Configure/repair the AI node in n8n UI
3. Run high-temp test
4. Run low-temp test
5. Save evidence
6. Then audit-review
7. Then report-scribe
8. Then commit

What you actually run

You do **not** run this text directly:

```
Temperature event -> n8n Webhook -> minimal LLM decision step...
```

That is the design explanation.

What you actually do is import this file into n8n:

```
cognitive_logic/workflows/agent-minimal.json
```

Then test it through the n8n webhook, like Step 6.

Step 7 full runtime test workflow

1. Start middleware

PowerShell, repo root:

```
cd "C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
python -m middleware.api.app
```

Leave it running.

2. Start n8n

Second PowerShell:

```
cd "C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub  
-thesis\infrastructure\docker"  
docker compose up -d
```

Open:

```
http://localhost:5678
```

3. Import Step 7 workflow

In n8n:

```
Workflows -> Import from file
```

Select:

```
cognitive_logic/workflows/agent-minimal.json
```

Now one of two things will happen.

Case A — it imports cleanly

Good. Then inspect the nodes.

You should see something like:

```
Webhook  
-> Prepare agent input  
-> Minimal LLM / AI decision  
-> Parse structured action  
-> fan_on / fan_off routing  
-> POST middleware fan on/off
```

Then click the AI/LLM node.

Check:

```
model/provider credential is selected  
prompt follows system-prompt-v1.md  
output expected is JSON  
no memory node is connected
```

The prompt should force output like:

```
{  
  "action_id": "fan_on",  
  "target": "fan_1",  
  "reason": "temperature is at or above threshold",  
  "requires_approval": false  
}
```

or:

```
{
  "action_id": "fan_off",
  "target": "fan_1",
  "reason": "temperature is below threshold",
  "requires_approval": false
}
```

Case B — import fails or AI node is broken

This is possible and not surprising.

Then Step 7 is still useful, but you manually rebuild the workflow in n8n UI using the `.md` file as the guide.

If this happens, tell Codex:

The Step 7 n8n import failed or the AI node did not import correctly.

Create a manual n8n recreation guide for agent-minimal.md:

- exact nodes to create
- node order
- what each node should do
- where to paste system-prompt-v1.md
- expected structured JSON output
- how to route fan_on/fan_off
- what screenshots to save

Do not change scope.

Do not implement Step 8 safety.

4. Configure the AI model

This is the part Step 6 did not need.

In the AI node, you need one working model credential/provider.

Use whatever single provider you planned for this thesis. Do **not** compare models.

The thesis rule is:

```
one model/provider as implementation parameter
not model benchmarking
```

So choose one and move on.

If you do not have an API key/model available yet, then Step 7 runtime test is blocked. In that case, document:

Step 7 import/configuration prepared, runtime verification pending due to missing model credential.

But if you have credentials, continue.

5. Test high temperature

Click **Execute workflow** in n8n.

Send this in PowerShell:

```
$webhookUrl = "http://localhost:5678/webhook-test/agent-minimal"

$highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -
Body $highBody
```

But check the actual webhook path in n8n. If the Webhook node path is not `agent-minimal`, use the URL shown by n8n.

Expected result:

```
AI outputs structured fan_on action
workflow routes to fan_on
POST middleware fan on executes
middleware response shows:
status = ok
action = fan_on
fan = on
simulated = true
```

Save screenshots:

```
step-07-high-temp-agent-output.png
step-07-high-temp-fan-on-canvas.png
step-07-high-temp-fan-on-http-output.png
```

6. Test low temperature

Click **Execute workflow** again.

```
$webhookUrl = "http://localhost:5678/webhook-test/agent-minimal"

$lowBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -
Body $lowBody
```

Expected result:

```
AI outputs structured fan_off action
workflow routes to fan_off
POST middleware fan off executes
```

```
middleware response shows:
status = ok
action = fan_off
fan = off
simulated = true
```

Save screenshots:

```
step-07-low-temp-agent-output.png
step-07-low-temp-fan-off-canvas.png
step-07-low-temp-fan-off-http-output.png
```

7. Evidence folder

Save Step 7 evidence here:

```
cognitive_logic/workflows/evidence/step-07/
```

Suggested files:

```
step-07-agent-workflow-canvas.png
step-07-ai-node-config.png
step-07-system-prompt-config.png
step-07-high-temp-agent-output.png
step-07-high-temp-fan-on-output.png
step-07-low-temp-agent-output.png
step-07-low-temp-fan-off-output.png
step-07-runtime-verification.md
```

The `step-07-runtime-verification.md` should say clearly whether runtime verification succeeded or was blocked by credentials/import issues.

8. Then audit-review

After runtime evidence exists, then audit-review.

Not before, unless you only want static audit.

So you were right to challenge that.

Correct Step 7 workflow

```
Step-runner creates artifacts
You import/test in n8n
Save evidence
Update runtime status
Audit-review checks it
Repair-fix only if needed
Report-scribe writes Step 7 note
Commit
```

Bottom line

Yes, ideally we do a full test run like Step 6.

The only reason I hesitated is because Step 7 depends on actual n8n AI/model credential setup. But if you can configure the AI node, then absolutely: test both `31.4 -> fan_on` and `24.5 -> fan_off` before calling Step 7 done.

▼ step 7 workflow

What the Step 7 workflow is supposed to be

The intended workflow is:

```
Temperature event
-> n8n Webhook
-> prepare event for agent
-> minimal LLM/AI decision step
-> structured action output
-> route fan_on / fan_off
-> call middleware endpoint
```

More concretely:

```
Receive temperature event
-> Prepare agent input
-> Minimal AI decision node
    outputs:
    {
      "action_id": "fan_on" or "fan_off",
      "target": "fan_1",
      "reason": "...",
      "requires_approval": false
    }
-> Check action_id
    fan_on -> POST http://host.docker.internal:8000/fan/on
    fan_off -> POST http://host.docker.internal:8000/fan/off
```

The important difference from Step 6:

```
Step 6: deterministic IF rule decides fan_on/fan_off
Step 7: minimal AI/LLM step decides fan_on/fan_off using strict structured output
```

What Step 7 should NOT include

Codex seems to have respected this:

```
no broad ReAct system
no multi-agent architecture
no multiple memory strategies
no model benchmarking
no Step 8 safety enforcement
no measurement harness
no real GPIO
no Raspberry Pi
```

Good.

▼ high temp

▼ seperate terminal

Step 7 full runtime test workflow

1. Start middleware

PowerShell, repo root:

```
cd"C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"  
python-mmiddleware.api.app
```

Leave it running.

2. Start n8n

Second PowerShell:

```
cd"C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"  
dockercomposeup-d
```

Open:

```
http://localhost:5678
```

3. Import Step 7 workflow

In n8n:

Workflows -> Import from file

Select:

cognitive_logic/workflows/agent-minimal.json

▼ terminal high temp

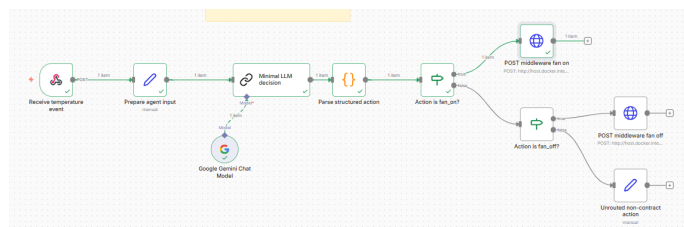
```
$webhookUrl = "http://localhost:5678/webhook-test/agent-minimal"
```

```
$highBody = @{  
  sensor_id = "temp_sensor_1"  
  timestamp = "2026-04-25T20:00:00Z"  
  type = "temperature"  
  value = 31.4  
  unit = "C"  
} | ConvertTo-Json
```

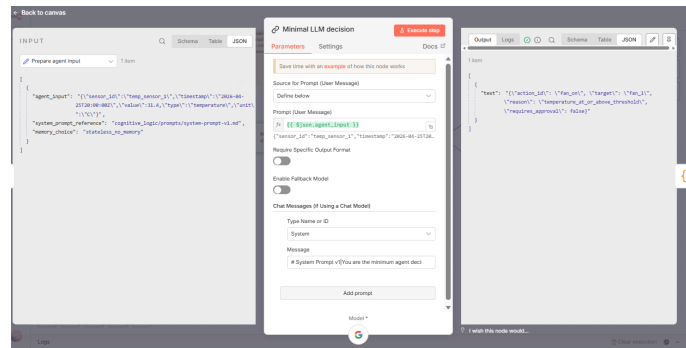
```
Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $highBody
```

- -

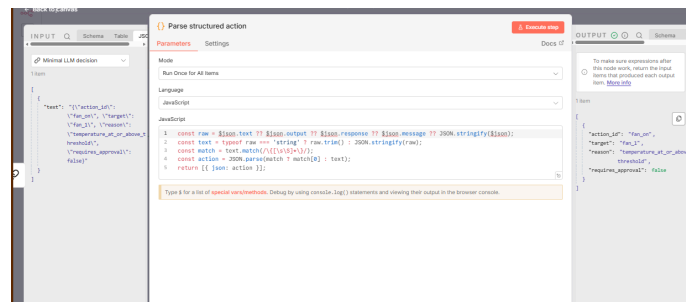
▼ high temp n8n



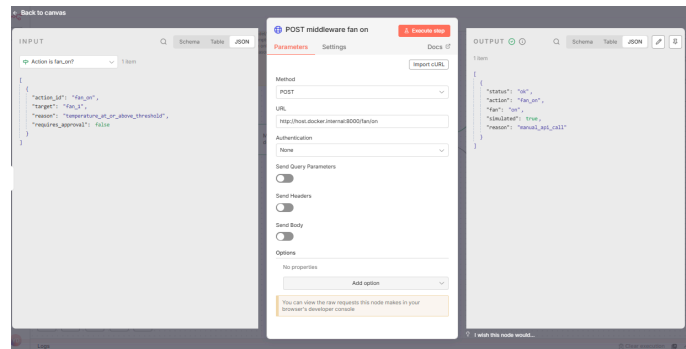
▼ LLM node - Save screenshot showing the LLM output. You want to see structured JSON or text containing:



▼ parse structure action



▼ post middleware fan on



▼ low temp

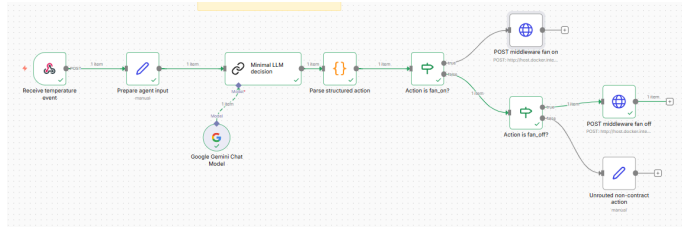
▼ terminal

```
$webhookUrl = "http://localhost:5678/webhook-test/agent-minimal"

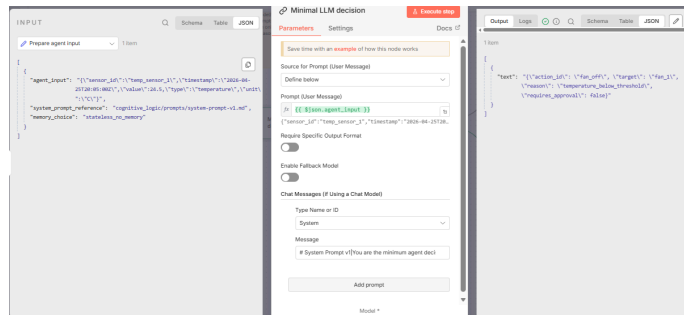
$lowBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json
```

```
Invoke-RestMethod -Method Post -Uri $webhookUrl -ContentType "application/json" -Body $lowBody
```

- -
- ▼ low temp tree



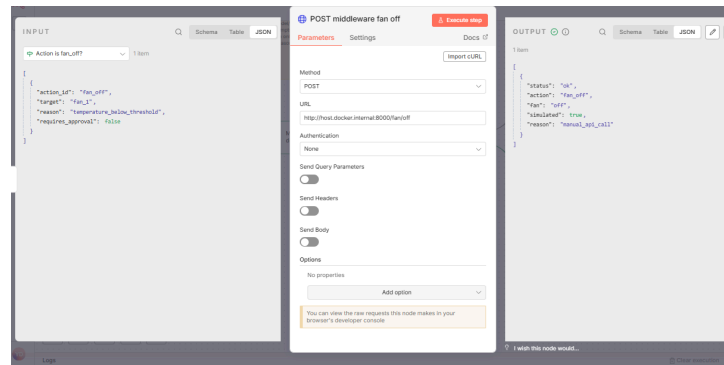
- ▼ minimal llm



- ▼ parse structure action



▼ post middleware fan on



-

▼ llm agent in n8n prompt

```
# System Prompt v1

You are the minimum agent decision step for the new-yacoub thesis IoT workflow.

Your only task is to choose one fan action for one valid temperature event.

Return only one JSON object. Do not return Markdown, commentary, code fences,
or extra text.

The JSON object must have exactly these fields:
```

▼ chapter feeds from and deliverables

Chapter feeds from

Chapter 4 — Choice of approach feeds from:

```
docs/decisions.md
cognitive_logic/memory/memory-choice-v1.md
```

Chapter 5 — Implementation feeds from:

```
cognitive_logic/workflows/agent-minimal.json
cognitive_logic/prompts/system-prompt-v1.md
cognitive_logic/memory/memory-choice-v1.md
```

Chapter 7 — Discussion feeds from:

```
why Obid was minimized
agent limitations
prompt/memory limitations
```

Chapter deliverables

This step helps deliver:

Chapter 4

- full multi-agent vs minimal workflow reasoning
- why model benchmarking is out of scope

Chapter 5

- minimal agent workflow
- prompt and memory design

Chapter 7

- discussion of minimized Obid layer
- limitations of the agent-enhanced path

▼ step 7 rundown

Here is the polished integration of your notes and the AI dump for Step 7. This step is critical because it explicitly demonstrates how you controlled the "AI brain" to prevent it from exceeding your narrowed thesis scope.

Step 7 Rundown: The Minimal Agent Workflow & AI Collaboration Log

▼ 🎯 The Goal

Introduce just enough workflow intelligence to make the integrated system valid, proving the system can support an AI-enhanced path *without* bloating into the broader multi-agent architecture originally planned in the Obid thesis.

- **What this proves:** This proves that an LLM can be safely dropped into the n8n execution layer, provided it is strictly constrained by the shared contract (Step 5).

▼ 🤖 AI Collaboration Workflow (How it was built)

This section documents the specific division of labor between the AI assistant and the human researcher.

1. **AI Task (Prompt & Architecture Design):** The AI `step-runner` was instructed to build the agent workflow (`agent-minimal.json`). Crucially, the AI generated two governing documents:
 - `memory-choice-v1.md` : The AI deliberately locked the design to "stateless execution" (no memory) to keep the behavior predictable and measurable against the Step 6 baseline.
 - `system-prompt-v1.md` : The AI wrote a highly restrictive system prompt forcing the n8n LLM node to return *only* a JSON object matching the `fan_on` / `fan_off` contract, explicitly commanding the model not to invent fake hardware instructions.
2. **AI Task (Test Protocol & Scope Enforcement):** Because Step 7 requires a real LLM API key, the AI orchestrator provided the human with a strict testing protocol. It explicitly warned the human **not** to compare different models, actively enforcing the thesis rule that the LLM is just a simple "implementation parameter," not a benchmarking variable.
3. **Human Task (Execution & Credential Setup):** The human researcher imported the workflow into the local n8n UI and configured the AI node with a single model credential. The human then manually triggered the workflow from the n8n canvas.
4. **Human Task (Evidence Capture):** Using PowerShell, the human injected the high (`31.4°C`) and low (`24.5°C`) payloads. The human captured the critical visual evidence: screenshots proving the LLM node successfully ingested the temperature, obeyed the system prompt, and output the perfectly structured JSON strings (`{"action_id": "fan_on"...}`) which n8n then successfully routed to the local middleware.

▼ 1. The Code Maps to the Evidence

The files the AI created directly line up with the thesis requirements to explain *how* the intelligence is constrained.

- **Evidence: Workflow, Prompt, and Memory Rationale**

- `cognitive_logic/workflows/agent-minimal.json`
- `cognitive_logic/prompts/system-prompt-v1.md`
- `cognitive_logic/memory/memory-choice-v1.md`
- *Why it matters:* This feeds directly into **Chapter 4 (Choice of approach)** and **Chapter 5 (Implementation)**. The `memory-choice-v1.md` file serves as the written justification for why a stateless design fits the edge/resource constraints of your thesis, and the system prompt proves how you forced the AI to obey the Step 5 JSON contract.

▼ 2. The Actions Map to the Evidence

The physical testing proved the AI can reliably drive the physical API endpoints.

- **Evidence: Structured Agent Output Logs & Screenshots**

- *Action:* Human clicked "Execute workflow" in n8n, fired PowerShell webhooks, and inspected the LLM node output inside n8n.
- *Artifact Produced:* Screenshots of the "Minimal LLM decision node" generating raw JSON, the "Parse structured action" node successfully reading it, and the `POST middleware` node confirming the `{"simulated": true}` response.
- *Why it matters:* This feeds directly into **Chapter 7 (Discussion)**. The screenshots showing the raw JSON output prove that the system works, but they also serve as the foundation for discussing the limitations and risks of using LLMs for physical actions (which tees up the necessity of Step 8: The Safety Layer).

📝 Summary

Step 7 successfully bridges the gap between deterministic code and AI reasoning. The AI assistant was utilized to design the rigid constraints (stateless memory + strict JSON prompting), while the human researcher handled the credential injection, execution, and visual verification required to prove the "brain" successfully controlled the "hardware" via the established contract.

▼ Goal

Add only enough workflow intelligence to make the integrated system valid.

▼ Work to complete

▼ 2

Work to complete

Create:

```
cognitive_logic/workflows/agent-minimal.json
cognitive_logic/workflows/agent-minimal.md
cognitive_logic/prompts/system-prompt-v1.md
cognitive_logic/memory/memory-choice-v1.md
```

Keep it limited to:

- one workflow
- one prompt set
- one memory choice
- one decision path
- structured output following schema

Create:

- `cognitive_logic/workflows/agent-minimal.json`
- `cognitive_logic/prompts/system-prompt-v1.md`
- `cognitive_logic/memory/memory-choice-v1.md`

Keep it limited to:

- one workflow
- one prompt set
- one memory choice
- one decision path
- structured output following the shared contract

▼ What this proves

This proves the system can support a minimal agent-enhanced path without becoming a full Obid thesis implementation.

▼ Evidence produced

▼ 2

Evidence produced

- workflow export
- prompt file
- memory rationale
- structured output logs
- screenshots
- workflow export
- prompt file
- memory rationale
- logs showing structured agent output

▼ Step 8 — (`safetely layer`) Add the minimum safety layer, incase the step 7 AI agent goes A-wall
chapter 5 mplementation

▼ AI

▼ cx step runner prompt

▼ q

Step 8 — Add the minimum safety layer.

Use these files as source of truth:

- [AGENTS.md](#)
- [README.md](#)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/ongoing/integration-contract.md
- shared_interfaces/json-schema/agent-action.schema.json
- shared_interfaces/examples/fan-on.example.json
- shared_interfaces/examples/blocked-action.example.json
- cognitive_logic/workflows/agent-minimal.md
- cognitive_logic/prompts/system-prompt-v1.md
- cognitive_logic/memory/memory-choice-v1.md

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Obid minimum integration only
- Step 8 only
- minimum safety layer only
- one policy
- one validation/parser description
- one human-in-the-loop approval description
- one allowed case
- one blocked case
- one risky/unknown case requiring approval or rejection
- no broad guardrail research
- no multi-agent safety architecture
- no MCP
- no n8n core modification
- no real GPIO
- no Raspberry Pi deployment

- no measurement harness yet
- no Step 9 evaluation scripts yet

Goal:

Make the system safe enough to be academically defensible by documenting and testing the minimum safety boundary between agent output and middleware action.

Task:

Create the minimum safety-layer artifacts.

Create:

- safety_layer/policies/action-policy-v1.md
- safety_layer/parsers/output-validation-v1.md
- safety_layer/approvals/hitl-v1.md
- safety_layer/examples/allowed-case.md
- safety_layer/examples/blocked-case.md
- safety_layer/examples/risky-approval-case.md

Requirements:

1. action-policy-v1.md must define the minimum allowed action policy:
 - allowed action_id values:
 - fan_on
 - fan_off
 - allowed target:
 - fan_1
 - required fields:
 - action_id
 - target
 - reason
 - requires_approval
 - action must match shared_interfaces/json-schema/agent-action.schema.json
 - unknown actions or unknown targets are not allowed to directly reach middleware
 - malformed output must be blocked
 - risky/unknown output must require human approval or be rejected
 - Step 8 must clearly state that direct hardware access is never allowed
2. output-validation-v1.md must describe the minimum validation/parser logic:
 - parse agent output as JSON
 - verify required fields exist
 - verify action_id is fan_on or fan_off
 - verify target is fan_1
 - verify requires_approval is boolean
 - reject malformed JSON
 - reject schema-invalid action

- reject unknown action/target
 - pass allowed action forward only after validation
 - mention that this is a minimal validation design, not broad guardrail research
3. hitl-v1.md must describe the human-in-the-loop checkpoint:
 - when approval is required
 - what the human reviewer checks
 - allowed reviewer decisions:
 - approve
 - reject
 - what happens after approval
 - what happens after rejection
 - keep it lightweight and thesis-scoped
 4. allowed-case.md must document one valid allowed action:
 - example action:


```
{
  "action_id": "fan_on",
  "target": "fan_1",
  "reason": "temperature_at_or_above_threshold",
  "requires_approval": false
}
```
 - explain why it is allowed
 - explain which middleware endpoint it can reach after validation
 5. blocked-case.md must document one malformed or invalid action:
 - use either malformed JSON or missing required fields
 - explain why it is blocked
 - explain that it must not reach middleware
 6. risky-approval-case.md must document one risky or unknown action:
 - example:


```
{
  "action_id": "open_window",
  "target": "unknown_device",
  "reason": "agent_invented_action",
  "requires_approval": true
}
```
 - explain why it is not allowed to directly reach middleware
 - explain whether the minimum policy rejects it or routes it to human approval
 - keep the behavior consistent with action-policy-v1.md
 7. Do not implement a large runtime safety framework.
 8. Do not modify middleware behavior unless absolutely necessary.
 9. Do not modify the Step 7 agent workflow unless absolutely necessary.
 10. Do not implement Step 9 measurement harness.
 11. Do not claim real runtime enforcement exists unless you actually add and test it.

12. If this step is documentation/specification only, clearly state that runtime enforcement is represented as a minimum safety design and case documentation, not production-grade enforcement.

Optional update:

- Update docs/architecture/README.md only if needed to point to the safety_layer files.
- Update docs/ongoing/integration-contract.md only if needed to mention that Step 8 defines the validation/approval boundary.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to inspect the safety layer.
- tell me what evidence should be saved for Chapter 5, Chapter 6, and Chapter 7.
- clearly state whether Step 8 is runtime-enforced or documentation/specification-level only.

▼ a

▼ files

8 files changed +499 -0	Undo ↶
docs/architecture/README.md +13 -0	▼
docs/ongoing/integration-contract.md +13 -0	▼
safety_layer/approvals/hitl-v1.md +76 -0	▼
safety_layer/examples/allowed-case.md +47 -0	▼
safety_layer/examples/blocked-case.md +50 -0	▼
safety_layer/examples/risky-approval-case.md +64 -0	▼
safety_layer/parsers/output-validation-v1.md +110 -0	▼
safety_layer/policies/action-policy-v1.md +126 -0	▼

Step 8 minimum safety layer is done as a documentation/specification layer.

Created:

- [action-policy-v1.md](#)
- [output-validation-v1.md](#)
- [hitl-v1.md](#)
- [allowed-case.md](#)
- [blocked-case.md](#)
- [risky-approval-case.md](#)

Updated:

- [docs/architecture/README.md](#)
- [docs/ongoing/integration-contract.md](#)

Policy behavior:

- `fan_on` and `fan_off` are the only allowed actions.

- `fan_1` is the only allowed target.
- malformed output is blocked.
- unknown actions/targets are rejected for direct execution.
- valid actions with `requires_approval: true` go to HITL.
- direct hardware access is explicitly never allowed.

How to inspect:

```
Get-Content safety_layer\\policies\\action-policy-v1.md
Get-Content safety_layer\\parsers\\output-validation-v1.md
Get-Content safety_layer\\approvals\\hitl-v1.md
Get-Content safety_layer\\examples\\*.md
```

Evidence to save:

- Chapter 5: policy, parser design, HITL design, and the three case files.
- Chapter 6: allowed/blocked/risky case outcomes once later runtime evidence exists.
- Chapter 7: limitations, why this is minimum safety, and why full enforcement is deferred.

Important: Step 8 is **not runtime-enforced yet**. It is a minimum safety design and case documentation layer, not production-grade enforcement.

▼ audit-review

▼ q

Audit the Step 8 minimum safety layer changes only.

Check:

- Step 8 stays inside new-yacoub scope.
- The safety layer is minimum and thesis-scoped.
- action-policy-v1.md defines:
 - allowed action_id values: fan_on, fan_off
 - allowed target: fan_1
 - required fields: action_id, target, reason, requires_approval
 - malformed output is blocked
 - unknown actions/targets cannot directly reach middleware
 - valid actions with requires_approval=true go to HITL
 - direct hardware access is never allowed
- output-validation-v1.md describes:
 - parse JSON
 - check required fields
 - check allowed action_id
 - check allowed target
 - check requires_approval is boolean
 - reject malformed JSON
 - reject schema-invalid action

- reject unknown action/target
- hitl-v1.md describes:
 - when approval is required
 - what reviewer checks
 - approve/reject decisions
 - what happens after approval/rejection
- allowed-case.md documents one valid allowed action.
- blocked-case.md documents one malformed or invalid blocked action.
- risky-approval-case.md documents one risky/unknown action requiring approval or rejection.
- docs/architecture/README.md and docs/ongoing/integration-contract.md were updated only as needed.
- no middleware runtime behavior was changed.
- no Step 9 measurement harness was implemented.
- no MCP, n8n core changes, real GPIO, Raspberry Pi deployment, or broad guardrail research was added.
- the docs clearly state whether Step 8 is runtime-enforced or specification/documentation-level only.

Important:

- Do not demand full runtime enforcement unless the current Step 8 plan requires it.
- Do not suggest broader guardrail architecture unless something blocks Step 9.

Return:

1. blocking issues
2. non-blocking cleanup
3. Step 8 verdict
4. whether report-scribe can create the Step 8 report note

▼ a

Blocking Issues

No blocking issues found for Step 8.

The Step 8 safety layer stays inside new-yacoub scope and is clearly specification/documentation-level only. It does not falsely claim runtime enforcement.

Non-Blocking Cleanup

The working tree still has untracked Step 7 files/report note alongside the Step 8 files. That is not a Step 8 scope problem, but staging should be deliberate so Step 8 does not accidentally mix unrelated prior-step artifacts unless that is intended.

No root `.zip` archive artifacts are present now.

Step 8 Verdict

Step 8 passes audit.

The required files are present and cover the minimum safety layer:

- `action-policy-v1.md` defines allowed `fan_on` / `fan_off`, target `fan_1`, required fields, blocking rules, HITL routing for valid approval-required actions, and no direct hardware access.
- `output-validation-v1.md` describes JSON parsing, required-field checks, allowed action/target checks, boolean approval checks, and rejection of malformed/schema-invalid/unknown output.

- `hitl-v1.md` defines when approval is required, reviewer checks, approve/reject decisions, and post-decision behavior.
- The allowed, blocked, and risky approval examples are present.
- `docs/architecture/README.md` and `docs/ongoing/integration-contract.md` only add concise references to the Step 8 safety artifacts and clarify that runtime enforcement is not claimed yet.
- No middleware runtime behavior, Step 9 measurement harness, MCP, n8n core changes, real GPIO, Raspberry Pi deployment, or broad guardrail research was added.

Report-Scribe

Yes, report-scribe can create the Step 8 report note now.

▼ report scribe

▼ q

Create the report note for Step 8.

Create:

`docs/report-notes/step-08-minimum-safety-layer.md`

Use only evidence already present in the repo and the Step 8 safety files.

Use these sources:

- `safety_layer/policies/action-policy-v1.md`
- `safety_layer/parsers/output-validation-v1.md`
- `safety_layer/approvals/hitl-v1.md`
- `safety_layer/examples/allowed-case.md`
- `safety_layer/examples/blocked-case.md`
- `safety_layer/examples/risky-approval-case.md`
- `docs/architecture/README.md`
- `docs/ongoing/integration-contract.md`
- `shared_interfaces/json-schema/agent-action.schema.json`
- `docs/plans/new-yacoub-14-step-process.md`

Include:

- what was built
- why it matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent runtime results
- do not claim production-grade safety
- do not claim middleware runtime enforcement exists if it does not
- clearly document that Step 8 is a minimum safety design/specification layer
- document allowed, blocked, and risky/approval cases

- mention that runtime measurement/evaluation belongs to Step 9

▼ a

Step 08 - Minimum Safety Layer

What was built

- A minimum Step 8 safety design/specification layer was added under `safety_layer/`.
- The safety layer defines a narrow action policy for the first temperature-to-fan scenario.
- A parser/validation design describes how workflow or agent output should be checked before middleware routing.
- A lightweight human-in-the-loop approval design describes when a reviewer may approve or reject an action.
- Three example cases were documented:
 - allowed valid action
 - blocked malformed action
 - risky or unknown action requiring review and expected rejection

This step documents the intended safety boundary. It does not implement a production-grade safety framework.

Why it matters for the thesis

Step 8 makes the workflow-to-middleware boundary academically defensible by specifying that workflow-produced actions should not directly trigger middleware endpoints unless they pass validation and, when needed, human approval.

It supports the narrowed thesis scope by providing exactly one minimal validation/approval path for one action family: `fan_on` and `fan_off` targeting `fan_1`. This keeps the Obid-compatible layer small while giving the report concrete safety cases to discuss.

Evidence produced

- `safety_layer/policies/action-policy-v1.md`
- `safety_layer/parsers/output-validation-v1.md`
- `safety_layer/approvals/hitl-v1.md`
- `safety_layer/examples/allowed-case.md`
- `safety_layer/examples/blocked-case.md`
- `safety_layer/examples/risky-approval-case.md`
- `docs/architecture/README.md`
- `docs/ongoing/integration-contract.md`
- `shared_interfaces/json-schema/agent-action.schema.json`

The safety design is based on the Step 5 action schema:

- required fields: `action_id`, `target`, `reason`, `requires_approval`
- allowed actions: `fan_on`, `fan_off`

- allowed target: `fan_1`
- no additional fields allowed

Safety Cases

Allowed case:

- Example file: `safety_layer/examples/allowed-case.md`
- Input action: `fan_on`, target `fan_1`, non-empty reason, `requires_approval: false`
- Expected decision: allow after validation
- Expected middleware endpoint after validation: `POST /fan/on`

Blocked case:

- Example file: `safety_layer/examples/blocked-case.md`
- Input action includes `action_id` and `target` only
- Missing fields: `reason`, `requires_approval`
- Expected decision: block
- Expected middleware endpoint: none

Risky/approval case:

- Example file: `safety_layer/examples/risky-approval-case.md`
- Input action: `open_window`, target `unknown_device`, `requires_approval: true`
- Reason it is risky: unknown action, unknown target, outside the first scenario
- Expected minimum-policy behavior: reject for direct execution; it may be shown to a human reviewer for explanation/evidence
- Expected middleware endpoint: none

Report chapters it feeds

- Chapter 5 - Implementation: safety policy, validation design, HITL approval design, and documented safety cases.
- Chapter 6 - Results: later allowed/blocked/risky case outcomes once Step 9 runtime evaluation exists.
- Chapter 7 - Discussion: safety limitations, human approval tradeoffs, blocked-case interpretation, and why this is a minimum thesis-scoped safety layer.

Limitations or assumptions

- Step 8 is a minimum safety design/specification layer.
- The files document intended validation and approval behavior; they do not prove runtime enforcement.
- Middleware runtime enforcement is not claimed here.
- No production-grade safety framework is implemented.
- No runtime approval UI or operational approval system is implemented.
- Unknown actions and unknown targets are not allowed to execute under policy v1.
- Human approval cannot turn an unknown action or unknown target into a middleware call under policy v1.
- The middleware fan endpoints remain simulated at this stage.
- Runtime measurement and evaluation belong to Step 9.

- No latency, RAM, CPU, thermal/resource, or repeated-run safety outcome measurements are produced by this step.

Screenshots/logs still needed

- Runtime validation logs showing allowed actions passing validation.
- Runtime validation logs showing malformed actions being blocked.
- Runtime approval or rejection logs for risky/unknown output.
- Evidence that blocked and risky cases do not call `/fan/on` or `/fan/off`.
- Step 9 evaluation results for safe vs blocked action behavior.
- Any UI or workflow screenshots if a human approval checkpoint is later implemented in n8n or middleware.

▼ step 8 rundown

Here is the polished integration of your notes and the AI dump for Step 8. This step is particularly strong for your methodology because it highlights how you used AI to establish **academic boundaries and theoretical constraints** rather than just writing executable code

▼ 🎯 The Goal

Make the system safe enough to be academically defensible by documenting a strict safety boundary between the agent's output and the middleware.

- **What this proves:** This proves that the workflow output (the LLM's brain) cannot directly trigger physical middleware actions (the hardware) without first passing through strict validation and, if necessary, human approval.
-

▼ 🤖 AI Collaboration Workflow (How it was built)

This section documents the specific division of labor between the AI assistant and the human researcher.

1. **AI Task (Specification Generation):** The AI `step-runner` was instructed to design the minimum safety layer. It generated the governing documents (`action-policy-v1.md`, `output-validation-v1.md`, and `hitl-v1.md`). **Crucially for academic integrity**, the AI explicitly documented this as a *specification layer* rather than falsely claiming it had built a production-grade runtime safety firewall. It designed the rules, not the final runtime enforcement (which is deferred).
 2. **AI Task (Case Documentation):** To provide concrete data for your results and discussion chapters, the AI generated three distinct test cases based on the Step 5 contract:
 - **Allowed Case:** A perfectly structured `fan_on` command.
 - **Blocked Case:** A malformed JSON payload missing required fields.
 - **Risky/Approval Case:** An example of an LLM hallucination (requesting to `open_window` on an `unknown_device`), demonstrating what happens when the agent tries to break out of the defined thesis scope.
 3. **AI Task (Peer-Review & Scope Audit):** The `audit-review` AI was deployed to inspect the first AI's work. It rigorously audited the files to ensure the rules strictly adhered to the narrowed scope—confirming that the safety layer did not accidentally bloat into a massive, multi-agent guardrail research project.
 4. **AI Task (Report Generation):** Finally, the `report-scribe` AI compiled the documentation into a cohesive report note, explicitly noting that while the *rules* exist, actual *runtime measurement* is deferred to Step 9.
-

▼ 📁 1. The Code Maps to the Evidence

The files the AI created directly line up with the thesis requirements to explain *how* the system remains safe.

• Evidence: Safety Policies & Parsers

- `safety_layer/policies/action-policy-v1.md`
- `safety_layer/parsers/output-validation-v1.md`
- `safety_layer/approvals/hitl-v1.md`
- *Why it matters:* This feeds directly into **Chapter 5 (Implementation)**. It proves you have formally defined how JSON parsing, required-field checks, and Human-in-the-Loop (HITL) routing should govern the system.

▼ 📁 2. The Actions Map to the Evidence

The test cases prove you have systematically categorized the potential behaviors of the AI agent.

• Evidence: The Safety Cases

- `safety_layer/examples/allowed-case.md`
- `safety_layer/examples/blocked-case.md`
- `safety_layer/examples/risky-approval-case.md`
- *Why it matters:* These cases form the backbone of **Chapter 6 (Results)** and **Chapter 7 (Discussion)**. By documenting a "risky case" (like the `open_window` hallucination), you successfully seed the ethical and societal discussion around physical AI actions, human approval tradeoffs, and the inherent reliability limitations of LLMs.

▼ chapters fed from and chapter deliverables

Chapter feeds from

Chapter 5 — Implementation feeds from:

```
safety_layer/  
shared_interfaces/examples/
```

Chapter 6 — Results feeds from:

```
allowed-case.md  
blocked-case.md  
approval logs
```

Chapter 7 — Discussion feeds from:

```
safety limitations  
blocked case interpretation  
human approval tradeoffs
```

Chapter deliverables

This step helps deliver:

Chapter 5

- safety layer implementation
- validation/approval design

Chapter 6

- safety test results

Chapter 7

- reliability discussion
- ethical/societal discussion around physical actions

▼ Goal

Make the system safe enough to be academically defendable.

▼ Work to complete

▼ 2

Work to complete

Create:

```
safety_layer/policies/action-policy-v1.md
safety_layer/parsers/output-validation-v1.md
safety_layer/approvals/hitl-v1.md
safety_layer/examples/allowed-case.md
safety_layer/examples/blocked-case.md
```

Required cases:

- one valid allowed action
- one malformed blocked action
- one risky/unknown action requiring approval or rejection

Create:

- `safety_layer/policies/action-policy-v1.md`
- `safety_layer/parsers/output-validation-v1.md`
- `safety_layer/approvals/hitl-v1.md`

Required safety cases:

- one valid action that is allowed
- one malformed action that is blocked
- one risky/unknown action that requires approval or is rejected

▼ What this proves

This proves that the workflow output cannot directly trigger middleware action without validation.

▼ Evidence produced

▼ 2

- policy file
- parser/validation notes
- allowed case
- blocked case

- approval screenshot/log
- good output example
- bad output example
- blocked case
- approval checkpoint screenshot/log

▼ (Phase 4: Testing & Writing)

▼ Step 9 — Build the measurement and evaluation harness for non-AI worker, and AI worker (4 terminals)
chapter 3, 5 6

▼ AI

▼ q

Step 9 — Build measurement and evaluation harness.

Use these files as source of truth:

- [AGENTS.md](#)
- [README.md](#)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/evaluation/evaluation-protocol.md
- docs/ongoing/integration-contract.md
- cognitive_logic/workflows/deterministic-baseline.md
- cognitive_logic/workflows/deterministic-baseline.json
- cognitive_logic/workflows/agent-minimal.md
- cognitive_logic/workflows/agent-minimal.json
- cognitive_logic/workflows/evidence/
- safety_layer/policies/action-policy-v1.md
- safety_layer/parsers/output-validation-v1.md
- safety_layer/approvals/hitl-v1.md
- safety_layer/examples/allowed-case.md
- safety_layer/examples/blocked-case.md
- safety_layer/examples/risky-approval-case.md
- middleware/api/routes.py
- middleware/tests/manual-test-notes.md

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Obid minimum integration only

- Step 9 only
- measurement/evaluation harness only
- no new workflow architecture
- no broad benchmarking
- no multiple models
- no MCP
- no n8n core modification
- no real GPIO
- no Raspberry Pi deployment in this step
- no Prometheus/Grafana
- no heavy observability stack

Goal:

Build a lightweight, repeatable measurement and evaluation harness that can produce report evidence for latency, success/failure, resource notes, baseline-vs-agent comparison, and safety outcomes.

Task:

Create or update the minimum evaluation structure and scripts.

Create:

- evaluation/datasets/
- evaluation/results/raw/
- evaluation/results/processed/
- evaluation/metrics/
- scripts/collect_metrics.py
- scripts/aggregate_results.py

Create if useful:

- evaluation/datasets/test-cases.json
- evaluation/README.md
- evaluation/metrics/metrics-schema.md

Update only if needed:

- docs/evaluation/evaluation-protocol.md
- README.md with a short pointer to the evaluation harness

Requirements for evaluation/datasets/test-cases.json:

Include minimal test cases for:

1. deterministic_high_temp:
 - mode: deterministic
 - value: 31.4
 - expected_action: fan_on
2. deterministic_low_temp:
 - mode: deterministic
 - value: 24.5
 - expected_action: fan_off

3. agent_high_temp:
 - mode: agent
 - value: 31.4
 - expected_action: fan_on
4. agent_low_temp:
 - mode: agent
 - value: 24.5
 - expected_action: fan_off
5. safety_allowed:
 - expected safety result: allowed
6. safety_blocked:
 - expected safety result: blocked
7. safety_risky_approval:
 - expected safety result: approval_required or rejected, matching the Step 8 policy

Requirements for scripts/collect_metrics.py:

- Use Python standard library only unless the repo already has a dependency system.
- Accept command-line arguments.
- Support at least:
 - -mode deterministic or agent or safety
 - -webhook-url
 - -runs
 - -dataset
 - -output
- For workflow modes:
 - send HTTP POST requests to the provided n8n webhook URL
 - measure request latency with time.perf_counter()
 - record success/failure
 - record expected_action
 - record observed response if available
 - write CSV rows to evaluation/results/raw/
- For resource metrics:
 - collect lightweight best-effort data only
 - if Docker is available, optionally call docker stats --no-stream for n8n container
 - if resource data cannot be collected, write clear "not_available" fields instead of failing
 - include RAM/CPU fields even if blank/not_available
- For thermal:
 - do not fake thermal data
 - write thermal_c as not_available on Windows/PC if not available

- document that Raspberry Pi thermal collection can later use vcgencmd in Step 10
- Do not require Prometheus/Grafana.
- Do not require Raspberry Pi.

Requirements for scripts/aggregate_results.py:

- Read raw CSV files from evaluation/results/raw/
- Produce processed summary CSVs in evaluation/results/processed/
- Produce at minimum:
 - summary_latency.csv
 - summary_resources.csv
 - safety_outcomes.csv
 - baseline_vs_agent.csv
- Summaries should include:
 - mode
 - test_case_id
 - run_count
 - success_count
 - failure_count
 - average_latency_ms
 - min_latency_ms
 - max_latency_ms
 - expected_action
 - observed_action if available
- Use Python standard library only.

Requirements for docs/evaluation/evaluation-protocol.md:

Update only if needed to reflect the actual harness:

- repeated run strategy
- deterministic baseline vs agent-enhanced comparison
- safety cases
- latency measurement method
- RAM/CPU collection method
- thermal collection status/path
- raw and processed result locations
- minimum run count
- limitations

Important:

- Do not invent measurement results.
- Do not create fake CSV result data unless clearly marked as example/template.
- If creating example result files, put them under a clearly named examples folder, not results/raw.
- Do not claim evaluation has been run unless you actually run it.

- Do not modify Step 6 or Step 7 workflows unless absolutely necessary.
- Do not implement Step 10 Raspberry Pi validation.
- Do not implement production-grade safety enforcement in this step.
- Keep this lightweight and thesis-report-focused.

Validation:

- Run Python syntax checks on the new scripts.
- If possible, run scripts with `--help`.
- If possible, run `aggregate_results.py` against an empty raw folder and make it fail gracefully or explain that raw data is needed.
- Do not run full measurement unless explicitly requested after the harness exists.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- tell me how to run a small deterministic measurement.
- tell me how to run a small agent measurement.
- tell me how to aggregate results.
- tell me what evidence should be saved for Chapter 3, Chapter 5, and Chapter 6.

▼ a

Step 9 harness is in place. I did not run a real measurement or create fake result CSVs.

Created:

- [`test-cases.json`](#)
- [`evaluation/README.md`](#)
- [`metrics-schema.md`](#)
- [`collect\_metrics.py`](#)
- [`aggregate\_results.py`](#)
- `.gitkeep` files for `evaluation/results/raw/` and `evaluation/results/processed/`

Updated:

- [`evaluation-protocol.md`](#)
- [`README.md`](#)

Validated:

- `test-cases.json` parses as JSON.
- both scripts pass `py_compile`.
- both scripts support `-help`.
- aggregator handles an empty raw folder gracefully in a temp directory.

Run a small deterministic measurement:

```
python scripts/collect_metrics.py --mode deterministic --webhook-url "<deterministic-test-webhook-url>" --runs 1
```

Run a small agent measurement:

```
python scripts/collect_metrics.py --mode agent --webhook-url "<agent-test-webhook-url>" --runs 1
```

Aggregate results:

```
python scripts/aggregate_results.py
```

Evidence to save:

- Chapter 3: commands used, run count, dataset file, and note that the harness is lightweight/standard-library.
- Chapter 5: scripts, metrics schema, raw/processed folder structure.
- Chapter 6: real raw CSVs, processed summaries, latency/resource/safety outcome tables once you actually run measurements.

▼ audit review

▼ repair-fix

Updated the stale status text in [README.md](#) so middleware, workflow exports, safety implementation, and the evaluation harness are listed as existing. The only remaining “does not exist yet” item there is `evaluation results`.

Also cleaned the affected dash rendering in [evaluation-protocol.md](#) by replacing the problematic em dashes with plain hyphens. No tests were run since this was documentation-only.

▼ q

Audit the Step 9 measurement and evaluation harness changes only.

Check:

- Step 9 stays inside new-yacoub scope.
- The harness is lightweight and thesis-focused.
- No fake measurement results were created.
- No broad benchmarking was added.
- No Prometheus/Grafana or heavy observability stack was added.
- No Raspberry Pi validation was implemented in this step.
- No MCP, n8n core modification, real GPIO, or new workflow architecture was added.

Inspect:

- `evaluation/datasets/test-cases.json`
- `evaluation/README.md`
- `evaluation/metrics/metrics-schema.md`
- `evaluation/results/raw/.gitkeep`
- `evaluation/results/processed/.gitkeep`
- `scripts/collect_metrics.py`
- `scripts/aggregate_results.py`
- `docs/evaluation/evaluation-protocol.md`
- [README.md](#)

Check test-cases.json includes:

- deterministic_high_temp → fan_on
- deterministic_low_temp → fan_off
- agent_high_temp → fan_on
- agent_low_temp → fan_off
- safety_allowed
- safety_blocked
- safety_risky_approval

Check collect_metrics.py:

- uses standard library only
- supports --mode
- supports --webhook-url
- supports --runs
- supports --dataset
- supports --output
- measures latency
- records success/failure
- records expected and observed action if available
- writes raw CSV output
- handles RAM/CPU/thermal as lightweight best-effort or not_available
- does not fake thermal/resource data

Check aggregate_results.py:

- reads raw CSV files
- writes processed summaries
- creates or supports:
 - summary_latency.csv
 - summary_resources.csv
 - safety_outcomes.csv
 - baseline_vs_agent.csv
- handles empty raw folder gracefully or clearly explains raw data is needed

Check docs/evaluation/evaluation-protocol.md:

- reflects the actual harness
- explains repeated run strategy
- explains deterministic vs agent comparison
- explains safety cases
- explains latency measurement
- explains RAM/CPU collection method
- explains thermal collection status/path

- explains raw and processed result locations
- does not claim results exist before measurements are run

Also check:

- README update is accurate and not outdated
- no zip/archive artifacts are included

Return:

1. blocking issues
2. non-blocking cleanup
3. Step 9 verdict
4. whether we can run a small real measurement after fixes, before report-scribe

▼ a

P2README status list is outdated

Dismiss

The README still says middleware implementation, workflow exports, and safety implementation do not exist yet, but the current repo now has those artifacts. This does not block running Step 9, but it should be updated so the Step 9 README section is not contradicted by older status text.

C:/Users/Jake_/OneDrive - Mittuniversitet/EDU/4. RED/1. kaniddat/new-yacoub-thesis/README.md:99-104

Blocking Issues

No blocking Step 9 issues found.

The harness is lightweight and thesis-scoped: standard-library Python, CSV files, .gitkeep placeholders only, no fake raw results, no Prometheus/Grafana, no Pi validation, no MCP, no real GPIO, no n8n core changes, and no new workflow architecture.

Non-Blocking Cleanup

The README has an outdated "does not exist yet" list. Also, docs/evaluation/evaluation-protocol.md has a few mojibake em dashes like "â€"; not functionally blocking, but worth cleaning before report polish.

evaluation/results/raw/ and evaluation/results/processed/ contain only .gitkeep, which is correct before real runs.

Step 9 Verdict

Step 9 passes audit. The dataset includes all required cases, collect_metrics.py supports the required flags and records latency/success/actions/resource best-effort, and aggregate_results.py writes the four expected summary CSVs and handles an empty raw folder clearly.

Small Measurement

Yes, we can run a small real measurement before report-scribe. Start with --runs 1 for deterministic and agent modes using the n8n test webhook URLs, then run python scripts/aggregate_results.py.

next time Git touches it
warning: in the working copy of 'safety_layer/examples/allowed-case.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'safety_layer/examples/blocked-case.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'safety_layer/examples/risky-approval-case.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'safety_layer/parsers/output-validation-v1.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'safety_layer/policies/action-policy-v1.md', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git commit -m "docs(safety): add minimum safety layer design"
[main e8a94fb] docs(safety): add minimum safety layer design
9 files changed, 591 insertions(+)
create mode 100644 docs/report-notes/step-08-minimum-safety-layer.md
create mode 100644 safety_layer/approvals/hitl-v1.md
create mode 100644 safety_layer/examples/allowed-case.md
create mode 100644 safety_layer/examples/blocked-case.md
create mode 100644 safety_layer/examples/risky-approval-case.md
create mode 100644 safety_layer/parsers/output-validation-v1.md
create mode 100644 safety_layer/policies/action-policy-v1.md
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git push
Enumerating objects: 27, done.
Counting objects: 100% (27/27), done.
Delta compression using up to 8 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (20/20), 8.12 KiB | 1.62 MiB/s, done.
Total 20 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 6 local objects.
To <https://github.com/Rumple12/new-yacoub-thesis.git>
cd76785..e8a94fb main → main
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -m py_compile scripts/collect_metrics.py scripts/aggregate_results.py
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts/collect_metrics.py --help
usage: collect_metrics.py [-h] --mode {deterministic,agent,safety} [--webhook-url WEBHOOK_URL] [--runs RUNS] [--dataset DATASET] [--output OUTPUT] [--case-id CASE_ID] [--timeout TIMEOUT] [--n8n-container N8N_CONTAINER]
Collect Step 9 evaluation metrics.
options:
-h, --help show this help message and exit
--mode {deterministic,agent,safety}
--webhook-url WEBHOOK_URL
n8n test webhook URL for workflow modes.
--runs RUNS
--dataset DATASET
--output OUTPUT
--case-id CASE_ID Optional single test case id to run.
--timeout TIMEOUT
--n8n-container N8N_CONTAINER
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python

```
scripts/aggregate_results.py --help
usage: aggregate_results.py [-h] [--raw-dir RAW_DIR] [--output-dir OUTPUT_DIR]
```

Aggregate Step 9 evaluation results.

options:

-h, --help show this help message and exit

--raw-dir RAW_DIR

--output-dir OUTPUT_DIR

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python
scripts/aggregate_results.py --help
```

```
usage: aggregate_results.py [-h] [--raw-dir RAW_DIR] [--output-dir OUTPUT_DIR]
```

Aggregate Step 9 evaluation results.

options:

-h, --help show this help message and exit

--raw-dir RAW_DIR

--output-dir OUTPUT_DIR

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -
c "import json; json.load(open('evaluation/datasets/test-cases.json', encoding='utf-8'))"; print('JSON
OK: evaluation/datasets/test-cases.json')
```

JSON OK: evaluation/datasets/test-cases.json

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git
status
```

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: [README.md](#)

modified: docs/evaluation/evaluation-protocol.md

Untracked files:

(use "git add <file>..." to include in what will be committed)

cognitive_logic/memory/

cognitive_logic/prompts/

cognitive_logic/workflows/agent-minimal.json

cognitive_logic/workflows/agent-minimal.md

cognitive_logic/workflows/evidence/step-07/

docs/report-notes/step-07-minimal-agent-workflow.md

evaluation.zip

evaluation/

scripts.zip

scripts/

no changes added to commit (use "git add" and/or "git commit -a")

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> cd
"C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Get-
Content evaluation\datasets\test-cases.json
```

```
{
  "version": 1,
  "description": "Minimal Step 9 evaluation cases for the new-yacoub thesis.",
  "test_cases": [
    {
      "id": "deterministic_high_temp",
```

```

"mode": "deterministic",
"sensor_event": {
  "sensor_id": "temp_sensor_1",
  "timestamp": "2026-04-25T20:00:00Z",
  "type": "temperature",
  "value": 31.4,
  "unit": "C"
},
"expected_action": "fan_on"
},
{
  "id": "deterministic_low_temp",
  "mode": "deterministic",
  "sensor_event": {
    "sensor_id": "temp_sensor_1",
    "timestamp": "2026-04-25T20:05:00Z",
    "type": "temperature",
    "value": 24.5,
    "unit": "C"
  },
  "expected_action": "fan_off"
},
{
  "id": "agent_high_temp",
  "mode": "agent",
  "sensor_event": {
    "sensor_id": "temp_sensor_1",
    "timestamp": "2026-04-25T20:00:00Z",
    "type": "temperature",
    "value": 31.4,
    "unit": "C"
  },
  "expected_action": "fan_on"
},
{
  "id": "agent_low_temp",
  "mode": "agent",
  "sensor_event": {
    "sensor_id": "temp_sensor_1",
    "timestamp": "2026-04-25T20:05:00Z",
    "type": "temperature",
    "value": 24.5,
    "unit": "C"
  },
  "expected_action": "fan_off"
},
{
  "id": "safety_allowed",
  "mode": "safety",
  "input_action": {
    "action_id": "fan_on",
    "target": "fan_1",
    "reason": "temperature_at_or_above_threshold",
    "requires_approval": false
  }
}

```



```

},
"expected_safety_result": "allowed"
},
{
"id": "safety_blocked",
"mode": "safety",
"input_action": {
"action_id": "fan_on",
"target": "fan_1"
},
"expected_safety_result": "blocked"
},
{
"id": "safety_risky_approval",
"mode": "safety",
"input_action": {
"action_id": "open_window",
"target": "unknown_device",
"reason": "agent_invented_action",
"requires_approval": true
},
"expected_safety_result": "rejected"
}
]
}

```

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Get-Content scripts\collect_metrics.py -TotalCount 80
 """Collect lightweight Step 9 evaluation metrics.

This script intentionally uses only the Python standard library.
 """

```

from future import annotations
import argparse
import csv
import json
import subprocess
import sys
import time
from datetime import UTC, datetime
from pathlib import Path
from typing import Any
from urllib.error import HTTPError, URLError
from urllib.request import Request, urlopen

FIELDNAMES = [
    "timestamp_utc",
    "mode",
    "test_case_id",
    "run_index",
    "webhook_url",
    "input_temperature",
    "expected_action",
    "observed_action",
    "expected_safety_result",

```

```

"observed_safety_result",
"http_status",
"success",
"latency_ms",
"cpu_percent",
"ram_usage",
"ram_percent",
"thermal_c",
"observed_response",
"error",
"notes",
]

def utc_now() → str:
    return datetime.now(UTC).replace(microsecond=0).isoformat().replace("+00:00", "Z")

def load_cases(path: Path, mode: str, case_id: str | None) → list[dict[str, Any]]:
    data = json.loads(path.read_text(encoding="utf-8"))
    cases = [case for case in data.get("test_cases", []) if case.get("mode") == mode]
    if case_id:
        cases = [case for case in cases if case.get("id") == case_id]
    if not cases:
        raise SystemExit(f"No test cases found for mode={mode!r} case_id={case_id!r}")
    return cases

def default_output_path(mode: str) → Path:
    stamp = datetime.now(UTC).strftime("%Y%m%dT%H%M%S")
    return Path("evaluation") / "results" / "raw" / f"run_{stamp}_{mode}.csv"

def resolve_output_path(value: str | None, mode: str) → Path:
    if not value:
        return default_output_path(mode)
    path = Path(value)
    if path.exists() and path.is_dir():
        return path / default_output_path(mode).name
    if str(value).endswith(("/", "\\")):
        return path / default_output_path(mode).name
    return path

def post_json(url: str, payload: dict[str, Any], timeout: float) → tuple[int | str, str, str]:
    body = json.dumps(payload).encode("utf-8")
    request = Request(
        url,
        data=body,
        headers={"Content-Type": "application/json"},
    )
    PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Get-Content scripts\aggregate_results.py -TotalCount 80
    """Aggregate Step 9 raw evaluation CSV files.

    This script intentionally uses only the Python standard library.
    """

from future import annotations
import argparse
import csv
import statistics
import sys
from collections import defaultdict

```

```

from pathlib import Path
from typing import Iterable

SUMMARY_FIELDS = [
    "mode",
    "test_case_id",
    "run_count",
    "success_count",
    "failure_count",
    "average_latency_ms",
    "min_latency_ms",
    "max_latency_ms",
    "expected_action",
    "observed_action",
]

RESOURCE_FIELDS = [
    "mode",
    "test_case_id",
    "run_count",
    "success_count",
    "failure_count",
    "average_latency_ms",
    "min_latency_ms",
    "max_latency_ms",
    "expected_action",
    "observed_action",
    "cpu_available_count",
    "ram_available_count",
    "thermal_available_count",
    "cpu_percent_values",
    "ram_usage_values",
    "ram_percent_values",
    "thermal_c_values",
]

SAFETY_FIELDS = [
    "mode",
    "test_case_id",
    "run_count",
    "success_count",
    "failure_count",
    "average_latency_ms",
    "min_latency_ms",
    "max_latency_ms",
    "expected_action",
    "observed_action",
    "expected_safety_result",
    "observed_safety_result",
]

def read_rows(raw_dir: Path) → list[dict[str, str]]:
    rows: list[dict[str, str]] = []
    for path in sorted(raw_dir.glob("*.csv")):
        with path.open("r", newline="", encoding="utf-8") as f:

```

```

rows.extend(csv.DictReader(f))
return rows

def write_csv(path: Path, fieldnames: list[str], rows: Iterable[dict[str, str]]) → None:
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open("w", newline="", encoding="utf-8") as f:
        writer = csv.DictWriter(f, fieldnames=fieldnames)
        writer.writeheader()

```

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -c "import json; json.load(open('evaluation/datasets/test-cases.json', encoding='utf-8')); print('JSON OK: evaluation/datasets/test-cases.json')"

JSON OK: evaluation/datasets/test-cases.json

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -m py_compile scripts/collect_metrics.py scripts/aggregate_results.py

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts/collect_metrics.py --help

usage: collect_metrics.py [-h] --mode {deterministic,agent,safety} [--webhook-url WEBHOOK_URL] [--runs RUNS] [--dataset DATASET] [--output OUTPUT] [--case-id CASE_ID] [--timeout TIMEOUT] [--n8n-container N8N_CONTAINER]

Collect Step 9 evaluation metrics.

options:

- h, --help show this help message and exit
- mode {deterministic,agent,safety}
- webhook-url WEBHOOK_URL

n8n test webhook URL for workflow modes.

- runs RUNS
- dataset DATASET
- output OUTPUT
- case-id CASE_ID Optional single test case id to run.
- timeout TIMEOUT
- n8n-container N8N_CONTAINER

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts/aggregate_results.py --help

usage: aggregate_results.py [-h] [--raw-dir RAW_DIR] [--output-dir OUTPUT_DIR]

Aggregate Step 9 evaluation results.

options:

- h, --help show this help message and exit
- raw-dir RAW_DIR
- output-dir OUTPUT_DIR

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> cd "C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python -m middleware.api.app

new-yacoub middleware running at <http://127.0.0.1:8000>

Press Ctrl+C to stop.

127.0.0.1 - "GET /status HTTP/1.1" 200 -

- cd "C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
- python -c "import json; json.load(open('evaluation/datasets/test-cases.json', encoding='utf-8')); print('JSON OK: evaluation/datasets/test-cases.json')"
- python -m py_compile scripts/collect_metrics.py scripts/aggregate_results.py

- python scripts/collect_metrics.py --help

▼ terminal 2 - start middleware

Part C — start middleware

Open **Terminal 2**.

From repo root:

```
cd "C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
python-mmiddleware.api.app
```

Keep this terminal open.

Expected:

```
new-yacoub middleware running at http://127.0.0.1:8000
Press Ctrl+C to stop.
```

Quick check from another terminal:

```
Invoke-WebRequest -Method Get -Uri "http://127.0.0.1:8000/status"
```

▼ terminal 3 - start n8n

- cd "C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"cd
- "C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"
- <http://localhost:5678>

▼ terminal 4 - invoke n8n

▼ high temp

```
http://localhost:5678/webhook-test/deterministic-baseline

$highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json

Invoke-WebRequest -Method Post -Uri $deterministicUrl -ContentType "application/json" -Body $highBody
```

▼ low temp

▼

▼ terminal 1

Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\Jake_> cd "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\inf
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> docker compose up -d
```

```
[+] up 1/1
```

```
✓ Container new-yacoub-n8n Started 0.8s
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-
```

```
thesis\infrastructure\docker> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> ..
```

```
.. : The term '..' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
```

```
At line:1 char:1
```

- ..
- ~~
 - CategoryInfo : ObjectNotFound: (..:String) [], CommandNotFoundException
 - FullyQualifiedErrorId : CommandNotFoundException

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker> cd..
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure> cd..
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts\collect_metrics.py --mode deterministic --webhook-url
```

```
"http://localhost:5678/webhook/deterministic-baseline" --runs 1
```

```
Wrote 2 rows to evaluation\results\raw\run_20260429T165702Z_deterministic.csv
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts\collect_metrics.py --mode deterministic --webhook-url
```

```
"http://localhost:5678/webhook/deterministic-baseline" --runs 1
```

```
Wrote 2 rows to evaluation\results\raw\run_20260429T165733Z_deterministic.csv
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts\collect_metrics.py --mode agent --webhook-url "http://localhost:5678/webhook/agent-minimal" --runs 1
```

```
Wrote 2 rows to evaluation\results\raw\run_20260429T165859Z_agent.csv
```

```
PS C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> python scripts\collect_metrics.py --mode agent --webhook-url "http://localhost:5678/webhook/agent-minimal" --runs 1
```

```
Traceback (most recent call last):
```

```
File "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\scripts\collect_metrics.py", line 306, in <module>
```

```
sys.exit(main())
```

```
~~~~^
```

```
File "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\scripts\collect_metrics.py", line 289, in main
```

```
workflow_row(
```

```

~~~~~^
mode=args.mode,
~~~~~
...<4 lines>...
n8n_container=args.n8n_container,
~~~~~
)
^
File "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\scripts\collect_metrics.py", line 188, in workflow_row
status, response_text, error = post_json(webhook_url, payload, timeout)
~~~~~
File "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\scripts\collect_metrics.py", line 84, in post_json
with urlopen(request, timeout=timeout) as response:
~~~~~
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 189, in urlopen
return opener.open(url, data, timeout)
~~~~~
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 489, in open
response = self._open(req, data)
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 506, in _open
result = self._call_chain(self.handle_open, protocol, protocol +
'_open', req)
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 466, in call_chain
result = func(*args)
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 1348, in http_open
return self.do_open(http.client.HTTPConnection, req)
~~~~~
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\urllib\requ
line 1323, in do_open
r = h.getresponse()
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\http\clien
line 1450, in getresponse
response.begin()
~~~~~^^
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.3568.0_x64__qbz5n2kfra8p0\Lib\http\clien
line 336, in begin
version, status, reason = self.read_status()
~~~~~^^
File "C:\Program

```


Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: [README.md](#)

modified: docs/evaluation/evaluation-protocol.md

Untracked files:

(use "git add <file>..." to include in what will be committed)

cognitive_logic/memory/

cognitive_logic/prompts/

cognitive_logic/workflows/agent-minimal.json

cognitive_logic/workflows/agent-minimal.md

cognitive_logic/workflows/evidence/step-07/

docs/report-notes/step-07-minimal-agent-workflow.md

docs/report-notes/step-09-measurement-evaluation-harness.md

evaluation/

scripts/

no changes added to commit (use "git add" and/or "git commit -a")

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git status

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: [README.md](#)

modified: docs/evaluation/evaluation-protocol.md

Untracked files:

(use "git add <file>..." to include in what will be committed)

cognitive_logic/memory/

cognitive_logic/prompts/

cognitive_logic/workflows/agent-minimal.json

cognitive_logic/workflows/agent-minimal.md

cognitive_logic/workflows/evidence/step-07/

docs/report-notes/step-07-minimal-agent-workflow.md

docs/report-notes/step-09-measurement-evaluation-harness.md

evaluation/

scripts/

no changes added to commit (use "git add" and/or "git commit -a")

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git status

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: [README.md](#)

modified: docs/evaluation/evaluation-protocol.md

Untracked files:

(use "git add <file>..." to include in what will be committed)

cognitive_logic/memory/

cognitive_logic/prompts/

cognitive_logic/workflows/agent-minimal.json

cognitive_logic/workflows/agent-minimal.md

cognitive_logic/workflows/evidence/step-07/
docs/report-notes/step-07-minimal-agent-workflow.md
docs/report-notes/step-09-measurement-evaluation-harness.md
evaluation/
scripts/

no changes added to commit (use "git add" and/or "git commit -a")

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> cd  
"C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Remove-Item  
docs.zip -Force -ErrorAction SilentlyContinue
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> -Remove-Item  
evaluation.zip -Force -ErrorAction SilentlyContinue
```

-Remove-Item : The term '-Remove-Item' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

At line:1 char:1

- Remove-Item evaluation.zip -Force -ErrorAction SilentlyContinue
- + CategoryInfo : ObjectNotFound: (-Remove-Item:String) [], CommandNotFoundException + FullyQualifiedErrorId : CommandNotFoundException

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Remove-Item  
evaluation.zip -Force -ErrorAction SilentlyContinue
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Remove-Item  
scripts.zip -Force -ErrorAction SilentlyContinue
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Get-ChildItem  
-Recurse -Directory -Filter pycache | Remove-Item -Recurse -Force
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> Get-ChildItem  
-Recurse -Directory -Filter pycache | Remove-Item -Recurse -Force
```

```
PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add  
README.md docs/evaluation/evaluation-protocol.md evaluation/README.md evaluation/datasets/test-  
cases.json evaluation/metrics/metrics-schema.md evaluation/results/raw evaluation/results/processed  
evaluation/results/step-09-small-measurement-notes.md scripts/collect_metrics.py
```

```
scripts/aggregate_results.py docs/report-notes/step-09-measurement-evaluation-harness.md
```

warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'docs/evaluation/evaluation-protocol.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'docs/report-notes/step-09-measurement-evaluation-harness.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/README.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/datasets/test-cases.json', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/metrics/metrics-schema.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/results/processed/.gitkeep', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/results/raw/.gitkeep', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'evaluation/results/step-09-small-measurement-notes.md', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'scripts/aggregate_results.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'scripts/collect_metrics.py', LF will be replaced by CRLF the next time Git touches it

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git status

On branch main

Your branch is up to date with 'origin/main'.

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: [README.md](#)

modified: docs/evaluation/evaluation-protocol.md

new file: docs/report-notes/step-09-measurement-evaluation-harness.md

new file: evaluation/README.md

new file: evaluation/datasets/test-cases.json

new file: evaluation/metrics/metrics-schema.md

new file: evaluation/results/processed/.gitkeep

new file: evaluation/results/processed/baseline_vs_agent.csv

new file: evaluation/results/processed/safety_outcomes.csv

new file: evaluation/results/processed/summary_latency.csv

new file: evaluation/results/processed/summary_resources.csv

new file: evaluation/results/raw/.gitkeep

new file: evaluation/results/raw/dump/run_20260429T165702Z_deterministic.csv

new file: evaluation/results/raw/run_20260429T165733Z_deterministic.csv

new file: evaluation/results/raw/run_20260429T165859Z_agent.csv

new file: evaluation/results/step-09-small-measurement-notes.md

new file: scripts/aggregate_results.py

new file: scripts/collect_metrics.py

Untracked files:

(use "git add <file>..." to include in what will be committed)

cognitive_logic/memory/

cognitive_logic/prompts/

cognitive_logic/workflows/agent-minimal.json

cognitive_logic/workflows/agent-minimal.md

cognitive_logic/workflows/evidence/step-07/

docs/report-notes/step-07-minimal-agent-workflow.md

PS C:\Users\Jake_\OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git commit -m

" step 9 feat(evaluation): add measurement harness and initial results"

[main d0e3f1b] step 9 feat(evaluation): add measurement harness and initial results

18 files changed, 963 insertions(+), 10 deletions(-)

create mode 100644 docs/report-notes/step-09-measurement-evaluation-harness.md

create mode 100644 evaluation/README.md

create mode 100644 evaluation/datasets/test-cases.json

create mode 100644 evaluation/metrics/metrics-schema.md

create mode 100644 evaluation/results/processed/.gitkeep

create mode 100644 evaluation/results/processed/baseline_vs_agent.csv

create mode 100644 evaluation/results/processed/safety_outcomes.csv

create mode 100644 evaluation/results/processed/summary_latency.csv

create mode 100644 evaluation/results/processed/summary_resources.csv

create mode 100644 evaluation/results/raw/.gitkeep

create mode 100644 evaluation/results/raw/dump/run_20260429T165702Z_deterministic.csv

create mode 100644 evaluation/results/raw/run_20260429T165733Z_deterministic.csv

create mode 100644 evaluation/results/raw/run_20260429T165859Z_agent.csv

create mode 100644 evaluation/results/step-09-small-measurement-notes.md

create mode 100644 scripts/aggregate_results.py

create mode 100644 scripts/collect_metrics.py

```

PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git push
Enumerating objects: 35, done.
Counting objects: 100% (35/35), done.
Delta compression using up to 8 threads
Compressing objects: 100% (26/26), done.
Writing objects: 100% (29/29), 13.75 KiB | 1.38 MiB/s, done.
Total 29 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 5 local objects.
To https://github.com/Rumple12/new-yacoub-thesis.git
e8a94fb..d0e3f1b main -> main
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
(use "git add <file>..." to include in what will be committed)
cognitive_logic/memory/
cognitive_logic/prompts/
cognitive_logic/workflows/agent-minimal.json
cognitive_logic/workflows/agent-minimal.md
cognitive_logic/workflows/evidence/step-07/
docs/report-notes/step-07-minimal-agent-workflow.md

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
cognitive_logic/memory/
warning: in the working copy of 'cognitive_logic/memory/memory-choice-v1.md', LF will be replaced by
CRLF the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
cognitive_logic/prompts/
warning: in the working copy of 'cognitive_logic/prompts/system-prompt-v1.md', LF will be replaced by
CRLF the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
cognitive_logic/workflows/agent-minimal.json
warning: in the working copy of 'cognitive_logic/workflows/agent-minimal.json', LF will be replaced by
CRLF the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
cognitive_logic/workflows/agent-minimal.md
warning: in the working copy of 'cognitive_logic/workflows/agent-minimal.md', LF will be replaced by CRLF
the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
cognitive_logic/workflows/evidence/step-07/
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git add
docs/report-notes/step-07-minimal-agent-workflow.md
warning: in the working copy of 'docs/report-notes/step-07-minimal-agent-workflow.md', LF will be
replaced by CRLF the next time Git touches it
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git commit -m
"step 7 feat(agent): implement Step 7 minimal agent workflow"
[main 42e27e7] step 7 feat(agent): implement Step 7 minimal agent workflow
14 files changed, 997 insertions(+)
create mode 100644 cognitive_logic/memory/memory-choice-v1.md
create mode 100644 cognitive_logic/prompts/system-prompt-v1.md
create mode 100644 cognitive_logic/workflows/agent-minimal.json
create mode 100644 cognitive_logic/workflows/agent-minimal.md
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-high-temp-agent-output.png

```

```

create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-high-temp-canvas.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-high-temp-fan-on-output.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-high-temp-parsed-action.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-low-temp-agent-output.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-low-temp-canvas.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-low-temp-fan-off-output.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-low-temp-parsed-action.png
create mode 100644 cognitive_logic/workflows/evidence/step-07/step-07-runtime-verification.md
create mode 100644 docs/report-notes/step-07-minimal-agent-workflow.md
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git push
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (24/24), 662.26 KiB | 14.09 MiB/s, done.
Total 24 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/Rumple12/new-yacoub-thesis.git
d0e3f1b..42e27e7 main -> main
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis>

```

▼ terminal 2

▼ agent

```

Windows PowerShell
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PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json"
-Body $highBody

status : ok
action : fan_on
fan     : on
simulated : True
reason  : manual_api_call

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
PS C:\Users\Jake_> http://localhost:5678/webhook-test/agent-minimal
http://localhost:5678/webhook-test/agent-minimal : The term 'http://localhost:5678/webhook-test/agent-minimal' is not recognized as
the name of a cmdlet, function, script file, or operable program.

```

Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

At line:1 char:1

- <http://localhost:5678/webhook-test/agent-minimal>

```
+ CategoryInfo          : ObjectNotFound: (http://localhost:5678/webhook-test/agent-minimal:String) [], CommandNotFoundException +
FullyQualifiedErrorId : CommandNotFoundException
```

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $highBody
```

```
Invoke-RestMethod : {"message":"Error in workflow"}
```

At line:1 char:1

- Invoke-RestMethod -Method Post -Uri \$agentUrl -ContentType "application/json" -Body \$highBody
-

```
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
PS C:\Users\Jake_> $agentUrl = "<http://localhost:5678/webhook-test/agent-minimal>"
PS C:\Users\Jake_> $highBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:00:00Z"
>>   type = "temperature"
>>   value = 31.4
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $highBody
```

```
status      : ok
action      : fan_on
fan          : on
simulated   : True
reason      : manual_api_call
```

```
PS C:\Users\Jake_> $lowBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:05:00Z"
>>   type = "temperature"
>>   value = 24.5
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType "application/json" -Body $lowBody
Invoke-RestMethod : Cannot validate argument on parameter 'Uri'. The argument is null or empty. Provide an argument that is not null or empty, and then try the command again.
At line:1 char:37
+ Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentTy
pe "a ...
+
+ CategoryInfo          : InvalidData: (:) [Invoke-RestMethod], ParameterBindingValidationException
```

```

+ FullyQualifiedErrorId : ParameterArgumentValidationError,Mi
crosoft.PowerShell.Commands.InvokeRestMethodCommand

PS C:\Users\Jake_> $agentUrl = "<http://localhost:5678/webhook/agent-minimal>"
PS C:\Users\Jake_>
PS C:\Users\Jake_> $lowBody = @{
>> sensor_id = "temp_sensor_1"
>> timestamp = "2026-04-25T20:05:00Z"
>> type = "temperature"
>> value = 24.5
>> unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> $lowBody = @{
>> sensor_id = "temp_sensor_1"
>> timestamp = "2026-04-25T20:05:00Z"
>> type = "temperature"
>> value = 24.5
>> unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $lowBody

status      : ok
action      : fan_off
fan         : off
simulated   : True
reason      : manual_api_call

PS C:\Users\Jake_> $agentUrl = "<http://localhost:5678/webhook/agent-minimal>"
PS C:\Users\Jake_> $lowBody = @{
>> sensor_id = "temp_sensor_1"
>> timestamp = "2026-04-25T20:05:00Z"
>> type = "temperature"
>> value = 24.5
>> unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $lowBody
Invoke-RestMethod : {"message":"Error in workflow"}
At line:1 char:1
+ Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "appl
icati ...
+ ~~~~~

+ CategoryInfo          : InvalidOperation: (System.Net.HttpW

```

```

ebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Micro
soft.PowerShell.Commands.InvokeRestMethodCommand
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json"
-Body $highBody

status      : ok
action      : fan_on
fan         : on

```

```

simulated : True
reason   : manual_api_call

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json"
-Body $highBody

headers   : @{user-agent=Mozilla/5.0 (Windows NT; Windows NT 1
0.0; sv-SE) WindowsPowerShell/5.1.26100.8115; cont
ent-type=application/json; host=localhost:5678; co
ntent-length=151}
params    :
query     :
body      : @{sensor_id=temp_sensor_1; timestamp=2026-04-25T20
:00:00Z; value=31,4; type=temperature; unit=C}
webhookUrl : http://localhost:5678/webhook-test/agent-minimal
executionMode : test

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json"
-Body $highBody

status   : ok
action   : fan_on
fan      : on
simulated : True
reason   : manual_api_call

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json"
-Body $highBody
Invoke-RestMethod : {"message":"Error in workflow"}
At line:1 char:1

```


- Invoke-RestMethod -Method Post -Uri \$agentUri -ContentType "applicati ...
- + CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebExc eption + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $highBody
```

```
Invoke-RestMethod : {"code":404,"message":"The requested webhook \"agent-minimal\" is not
registered.", "hint":"Click th
e 'Execute workflow' button on the canvas, then try again. (In test mode, the webhook only works for
one call after you
click this button)"}
At line:1 char:1
```

- Invoke-RestMethod -Method Post -Uri \$agentUri -ContentType "applicati ...
- + CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebExc eption + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $lowBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json"
-Body $lowBody
```

```
status : ok
action : fan_off
fan : off
simulated : True
reason : manual_api_call
```

▼ determinisk

Windows PowerShell

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```
PS C:\WINDOWS\system32> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"
```

```
PS C:\WINDOWS\system32> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType "application/json" -Body $highBody
```

```
Invoke-RestMethod : {"code":404,"message":"The requested webhook \
"POST deterministic-baseline" is not registered.","hint":"The wor
kflow must be active for a production URL to run successfully. You
can activate the workflow using the toggle in the top-right of th
e editor. Note that unlike test URL calls, production URL calls ar
en't shown on the canvas (only in the executions list)"}
At line:1 char:1
```

- Invoke-RestMethod -Method Post -Uri \$deterministicUrl -ContentTy
pe "a ...
-

```
+ CategoryInfo          : InvalidOperation: (System.Net.HttpW
ebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Micro
soft.PowerShell.Commands.InvokeRestMethodCommand
PS C:\WINDOWS\system32> $deterministicUrl = "<http://localhost:5678/webhook/deterministic-baseline>"
PS C:\WINDOWS\system32> $highBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:00:00Z"
>>   type = "temperature"
>>   value = 31.4
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType "application/json" -B
ody $highBody
Invoke-RestMethod : {"code":404,"message":"The requested webhook \
"POST deterministic-baseline" is not registered.","hint":"The wor
kflow must be active for a production URL to run successfully. You
can activate the workflow using the toggle in the top-right of th
e editor. Note that unlike test URL calls, production URL calls ar
en't shown on the canvas (only in the executions list)"}
At line:1 char:1
+ Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentTy
pe "a ...
+ ~~~~~
```

```
+ CategoryInfo          : InvalidOperation: (System.Net.HttpW
```

```
ebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Micro
soft.PowerShell.Commands.InvokeRestMethodCommand
```

```

PS C:\WINDOWS\system32> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"
PS C:\WINDOWS\system32> $highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType
"application/json" -Body $highBody

```

```

status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call

```

```

PS C:\WINDOWS\system32> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType
"application/json" -Body $highBody

```

```

status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call

```

```

PS C:\WINDOWS\system32> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"
PS C:\WINDOWS\system32> $highBody = @{

```

```

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json

```

```

PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> $highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C^C"

```

```

PS C:\WINDOWS\system32> $deterministicUrl = "http://localhost:5678/webhook/deterministic-baseline"

```

```

PS C:\WINDOWS\system32> $highBody = @{
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json

```

```
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUri -ContentType  
"application/json" -Body $highBody
```

```
status : ok  
action : fan_on  
fan : on  
simulated : True  
reason : manual_api_call
```

```
PS C:\WINDOWS\system32> $deterministicUri = "http://localhost:5678/webhook-test/deterministic-baseline"
```

```
PS C:\WINDOWS\system32> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"  
    timestamp = "2026-04-25T20:00:00Z"  
    type = "temperature"  
    value = 31.4  
    unit = "C"  
} | ConvertTo-Json
```

```
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUri -ContentType  
"application/json" -Body $highBody
```

```
headers : @{user-agent=Mozilla/5.0 (Windows NT; Windows NT 1  
0.0; sv-SE) WindowsPowerShell/5.1.26100.8115; content-type=application/json; host=localhost:5678; content-length=151}  
params :  
query :  
body : @{sensor_id=temp_sensor_1; timestamp=2026-04-25T20:00:00Z; value=31.4; type=temperature; unit=C}  
webhookUrl : http://localhost:5678/webhook-test/deterministic-baseline  
executionMode : test
```

```
PS C:\WINDOWS\system32> $deterministicUri = "http://localhost:5678/webhook-test/deterministic-baseline"
```

```
PS C:\WINDOWS\system32> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"  
    timestamp = "2026-04-25T20:00:00Z"  
    type = "temperature"  
    value = 31.4  
    unit = "C"  
} | ConvertTo-Json
```

```
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUri -ContentType  
"application/json" -Body $highBody
```

```
status : ok  
action : fan_on  
fan : on  
simulated : True  
reason : manual_api_call
```

```
PS C:\WINDOWS\system32> $lowBody = @{
```

```
    sensor_id = "temp_sensor_1"  
    timestamp = "2026-04-25T20:05:00Z"  
    type = "temperature"
```

```

    value = 24.5
    unit = "C"
  } | ConvertTo-Json
PS C:\WINDOWS\system32> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType
"application/json" -Body $lowBody

```

```

status : ok
action : fan_off
fan : off
simulated : True
reason : manual_api_call

```

```

PS C:\WINDOWS\system32> python scripts/collect_metrics.py --mode deterministic --webhook-url
"http://localhost:5678/webhook/deterministic-baseline" --runs 1
C:\Users\Jake\AppData\Local\Microsoft\WindowsApps\python.exe: can't open file
'C:\WINDOWS\system32\scripts\collect_metrics.py': [Errno 2] No such file or directory
PS C:\WINDOWS\system32> python scripts/collect_metrics.py --mode deterministic --webhook-url
"http://localhost:5678/webhook/deterministic-baseline" --runs 1
C:\Users\Jake\AppData\Local\Microsoft\WindowsApps\python.exe: can't open file
'C:\WINDOWS\system32\scripts\collect_metrics.py': [Errno 2] No such file or directory
PS C:\WINDOWS\system32> python scripts/collect_metrics.py --mode deterministic --webhook-url
"http://localhost:5678/webhook/deterministic-baseline" --runs 1
C:\Users\Jake\AppData\Local\Microsoft\WindowsApps\python.exe: can't open file
'C:\WINDOWS\system32\scripts\collect_metrics.py': [Errno 2] No such file or directory
PS C:\WINDOWS\system32>

```

▼ terminal 3



Windows PowerShell

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```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
  } | ConvertTo-Json

```

```

PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
Body $highBody

```

```

status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call

```

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> http://localhost:5678/webhook-test/agent-minimal
```

```

http://localhost:5678/webhook-test/agent-minimal : The term 'http:
//localhost:5678/webhook-test/agent-minimal' is not recognized as

```

the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify
that the path is correct and try again.

At line:1 char:1

- <http://localhost:5678/webhook-test/agent-minimal>
- + CategoryInfo : ObjectNotFound: (http://localhost:5678/webhook-test/agent-minimal:String) [], CommandNotFoundException + FullyQualifiedErrorId : CommandNotFoundException

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $highBody
```

```
Invoke-RestMethod : {"message":"Error in workflow"}
```

At line:1 char:1

- Invoke-RestMethod -Method Post -Uri \$agentUrl -ContentType "application/json" -Body \$highBody
-

```
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
PS C:\Users\Jake_> $highBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:00:00Z"
>>   type = "temperature"
>>   value = 31.4
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $highBody

status      : ok
action      : fan_on
fan         : on
simulated   : True
reason      : manual_api_call

PS C:\Users\Jake_> $lowBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:05:00Z"
>>   type = "temperature"
>>   value = 24.5
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentType "application/json" -Body $lowBody
Invoke-RestMethod : Cannot validate argument on parameter 'Uri'. The argument is null or empty. Provide an argument that is not null or empty, and then try the command again.
At line:1 char:37
+ Invoke-RestMethod -Method Post -Uri $deterministicUrl -ContentTy
pe "a ...
+
+ CategoryInfo          : InvalidData: (:) [Invoke-RestMethod]
```

```

    }, ParameterBindingValidationException
    + FullyQualifiedErrorId : ParameterArgumentValidationError,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

PS C:\Users\Jake_> $agentUrl = "<http://localhost:5678/webhook/agent-minimal>"
PS C:\Users\Jake_>
PS C:\Users\Jake_> $lowBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:05:00Z"
>>   type = "temperature"
>>   value = 24.5
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> $lowBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:05:00Z"
>>   type = "temperature"
>>   value = 24.5
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $lowBody

status      : ok
action      : fan_off
fan          : off
simulated   : True
reason      : manual_api_call

PS C:\Users\Jake_> $agentUrl = "<http://localhost:5678/webhook/agent-minimal>"
PS C:\Users\Jake_> $lowBody = @{
>>   sensor_id = "temp_sensor_1"
>>   timestamp = "2026-04-25T20:05:00Z"
>>   type = "temperature"
>>   value = 24.5
>>   unit = "C"
>> } | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -Body $lowBody
Invoke-RestMethod : {"message":"Error in workflow"}
At line:1 char:1
+ Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "appl
icati ...
+ ~~~~~

```

```

+ CategoryInfo          : InvalidOperation: (System.Net.HttpW

```

```

ebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

```

```

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json

```

```

PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
Body $highBody

```

```

status      : ok
action      : fan_on
fan         : on

```



```

simulated : True
reason : manual_api_call

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
    Body $highBody

headers : @{"user-agent":Mozilla/5.0 (Windows NT; Windows NT 1
0.0; sv-SE) WindowsPowerShell/5.1.26100.8115; cont
ent-type=application/json; host=localhost:5678; co
ntent-length=151}
params :
query :
body : @{"sensor_id=temp_sensor_1; timestamp=2026-04-25T20
:00:00Z; value=31,4; type=temperature; unit=C}
webhookUrl : http://localhost:5678/webhook-test/agent-minimal
executionMode : test

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
    Body $highBody

status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call

PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
PS C:\Users\Jake_> $highBody = @{

    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
    Body $highBody
Invoke-RestMethod : {"message":"Error in workflow"}
At line:1 char:1

```

- Invoke-RestMethod -Method Post -Uri \$agentUrl -ContentType "applicati ...
- + CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebExc eption + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUrl = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUrl -ContentType "application/json" -
Body $highBody
```

```
Invoke-RestMethod : {"code":404,"message":"The requested webhook \"agent-minimal\" is not
registered.", "hint":"Click th
e 'Execute workflow' button on the canvas, then try again. (In test mode, the webhook only works for one
call after you
click this button)"}
At line:1 char:1
```

- Invoke-RestMethod -Method Post -Uri \$agentUri -ContentType "applicati ...
- + CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebExc eption + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook-test/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json" -
Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $highBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:00:00Z"
    type = "temperature"
    value = 31.4
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json" -
Body $highBody
```

```
status : ok
action : fan_on
fan : on
simulated : True
reason : manual_api_call
```

```
PS C:\Users\Jake_> $agentUri = "http://localhost:5678/webhook/agent-minimal"
```

```
PS C:\Users\Jake_> $lowBody = @{
```

```
    sensor_id = "temp_sensor_1"
    timestamp = "2026-04-25T20:05:00Z"
    type = "temperature"
    value = 24.5
    unit = "C"
} | ConvertTo-Json
```

```
PS C:\Users\Jake_> Invoke-RestMethod -Method Post -Uri $agentUri -ContentType "application/json" -
Body $lowBody
```

```
status : ok
action : fan_off
fan : off
simulated : True
reason : manual_api_call
```

▼ terminal 4

Windows PowerShell

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PS C:\WINDOWS\system32> Invoke-RestMethod -Method Get -Uri "<http://127.0.0.1:8000/status>"

status message state

ok middleware running @{fan=off; last_sensor_event=; hardware=simulated}

PS C:\WINDOWS\system32> docker compose up -d

no configuration file provided: not found

What's next:

Debug this Compose error with Gordon → docker ai "help me fix this compose error"

python scripts/aggregate_results.py --help

▼ report scribe

Step 09 - Measurement Evaluation Harness

What was built

- A lightweight Step 9 measurement and evaluation harness was added.
- The harness uses standard-library Python scripts and CSV files rather than a heavy observability stack.
- scripts/collect_metrics.py collects workflow run data for deterministic, agent, and safety modes.
- scripts/aggregate_results.py aggregates raw CSV rows into processed summary CSV files.
- evaluation/datasets/test-cases.json defines deterministic, agent, and safety test cases.
- evaluation/metrics/metrics-schema.md documents the raw and processed CSV fields.
- Real raw CSV files and processed summaries now exist under evaluation/results/.

Why it matters for the thesis

This step turns the implementation from a technical demo into something that can produce report evidence. It supports the thesis research questions about latency, RAM, CPU, thermal/resource behavior, safety outcomes, and deterministic-vs-agent comparison.

It also follows decision D-004 by using lightweight local logging and Docker/runtime statistics instead of Prometheus, Grafana, or another broad monitoring stack.

Evidence produced

- docs/evaluation/evaluation-protocol.md
- evaluation/README.md
- evaluation/datasets/test-cases.json
- evaluation/metrics/metrics-schema.md
- evaluation/results/step-09-small-measurement-notes.md
- scripts/collect_metrics.py
- scripts/aggregate_results.py

Raw CSV evidence:

- evaluation/results/raw/run_20260429T165733Z_deterministic.csv
- evaluation/results/raw/run_20260429T165859Z_agent.csv
- evaluation/results/raw/dump/run_20260429T165702Z_deterministic.csv

Processed CSV evidence:

- evaluation/results/processed/summary_latency.csv
- evaluation/results/processed/summary_resources.csv
- evaluation/results/processed/safety_outcomes.csv
- evaluation/results/processed/baseline_vs_agent.csv

The small measurement notes record these commands:

```
python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook/deterministic-baseline" --runs 1
python scripts/collect_metrics.py --mode agent --webhook-url "http://localhost:5678/webhook/agent-minimal" --runs 1
python scripts/aggregate_results.py
```

The successful small measurement run included both workflow modes:

- deterministic mode: high-temperature fan_on and low-temperature fan_off
- agent mode: high-temperature fan_on and low-temperature fan_off

An earlier deterministic run returned HTTP 404 because the production webhook was not registered. That file is stored under evaluation/results/raw/dump/ and should be treated as setup/misconfigured-webhook evidence, not final result evidence.

Report chapters it feeds

- Chapter 3 - Methodology: evaluation method, repeatability, run commands, and data collection format.
- Chapter 5 - Implementation: measurement harness, dataset, metrics schema, collection script, and aggregation script.
- Chapter 6 - Results: raw and processed CSV evidence for later latency, resource, and baseline-vs-agent tables.
- Chapter 7 - Discussion: measurement limitations, setup errors, Windows/PC thermal limitation, and best-effort resource collection.

Limitations or assumptions

- Final evaluation is not complete.
- The current successful run count is small: one run per workflow mode for initial harness verification.
- Larger repeated runs can be done later for final report tables, following the protocol target of at least 3 runs per test case per mode if time is limited, or 5 if time allows.
- The processed summaries are current small-run artifacts and should be reviewed before final report use because the earlier deterministic 404 setup attempt is reflected in the generated deterministic counts.
- Safety cases are defined in the dataset, but the current safety_outcomes.csv does not demonstrate runtime safety enforcement.
- Thermal data is currently not_available on the Windows/PC setup.
- CPU and RAM values are best-effort Docker/runtime values collected through docker stats --no-stream when available.
- No Raspberry Pi validation was started in this step.
- No charts or diagrams were created in this step.

Screenshots/logs still needed

- Larger repeated raw CSV runs for deterministic and agent modes.
- Runtime safety-mode evidence after enforcement exists, if safety results are to be claimed.
- Final cleaned processed summaries for report tables.
- Terminal logs or screenshots showing the collection and aggregation commands running.
- Thermal collection evidence if Raspberry Pi validation is later performed.
- Charts or diagrams later, after final result CSVs are stable.

▼ report urndown

▼ 🎯 The Goal

Produce the concrete evidence that will make the thesis report real.

- **What this proves:** This creates the actual data (latency, resource usage, success rates) required to compare the deterministic baseline against the agent-enhanced workflow, forming the core of your Results chapter.

▼ 🤖 AI Collaboration Workflow (How it was built)

This section documents how the evaluation harness was built and the messy, real-world troubleshooting required to gather the first data points.

1. **AI Task (Harness Generation):** The AI `step-runner` was instructed to build a lightweight, repeatable evaluation harness. True to the narrowed thesis scope (Decision D-004), it avoided heavy observability stacks like Prometheus or Grafana. Instead, it built standard-library Python scripts (`collect_metrics.py` and `aggregate_results.py`) and a strict `test-cases.json` dataset.
2. **AI Task (Scope & Reality Check):** The `audit-review` AI audited the harness to ensure academic integrity. It verified that **no fake CSV data** was generated. It also ensured the scripts gracefully handled missing data (e.g., logging `not_available` for thermal data since you are currently running on a Windows PC, not the Raspberry Pi).
3. **Human/AI Troubleshooting (The Run Logs):** The raw terminal logs show the real-world friction of gathering the first data points.
 - We booted the middleware, Docker n8n, and the metric scripts.
 - We hit `HTTP 404` errors trying to call unregistered production webhooks and `Error in workflow` exceptions.
 - After adjusting between test/production webhook URLs and validating the JSON payloads, we successfully recorded our first real measurements (one high-temp and one low-temp run for both deterministic and agent modes).
4. **AI Task (Report Generation):** The `report-scribe` AI finalized the Step 9 report note. It successfully aggregated the raw data into processed summaries (`summary_latency.csv` , etc.). Crucially, it documented the limitations: this was a small 1-run test, thermal data is missing, and larger repeated runs (3-5x) are still needed for the final thesis tables.

▼ 📁 1. The Code Maps to the Evidence

The scripts and schemas the AI created act as the methodology framework for your thesis.

- **Evidence: The Evaluation Tools**
 - `scripts/collect_metrics.py`
 - `scripts/aggregate_results.py`

- `evaluation/datasets/test-cases.json`
- `evaluation/metrics/metrics-schema.md`
- *Why it matters:* This directly feeds **Chapter 3 (Methodology)** and **Chapter 5 (Implementation)**. It proves you built a systematic, repeatable way to measure your system rather than just guessing.

▼ 2. The Actions Map to the Evidence

The actual execution of those scripts produced the raw scientific data.

• Evidence: The Result Data

- `evaluation/results/raw/run_..._deterministic.csv`
- `evaluation/results/raw/run_..._agent.csv`
- `evaluation/results/processed/summary_latency.csv`
- `evaluation/results/processed/baseline_vs_agent.csv`
- *Why it matters:* This is the goldmine for **Chapter 6 (Results)**. These CSVs will literally become the tables and charts in your final report, showing exactly how much latency the AI agent adds compared to the deterministic baseline. It also feeds **Chapter 7 (Discussion)**, where you will discuss the `404` webhook setup errors and the PC-vs-Raspberry Pi thermal limitations.

▼ 3 new full step 9

Step 9 — Build measurement and evaluation harness

Goal

Produce repeatable evidence for the report.

Work to complete

Create:

```
docs/evaluation/evaluation-protocol.md
evaluation/datasets/
evaluation/results/raw/
evaluation/results/processed/
evaluation/metrics/
scripts/collect_metrics.py
scripts/aggregate_results.py
```

Run both modes:

1. deterministic baseline
2. agent-enhanced workflow with validation/approval

Measure:

- success/failure
- latency
- RAM
- CPU
- thermal data or thermal collection path
- safe vs blocked action behavior

Evidence produced

```
evaluation/results/raw/run_01.csv
evaluation/results/raw/run_02.csv
evaluation/results/summary_latency.csv
evaluation/results/summary_resources.csv
evaluation/results/safety_outcomes.csv
evaluation/results/baseline_vs_agent.csv
```

Chapter feeds from

Chapter 3 — Methodology feeds from:

```
docs/evaluation/evaluation-protocol.md
scripts/collect_metrics.py
scripts/aggregate_results.py
```

Chapter 5 — Implementation feeds from:

```
measurement setup
metrics scripts
```

Chapter 6 — Results feeds from:

```
evaluation/results/
charts
tables
```

Chapter deliverables

This step helps deliver:

Chapter 3

- evaluation method
- measurement method
- repeatability

Chapter 5

- measurement/evaluation setup subsection

Chapter 6

- latency results
- RAM/CPU results
- safety outcomes
- baseline-vs-agent comparison

▼ Goal

Produce the evidence that will make the report real.

▼ Work to complete

In `evaluation/`, create:

- `evaluation/datasets/`
- `evaluation/results/raw/`

- `evaluation/results/processed/`
- `scripts/collect_metrics.py`
- `scripts/aggregate_results.py`

Run the same scenario in two modes:

1. deterministic baseline
2. agent-enhanced workflow with validation/approval

Measure:

- success/failure
- end-to-end latency
- RAM usage
- CPU usage
- thermal data or thermal collection path
- safe vs blocked action behavior

▼ What this proves

This creates the thesis results.

▼ Evidence produced

- raw CSV logs
- processed summaries
- result tables
- charts if useful
- baseline-vs-agent comparison

Suggested result files:

- `evaluation/results/raw/run_01.csv`
- `evaluation/results/raw/run_02.csv`
- `evaluation/results/summary_latency.csv`
- `evaluation/results/summary_resources.csv`
- `evaluation/results/safety_outcomes.csv`
- `evaluation/results/baseline_vs_agent.csv`

▼ Step 10 — Add Raspberry Pi validation if the core system is stable
 chapter 4, Choice of approach
 chapter 5, mplementation

chapter 6, Results
chapter 7 Discussion

▼ AI

▼ codex step runner

▼ q

```
Current step:
Step 10 – Add Raspberry Pi validation if the core system is stable.

Use these files as source of truth:

- [AGENTS.md](http://agents.md/)
- [README.md](http://readme.md/)
- docs/plans/implementation-plan.md
- docs/plans/new-yacoub-14-step-process.md
- docs/decisions.md
- docs/architecture/README.md
- docs/evaluation/evaluation-protocol.md
- docs/ongoing/integration-contract.md
- infrastructure/docker/docker-compose.yml
- infrastructure/docker/README.md
- middleware/
- cognitive_logic/workflows/deterministic-baseline.md
- cognitive_logic/workflows/agent-minimal.md
- evaluation/README.md
- evaluation/metrics/metrics-schema.md
- evaluation/results/raw/
- evaluation/results/processed/
- scripts/collect_metrics.py
- scripts/aggregate_results.py

Scope:

- new-yacoub thesis
- Yacoub complete in narrowed form
- Step 10 only
- Raspberry Pi validation is Tier 1.5 and optional/supporting
- do not redesign the local system
- do not change the core middleware/workflow architecture
- do not add multiple devices
- do not add MCP
- do not modify n8n core
- do not add Prometheus/Grafana
- do not create fake Pi results
- do not block the already-working local Tier 1 system

Goal:
Strengthen Yacoub's edge-computing angle by preparing a thin Raspberry Pi validation path that can run the same or simplified scenario and collect limited hardware/resource evidence if the Pi is available.58i8m 756

Task:
Create the Raspberry Pi validation documentation and result structure.

Create:

- infrastructure/os/raspberry-pi-notes.md
- infrastructure/docker/pi-deployment-notes.md
- evaluation/results/pi-validation/.gitkeep
- evaluation/results/pi-validation/README.md

Create if useful:

- evaluation/results/pi-validation/pi-validation-template.csv
- evaluation/results/pi-validation/pi-validation-notes-template.md

Update only if needed:

- docs/evaluation/evaluation-protocol.md
- docs/decisions.md
- [README.md](http://readme.md/) with a short pointer to the Pi validation notes
```

Requirements for infrastructure/os/raspberry-pi-notes.md:

- explain the purpose of Pi validation as Tier 1.5
- explain that PC-first local validation remains the primary working system
- list expected Pi prerequisites:
 - Raspberry Pi available
 - Raspberry Pi OS or compatible Linux
 - Docker or Python runtime depending on chosen validation path
 - network access to n8n/middleware if split across devices
- include basic commands/checks:
 - `uname -a`
 - `python --version`
 - `docker --version` if Docker is used
 - `free -h`
 - `top` or `ps`
 - `vcgencmd measure_temp` if available
- explain thermal collection:
 - use `vcgencmd measure_temp` if available
 - otherwise mark thermal as `not_available`
- document limitations:
 - limited run count
 - simplified scenario allowed
 - Pi validation must not expand into full deployment research

Requirements for infrastructure/docker/pi-deployment-notes.md:

- explain possible deployment choices:
 1. run middleware on Pi and n8n on PC
 2. run both middleware and n8n on Pi only if resources allow
 3. run only a simplified measurement on Pi if Docker/n8n is too heavy
- explain recommended first Pi validation path:
 - middleware on Pi
 - n8n still on PC or Docker host
 - same webhook/action scenario
- mention expected URL/network changes:
 - `host.docker.internal` may not apply on Pi/Linux
 - use actual Pi IP address or Docker network host mapping if needed
- include caution that Docker Compose file may need adaptation for arm64 and available memory
- do not add a new Docker Compose file unless absolutely necessary

Requirements for evaluation/results/pi-validation/README.md:

- explain what Pi evidence should be saved here
- list expected files:
 - `[pi-validation-notes.md](http://pi-validation-notes.md/)`
 - `pi-run-01.csv`
 - `pi-resource-observations.csv`
 - screenshots or terminal logs if available
- explain that no fake Pi measurements should be added
- explain how Pi results should be interpreted:
 - optional validation
 - local PC result remains baseline if Pi cannot be completed

Requirements for template files:

- `pi-validation-template.csv` should be clearly marked as a template, not real results
- include columns such as:
 - `timestamp`
 - `device`
 - `scenario`
 - `mode`
 - `test_case_id`
 - `expected_action`
 - `observed_action`
 - `success`
 - `latency_ms`
 - `cpu_percent`
 - `ram_mb`
 - `thermal_c`
 - `notes`
- `[pi-validation-notes-template.md](http://pi-validation-notes-template.md/)` should include sections for:
 - hardware used
 - OS/version
 - deployment choice
 - commands run
 - observed limitations
 - whether validation succeeded or was skipped

Requirements for docs/evaluation/evaluation-protocol.md:

- update only if needed to mention Pi validation as optional Tier 1.5
- do not claim Pi results exist
- clearly separate PC/local results from optional Pi validation

Requirements for docs/decisions.md:

- update only if useful with a short decision:
 - PC-first remains primary
 - Pi validation is optional Tier 1.5
 - Pi validation must not block final thesis completion

Important:

- Do not invent Pi results.
- Do not create fake measurement CSVs as real results.
- Do not claim the system was deployed on Pi unless it actually was.
- Do not implement Step 11 freeze.
- Do not implement new architecture.
- Do not modify middleware/workflows unless strictly necessary for documentation clarity.

Validation:

- No runtime test is required unless a Raspberry Pi is actually available in this environment.
- If no Pi is available, clearly mark Step 10 as documentation/preparation only.
- Check that Markdown files are clean and readable.
- Check that no fake Pi result rows are placed under evaluation/results/pi-validation/ as real data.

Before editing:

- summarize the planned file changes.

After editing:

- summarize exactly what changed.
- state clearly whether Step 10 is:
 - Pi runtime verified
 - or Pi validation prepared/documented only
- tell me what I should do manually if I have a Raspberry Pi available.
- tell me what evidence should be saved for Chapter 4, Chapter 5, Chapter 6, and Chapter 7.

▼ a

▼ files

9 files changed +421 -0	Undo ↶
README.md +13 -0	▼
docs/decisions.md +27 -0	▼
docs/evaluation/evaluation-protocol.md +19 -0	▼
evaluation/results/pi-validation/.gitkeep +1 -0	▼
evaluation/results/pi-validation/README.md +48 -0	▼
evaluation/results/pi-validation/pi-validation-notes-template.md +86 -0	▼
evaluation/results/pi-validation/pi-validation-template.csv +1 -0	▼
infrastructure/docker/pi-deployment-notes.md +116 -0	▼
infrastructure/os/raspberry-pi-notes.md +110 -0	▼

Step 10 is prepared/documented only, not Pi runtime verified. No Raspberry Pi was available or tested, and I did not create fake Pi results.

Created:

- [raspberry-pi-notes.md](#)

- [pi-deployment-notes.md](#)
- [evaluation/results/pi-validation/README.md](#)
- [pi-validation-template.csv](#)
- [pi-validation-notes-template.md](#)
- `.gitkeep` for `evaluation/results/pi-validation/`

Updated:

- [docs/evaluation/evaluation-protocol.md](#)
- [docs/decisions.md](#)
- [README.md](#)

Validation:

- Confirmed `pi-validation-template.csv` has only a header row.
- Confirmed docs clearly say optional Tier 1.5, local PC remains primary baseline, and no fake Pi measurements should be added.

If you have a Pi available:

1. Start with middleware on Pi and n8n on PC.
2. On Pi, record `uname -a`, `python --version`, `free -h`, `top` or `ps`, and `vcgencmd measure_temp`.
3. Change n8n HTTP URLs from `host.docker.internal` to `http://<pi-ip>:8000/fan/on` and `/fan/off`.
4. Run a small high/low temp scenario.
5. Save real notes/CSV/logs under `evaluation/results/pi-validation/`.

Evidence to save:

- Chapter 4: why PC-first remains primary and Pi is optional Tier 1.5.
- Chapter 5: deployment choice and commands used.
- Chapter 6: real Pi CSVs/logs if run.
- Chapter 7: limitations if Pi validation is skipped or simplified.

-

▼ no pi workflow

Part 1 — Documentation-only Step 10 workflow

Use this if you are **not running the Pi today**.

1. Inspect the files Codex created

From repo root:

```
Get-Content infrastructure\os\raspberry-pi-notes.md
Get-Content infrastructure\docker\pi-deployment-notes.md
Get-Content evaluation\results\pi-validation\README.md
Get-Content evaluation\results\pi-validation\pi-validation-template.csv
Get-Content evaluation\results\pi-validation\pi-validation-notes-template.md
```

Check that they clearly say:

```
Pi validation is optional Tier 1.5
PC-first remains primary
```

no fake Pi results exist

2. Audit-review prompt

Send this to `audit-review` if you are only documenting Step 10:

Audit the Step 10 Raspberry Pi validation preparation changes only.

Check:

- Step 10 stays inside new-yacoub scope.
- Pi validation is clearly optional Tier 1.5.
- PC/local validation remains the primary working system.
- No fake Pi measurements were created.
- pi-validation-template.csv contains only a template/header, not fake result rows.
- evaluation/results/pi-validation/README.md explains what real Pi evidence should be saved.
- infrastructure/os/raspberry-pi-notes.md gives useful Pi prerequisite and command notes.
- infrastructure/docker/pi-deployment-notes.md explains deployment choices and network URL changes.
- docs/evaluation/evaluation-protocol.md and docs/decisions.md do not claim Pi runtime results exist.
- README update is accurate.
- no middleware/workflow architecture was changed.
- no Step 11 freeze work was started.
- no MCP, Prometheus/Grafana, n8n core changes, real GPIO expansion, or multiple-device redesign was added.

Return:

1. blocking issues
2. non-blocking cleanup
3. Step 10 verdict
4. whether report-scribe can create the Step 10 note

Then report-scribe:

Create the report note for Step 10.

Create:

docs/report-notes/step-10-raspberry-pi-validation.md

Use only evidence already present in the repo.

Use these sources:

- infrastructure/os/raspberry-pi-notes.md
- infrastructure/docker/pi-deployment-notes.md
- evaluation/results/pi-validation/README.md
- evaluation/results/pi-validation/pi-validation-template.csv
- evaluation/results/pi-validation/pi-validation-notes-template.md
- docs/evaluation/evaluation-protocol.md
- docs/decisions.md
- README.md

Include:

- what was built
- why it matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent Pi results
- clearly state that Step 10 is prepared/documented only unless real Pi logs exist
- clearly state that PC/local validation remains the primary thesis baseline
- explain that Raspberry Pi validation is optional Tier 1.5
- mention what evidence should be added if Pi validation is later run

Then commit Step 10 docs.

▼ pi workflow - Part 2 — Real Raspberry Pi validation workflow

▼ pi specs

Ask for a **Raspberry Pi 4 Model B** or **Raspberry Pi 5**, ideally with **4 GB RAM or more**.

For your thesis, you do **not** need something extreme. You only need enough to run the middleware, basic Docker/Python tests, and collect lightweight CPU/RAM/thermal observations.

Minimum good request:

```
Raspberry Pi 4 Model B, 4 GB RAM  
microSD card, 32 GB or more  
power supply  
network access: Wi-Fi or Ethernet  
Raspberry Pi OS installed or installable
```

Even better if available:

```
Raspberry Pi 5, 4 GB or 8 GB RAM
```

Avoid asking for an old Pi like Pi 2 or Pi 3 unless there is no alternative. They may still run Python middleware, but n8n/Docker validation can become annoying or too slow.

Simple script to Stefan

Hej Stefan,

Jag skulle vilja fråga om det finns möjlighet att låna en Raspberry Pi för en begränsad valideringsdel i mitt examensarbete.

Mitt huvudsystem är redan tänkt att utvecklas och testas lokalt på PC först, men jag vill lägga till en mindre "Tier 1.5"-validering på edge-hårdvara. Syftet är inte att bygga om hela systemet för Raspberry Pi, utan att kunna visa att den enklare middleware-/IoT-delen kan köras på en mer resursbegränsad enhet och att jag kan samla enkla mätvärden som CPU, RAM och temperatur.

Det jag helst skulle behöva är:

- Raspberry Pi 4 Model B eller Raspberry Pi 5
- helst 4 GB RAM eller mer
- microSD-kort, minst 32 GB
- strömförsörjning
- möjlighet till Wi-Fi eller Ethernet
- Raspberry Pi OS, eller möjlighet att installera det

Raspberry Pi 4 med 4 GB RAM räcker troligen bra för mitt syfte. Om det finns en Raspberry Pi 5 tillgänglig fungerar det också, men det är inte ett krav.

Jag kommer använda den för en liten validering där middleware körs på Pi, medan n8n fortfarande kan köras på PC. Då kan jag testa samma temperatur/fan-scenari och dokumentera skillnader mot PC-miljön, till exempel resursanvändning och temperatur.

Målet är alltså att stärka edge computing-delen i arbetet utan att göra Raspberry Pi till ett nytt stort delprojekt.

Mvh

Yacoub

Even shorter version if you want to ask casually

Hej Stefan,

Finns det möjlighet att låna en Raspberry Pi till en mindre valideringsdel i mitt examensarbete?

Jag skulle helst behöva en Raspberry Pi 4 Model B eller Raspberry Pi 5, gärna med minst 4 GB RAM, microSD-kort, strömförsörjning och Wi-Fi/Ethernet.

Syftet är att göra en begränsad edge-validering: köra min Python-middleware på Pi, låta n8n på PC anropa Pi via nätverket, och samla enkla mätvärden som CPU, RAM och temperatur. Det är inte tänkt som en full ombyggnad av systemet, utan som en extra validering utöver den lokala PC-versionen.

Mvh

Yacoub

Use the shorter one if you are messaging him casually. Use the longer one if it is an email.



Use this only if you have the Pi ready.

Hardware/software goal

You are not trying to move the whole thesis to the Pi.

You are only proving:

```
The middleware/action side can run on Raspberry Pi
n8n can call the Pi middleware over the network
basic latency/resource/thermal evidence can be collected
```

Recommended setup:

```
PC:
  n8n Docker still running

Raspberry Pi:
  middleware running with python -m middleware.api.app

n8n workflow:
  calls http://<PI_IP>:8000/fan/on
  calls http://<PI_IP>:8000/fan/off
```

▼ 1. Prepare the Raspberry Pi

On the Pi terminal:

```
uname-a
python3--version
free-h
```

Check thermal command:

```
vcgencmd measure_temp
```

If `vcgencmd` does not exist, that is okay. Later write:


```
thermal_c = not_available
```

Check Pi IP address:

```
hostname-I
```

Write down the IP, for example:

```
192.168.1.45
```

▼ 2. Put the repo or middleware on the Pi

Simplest method: clone your repo on the Pi.

```
git clone https://github.com/Rumple12/new-yacoub-thesis.git
cd new-yacoub-thesis
```

If Git is not installed:

```
sudo apt update
sudo apt installgit-y
```

3. Run middleware on the Pi

From repo root on the Pi:

```
python3-m middleware.api.app
```

If it starts on `127.0.0.1`, n8n on your PC may not reach it.

For network access, run:

```
MIDDLEWARE_HOST=0.0.0.0 python3-m middleware.api.app
```

Expected:

```
new-yacoub middleware running at http://0.0.0.0:8000
```

Keep this terminal open.

4. Test Pi middleware from your PC

On your Windows PC PowerShell, replace `<PI_IP>`:

```
Invoke-RestMethod-MethodGet-Uri"http://<PI_IP>:8000/status"
```

Example:

```
Invoke-RestMethod-MethodGet-Uri"http://192.168.1.45:8000/status"
```

Then test fan endpoints:

```
Invoke-RestMethod-MethodPost-Uri"http://192.168.1.45:8000/fan/on"  
Invoke-RestMethod-MethodPost-Uri"http://192.168.1.45:8000/fan/off"
```

If this fails, likely causes:

```
Pi and PC are not on same network  
Pi firewall/network blocks port 8000  
middleware is only bound to 127.0.0.1 instead of 0.0.0.0  
wrong Pi IP
```

5. Open n8n on PC

On PC:

```
cd"C:\Users\Jake_\OneDrive - Mittuniversitetet\EDU\4. RED\1. kaniddat\new-yacoub-thesis\infrastructure\docker"  
dockercomposeup-d
```

Open:

```
http://localhost:5678
```

6. Make a Pi validation copy of the workflow

Do **not** destroy your working local workflow.

In n8n:

1. Open deterministic baseline workflow.
2. Duplicate it.
3. Rename copy to:

```
deterministic-baseline-pi-validation
```

Then update the HTTP Request node URLs:

```
http://192.168.1.45:8000/fan/on  
http://192.168.1.45:8000/fan/off
```

Use your actual Pi IP.

You can also do this with the agent workflow, but start with deterministic first because it is simpler.

7. Run deterministic Pi high-temp test

In n8n, click **Execute workflow** on the Pi validation workflow.

In PowerShell, use the Pi workflow webhook URL. It might be something like:

```
http://localhost:5678/webhook-test/deterministic-baseline-pi-validation
```

High temp:

```
$piWebhookUrl="http://localhost:5678/webhook-test/deterministic-baseline-pi-validation"

$highBody= @{
  sensor_id="temp_sensor_1"
  timestamp="2026-04-25T20:00:00Z"
  type="temperature"
  value=31.4
  unit="C"
}|ConvertTo-Json

Invoke-RestMethod-MethodPost-Uri$piWebhookUrl-ContentType"application/json"-Body$highBody
```

Expected:

```
n8n true branch
POST http://<PI_IP>:8000/fan/on
Pi middleware returns fan = on
```

Save screenshot.

8. Run deterministic Pi low-temp test

Click **Execute workflow** again.

```
$lowBody= @{
  sensor_id="temp_sensor_1"
  timestamp="2026-04-25T20:05:00Z"
  type="temperature"
  value=24.5
  unit="C"
}|ConvertTo-Json

Invoke-RestMethod-MethodPost-Uri$piWebhookUrl-ContentType"application/json"-Body$lowBody
```

Expected:

```
n8n false branch
POST http://<PI_IP>:8000/fan/off
Pi middleware returns fan = off
```

Save screenshot.

9. Collect Pi resource/thermal notes

On Pi, while middleware is running:

```
free-h
```

Check CPU/process:

```
top
```

or:

```
ps aux |grep python
```

Thermal:

```
vcgencmd measure_temp
```

Save these into:

```
evaluation/results/pi-validation/pi-validation-notes.md
```

10. Create real Pi CSV result

Do **not** overwrite the template. Create a real file:

```
evaluation/results/pi-validation/pi-run-01.csv
```

Use columns from the template.

Example rows, only if you actually observed them:

```
timestamp,device,scenario,mode,test_case_id,expected_action,observed_action,success,latency_ms,cpu_percent,ram_mb,thermal_c,notes
2026-04-25T20:00:00Z,raspberry_pi,temperature_fan,deterministic,pi_high_temp,fan_on,fan_on,true,not_measured,not_available,not_available,52.3,"manual Pi validation high-temp test"
2026-04-25T20:05:00Z,raspberry_pi,temperature_fan,deterministic,pi_low_temp,fan_off,fan_off,true,not_measured,not_available,not_available,52.6,"manual Pi validation low-temp test"
```

If latency is not measured, write:

```
not_measured
```

Do not invent numbers.

11. Evidence files to save

Put them here:

```
evaluation/results/pi-validation/
```

Suggested files:

```
pi-validation-notes.md
pi-run-01.csv
pi-high-temp-fan-on-canvas.png
pi-high-temp-fan-on-output.png
pi-low-temp-fan-off-canvas.png
pi-low-temp-fan-off-output.png
```

```
pi-terminal-uname.png
pi-terminal-free.png
pi-terminal-temp.png
```

Step 10 success levels

Level 1 — documentation only

```
Pi validation path documented
no fake results
templates created
```

This is already enough if time is tight.

Level 2 — Pi middleware reachable

```
PC can call http://<PI_IP>:8000/status
PC can call /fan/on and /fan/off on Pi
```

Good Tier 1.5 evidence.

Level 3 — n8n calls Pi middleware

```
n8n workflow high temp -> Pi /fan/on
n8n workflow low temp -> Pi /fan/off
```

Strong Tier 1.5 evidence.

Level 4 — Pi resource notes

```
free -h
top/ps
vcgencmd measure_temp
```

Best Step 10 evidence.

What to do next

If you have the Pi right now, start with:

```
hostname-I
MIDDLEWARE_HOST=0.0.0.0 python3-m middleware.api.app
```

Then from PC test:

```
Invoke-RestMethod-MethodGet-Uri"http://<PI_IP>:8000/status"
```

If that works, the rest becomes easy.

If you do **not** have the Pi ready, Step 10 can stay documentation-only, then audit-review + report-scribe + commit.

▼ explaining what we are doing with the raspberry pi

You're right — English.

Yes, what you wrote is **more minimized than the original broad Yacoub/Obid plan**, but that is exactly the point of the **new-yacoub** version.

It is not "weak." It is a controlled scope.

The original direction sounded like:

```
full n8n + agentic AI + middleware + Raspberry Pi + safety + evaluation + maybe broader Obid cognitive logic
```

Your current Stefan message says:

```
main system runs on PC
Raspberry Pi is used only for a small edge validation
middleware runs on Pi
n8n on PC calls the Pi over the network
collect CPU/RAM/temperature notes
```

That is **smaller**, yes. But it is also much more realistic and defensible.

The important framing is:

You are not saying:

```
I cannot do Raspberry Pi properly.
```

You are saying:

```
I am intentionally keeping the Raspberry Pi part as a limited validation step so it supports the thesis instead of becoming the whole thesis.
```

That is a strong academic framing.

Use this English explanation for yourself:

```
The Raspberry Pi is not the main implementation target. The main thesis system is developed and evaluated locally first so that the workflow, middleware, safety boundary, and measurements are stable. The Raspberry Pi is then used as a Tier 1.5 validation step to test part of the system under more resource-constrained edge conditions. This keeps the thesis focused while still preserving the edge-computing angle.
```

And if Stefan asks why not put everything on the Pi:

```
Because the goal is not to make Raspberry Pi deployment the entire thesis. The goal is to evaluate a minimal n8n-based IoT workflow with middleware, safety boundaries, and measurements. Running the middleware on the Pi is enough to validate the edge/device side, while keeping n8n on the PC avoids unnecessary deployment complexity.
```

So yes: it is minimized compared to the original plan, but it is minimized **correctly**.

I still don't get it, explain everything over again leading up to the raspberry pi. I literally don't understand what the project is about

Thought for 15s

You're confused because there are **two different things mixed together**:

1. **The system you are building**
2. **The thesis argument you are trying to prove**

Let's reset everything from zero.

1. What is the project actually about?

Your thesis is about this:

```
Can a small self-hosted n8n-based IoT system handle a simple physical action safely and measurably?
```

In human words:

You are building a small smart-IoT setup where a **temperature event** comes in, the system decides whether a **fan** should turn on or off, and then you measure whether this setup works.

The fan can be simulated. The sensor can be simulated. The important thing is the **architecture**:

```
temperature event
→ middleware
→ n8n workflow
→ decision
→ validation/safety boundary
→ fan action
→ measurement logs
```

That is the whole project.

It is not about making the world's best smart fan.

It is not about creating a huge agentic AI platform.

It is not about fully deploying everything on Raspberry Pi.

It is about building **one small complete system** and evaluating it properly.

2. The simple example

Imagine this scenario:

```
Temperature = 31.4°C
```

The system should decide:

```
31.4°C is hot
→ turn fan on
```

Another scenario:

```
Temperature = 24.5°C
```

The system should decide:

```
24.5°C is not hot
→ turn fan off
```

That is the entire physical behavior.

Everything else in the thesis exists to answer:

```
How does the system decide?
How safely does it decide?
How fast is it?
How much CPU/RAM does it use?
Can it run in a constrained/edge-like setup?
```

3. What are the parts of the system?

n8n

n8n is the workflow engine.

It is the place where the logic lives.

Think of n8n like this:

```
"If event comes in, do these steps in this order."
```

You built two n8n workflows:

```
Step 6: deterministic baseline workflow
Step 7: agent-enhanced workflow
```

Middleware

Middleware is your Python bridge.

It exposes endpoints like:

```
/status
/fan/on
/fan/off
```

n8n does **not** directly touch hardware.

n8n only says:

```
POST /fan/on
```

Then the middleware handles the action.

So middleware is the controlled doorway between workflow logic and the physical/simulated device.

Shared contract

The shared contract says:

```
{
  "action_id": "fan_on",
  "target": "fan_1",
  "reason": "temperature_at_or_above_threshold",
  "requires_approval": false
}
```

This prevents random messy AI output.

The system is not allowed to say:

```
open_window
start_heater
delete_files
run_shell_command
```

Only:

```
fan_on
fan_off
```

Target:

```
fan_1
```

That is the contract boundary.

Safety layer

The safety layer says:

```
If the action follows the contract, allow it.
If it is malformed, block it.
If it is unknown/risky, require approval or reject it.
```

So if the AI says:

```
{
  "action_id": "open_window",
  "target": "unknown_device"
}
```

The safety layer says:

```
No. This is outside the allowed contract.
```

Right now Step 8 is a **documented safety design**, not a full production runtime guardrail.

That is okay for your current scope.

Evaluation harness

Step 9 created scripts and CSVs to measure things:

```
success/failure
latency
CPU
RAM
thermal if available
baseline vs agent comparison
```

This is what turns the project into a thesis instead of just "I built something."

You can now say:

```
I ran the deterministic workflow.  
I ran the agent workflow.  
I collected result CSVs.  
I compared them.
```

4. Why are there two workflows?

Because the thesis needs a comparison.

Workflow 1: deterministic baseline

This is simple rule logic:

```
if temperature >= 30.0:  
    fan_on  
else:  
    fan_off
```

No AI.

This is your control group.

It answers:

```
What happens if we solve the problem with normal logic?
```

Workflow 2: agent-enhanced workflow

This uses an LLM/AI node.

It receives the temperature event and produces structured output:

```
{  
  "action_id": "fan_on",  
  "target": "fan_1",  
  "reason": "temperature_at_or_above_threshold",  
  "requires_approval": false  
}
```

This answers:

```
Can an agent-enhanced workflow make the same kind of decision while staying inside a strict contract?
```

You are **not** trying to prove that AI is better.

You are testing whether a minimal AI workflow can be integrated safely and measurably.

5. What is the thesis argument?

Your thesis basically says:

```
I built a small n8n-based IoT proof-of-concept.  
It has a middleware bridge, a contract boundary, a minimal agent path, and a safety design.  
I compared deterministic and agent-enhanced behavior.
```

```
I measured latency/resource behavior.  
This shows what is feasible, what works, and what limitations exist.
```

That is the academic story.

Not:

```
I made a huge AI smart home system.
```

But:

```
I evaluated a small controlled architecture for agentic IoT.
```

That is much stronger and more realistic.

6. Where does Raspberry Pi come in?

Now we finally reach Raspberry Pi.

The Raspberry Pi is **not the main project**.

The main project already works locally on your PC:

```
PC runs n8n  
PC runs middleware  
PC runs workflows  
PC collects measurements
```

That is your **Tier 1 system**.

The Raspberry Pi is only a **Tier 1.5 validation**.

It asks:

```
Can part of this system run on a more constrained edge device?
```

The Pi is there to strengthen the edge-computing angle, not to become the whole thesis.

7. Why not run everything on the Raspberry Pi?

Because that would explode the scope.

Running everything on Pi would mean dealing with:

```
Docker on Pi  
n8n on Pi  
LLM/API credentials  
networking  
performance issues  
storage issues  
thermal issues  
slow debugging  
possibly ARM compatibility problems
```

That becomes a separate deployment thesis.

You do **not** need that.

Instead, the smart minimized plan is:

```
PC runs n8n
Raspberry Pi runs Python middleware
n8n on PC calls the Pi over the network
```

So the Pi becomes the **edge device**.

8. The Raspberry Pi version of the system

Normal PC version:

```
n8n on PC
→ calls middleware on PC
→ /fan/on or /fan/off
```

Pi validation version:

```
n8n on PC
→ calls middleware on Raspberry Pi
→ /fan/on or /fan/off on Pi
```

So instead of n8n calling:

```
http://host.docker.internal:8000/fan/on
```

it calls:

```
http://<raspberry-pi-ip>:8000/fan/on
```

Example:

```
http://192.168.1.45:8000/fan/on
```

That proves:

```
The workflow can control an edge-side middleware endpoint over the network.
```

That is enough for Tier 1.5.

9. What exactly would run on the Raspberry Pi?

Minimum version:

```
Python middleware only
```

The Pi runs:

```
python3-m middleware.api.app
```

The Pi exposes:

```
/status  
/fan/on  
/fan/off
```

Your PC's n8n workflow calls those endpoints.

That is it.

You are not forced to run n8n on the Pi.

10. What evidence would the Pi give you?

You would save things like:

```
Pi model  
Pi OS version  
Python version  
RAM available  
CPU observation  
temperature observation  
whether /fan/on and /fan/off worked  
whether n8n could call Pi middleware  
basic latency if measured
```

Example:

```
Raspberry Pi 4, 4GB RAM  
Middleware started successfully  
PC could call http://192.168.1.45:8000/status  
n8n high-temp workflow triggered /fan/on on Pi  
n8n low-temp workflow triggered /fan/off on Pi  
Pi temperature was around X°C  
RAM usage was around Y MB
```

That gives you real edge-validation evidence.

11. What does this prove in the thesis?

It proves:

```
The architecture is not only a PC toy.  
The middleware/action side can be moved to a constrained edge device.  
The system can still communicate over the network.  
There are practical constraints around CPU/RAM/temperature.
```

But it does **not** claim:

```
The whole system is production-ready on Raspberry Pi.
```

That is an important difference.

12. How to explain it to Stefan

The honest explanation is:

```
My main implementation is local PC-first so I can keep the system stable and measurable.
```

The Raspberry Pi is an optional Tier 1.5 validation step. I want to use it to run the Python middleware on constrained edge hardware, while n8n remains on the PC. This lets me test the same temperature/fan scenario over the network and collect basic CPU, RAM, and thermal observations.

The goal is not to rebuild the whole system on Raspberry Pi. The goal is to strengthen the edge-computing part of the thesis with a limited hardware validation.

That is the cleanest framing.

13. Which Raspberry Pi do you need?

Ask for:

Raspberry Pi 4 Model B, 4 GB RAM or more

or:

Raspberry Pi 5, 4 GB or 8 GB

Minimum request:

Raspberry Pi 4 Model B
4 GB RAM
32 GB microSD card
power supply
Wi-Fi or Ethernet
Raspberry Pi OS

You do not need a cluster, sensors, relays, or extra hardware unless Stefan already has them.

14. The project in one sentence

Your thesis is:

A small evaluated n8n-based IoT system where temperature events trigger fan actions through middleware, with deterministic and agent-enhanced workflows compared, safety boundaries documented, measurements collected, and optional Raspberry Pi validation used to strengthen the edge-computing angle.

That is the whole thing.

15. The mental model

Think of it like this:

n8n = brain / workflow controller

AI node = optional smarter decision maker

middleware = controlled doorway to the device

contract = allowed language between AI and system

safety layer = guardrail before action

evaluation harness = proof collector

Raspberry Pi = optional edge device proving the system can touch constrained hardware

The Raspberry Pi is the **last supporting piece**, not the foundation.

Your foundation is already:

PC local system
middleware
workflows
contract
safety design
measurements

The Pi just says:

Can this idea survive contact with small edge hardware?

And even if you only document it or do a small validation, that still fits the new-yacoub scope.

stefan is not here, write me an email to him

Fråga om att låna Raspberry Pi till examensarbeteFråga om att låna Raspberry Pi till examensarbete

Hej Stefan,

Jag vill fråga om det finns möjlighet att låna en Raspberry Pi till en mindre valideringsdel i mitt examensarbete.

Mitt huvudsystem utvecklas och testas först lokalt på PC, så att jag har ett stabilt och mätbart system. Raspberry Pi skulle användas som en begränsad edge-validering, alltså inte som en full ombyggnad av hela systemet.

Tanken är att köra min Python-middleware på Raspberry Pi, medan n8n fortfarande körs på PC. Då kan n8n anropa Pi via nätverket och jag kan testa samma temperatur/fan-scenario, till exempel `/fan/on` och `/fan/off`, men med middleware-delen på mer resursbegränsad hårdvara.

Det jag helst skulle behöva är:

- Raspberry Pi 4 Model B eller Raspberry Pi 5
- gärna minst 4 GB RAM
- microSD-kort, helst minst 32 GB
- strömförsörjning
- Wi-Fi eller Ethernet
- Raspberry Pi OS, eller möjlighet att installera det

Syftet är att stärka edge computing-delen i arbetet genom att samla enklare observationer som CPU, RAM och temperatur på Raspberry Pi. Det är alltså tänkt som en extra validering utöver den lokala PC-versionen, inte som ett nytt stort delprojekt.

Mvh

Yacoub

in english

Question about borrowing a Raspberry Pi for my thesisQuestion about borrowing a Raspberry Pi for my thesis

Hi Stefan,

I wanted to ask if it would be possible to borrow a Raspberry Pi for a small validation part of my thesis.

My main system is being developed and tested locally on my PC first, so that I have a stable and measurable system. The Raspberry Pi would be used as a limited edge validation step, not as a full rebuild of the entire system.

The idea is to run my Python middleware on the Raspberry Pi, while n8n still runs on my PC. That way, n8n can call the Pi over the network and I can test the same temperature/fan scenario, for example `/fan/on` and `/fan/off`, but with the middleware part running on more resource-constrained hardware.

Ideally, I would need:

- Raspberry Pi 4 Model B or Raspberry Pi 5
- preferably at least 4 GB RAM
- microSD card, preferably at least 32 GB
- power supply
- Wi-Fi or Ethernet
- Raspberry Pi OS, or the possibility to install it

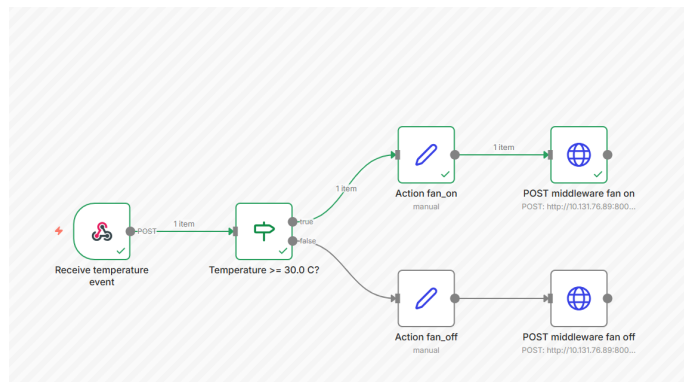
The purpose is to strengthen the edge-computing part of the thesis by collecting simple observations such as CPU, RAM, and temperature on the Raspberry Pi. So it is intended as an extra validation step in addition to the local PC version, not as a new large subproject.

Best regards,

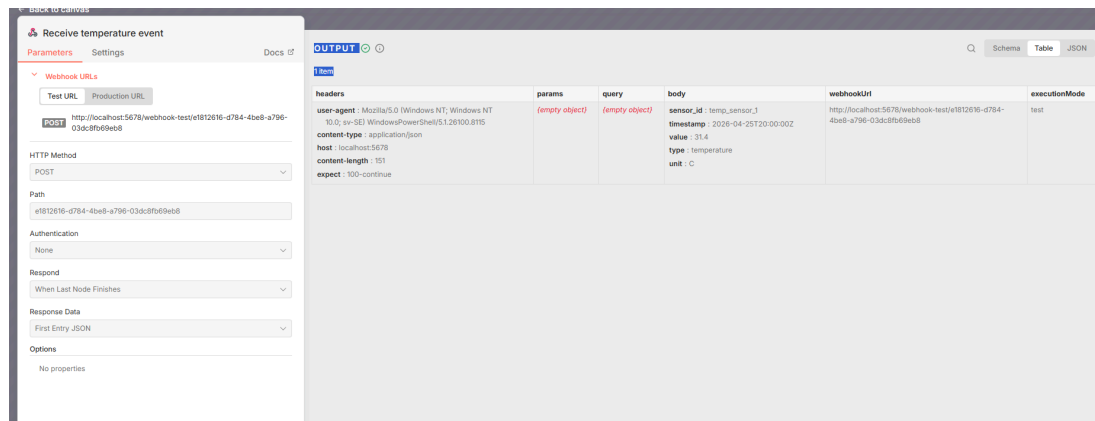
Yacoub

▼ n8n deterministic high temp pi

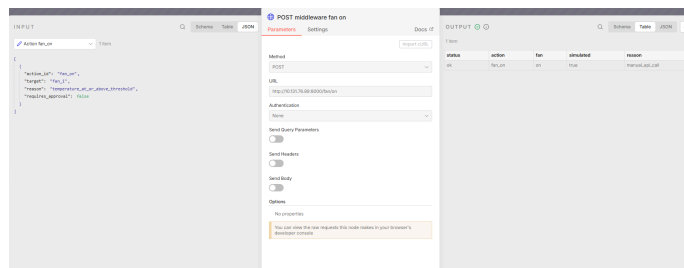
▼ high temp tree - 1. `pi-high-temp-fan-on-canvas.png`



▼ triangle



▼ globe - pi-high-temp-fan-on-output.png

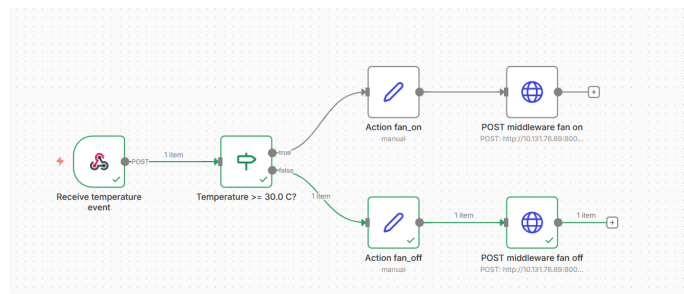


▼ terminal output - pi-terminal-high-temp-fan-on.png

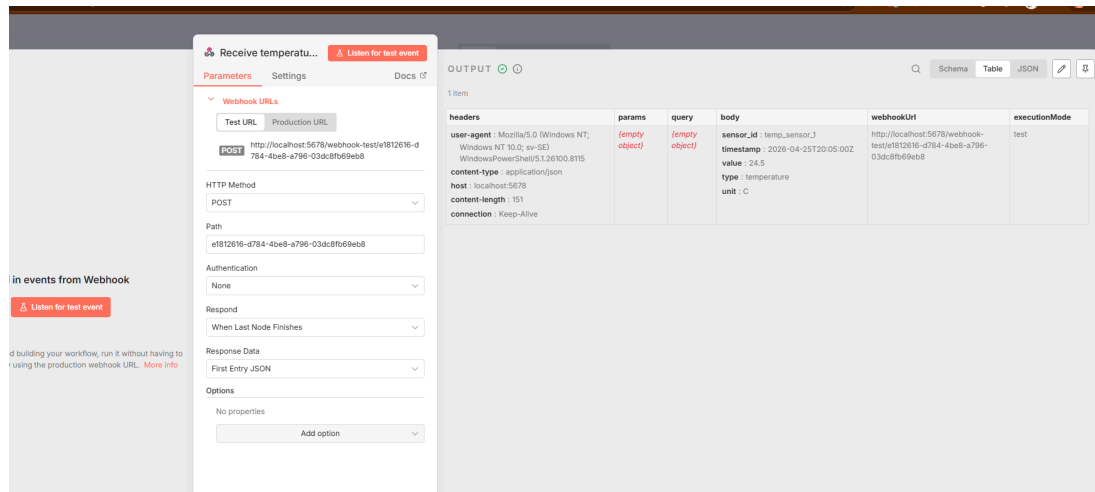


▼ n8n determinast low temp pi

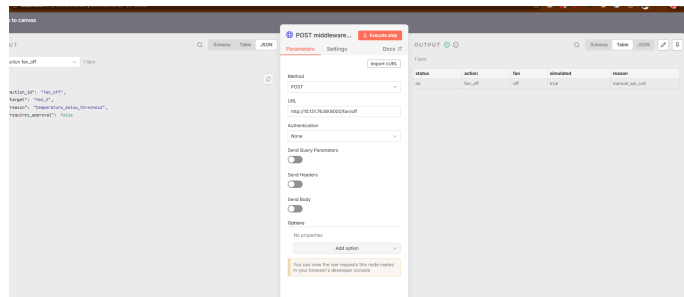
▼ low temp tree - [pi-low-temp-fan-on-canvas.png](#)



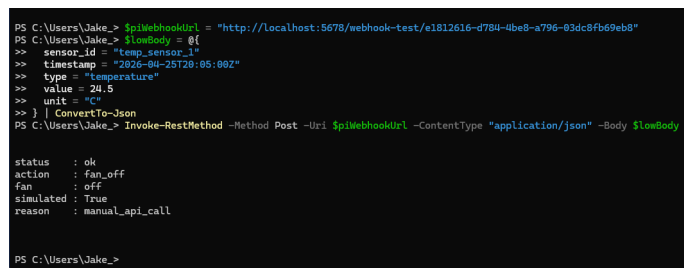
▼ triangle



▼ globe - pi-high-low-fan-on-output.png

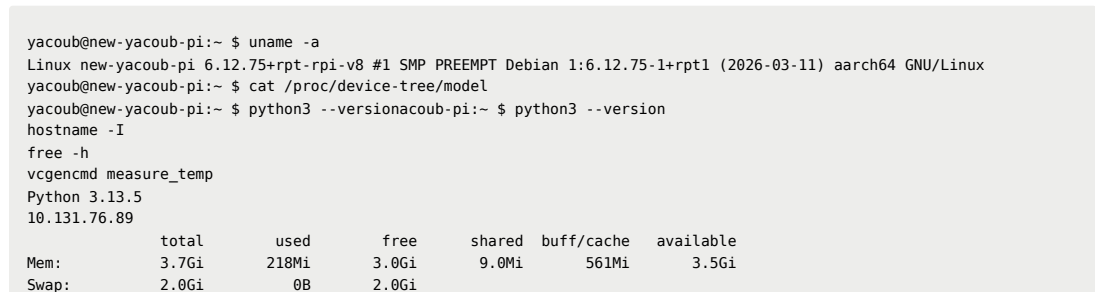


▼ terminal output - pi-terminal-low-temp-fan-on.png



▼ pi spec

▼ all pi sepcs



```
temp=45.7'C
yacoub@new-yacoub-pi:~$ ps aux | grep "middleware.api.app"
yacoub      1362  0.0  0.5 107156 21676 pts/0    S+   20:46   0:00 python3 -m middleware.api.app
yacoub      1641  0.0  0.0   6392  2024 pts/1    S+   22:32   0:00 grep --color=auto middleware.api.app
yacoub@new-yacoub-pi:~$ vcgencmd measure_temp
temp=45.7'C
yacoub@new-yacoub-pi:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi        208Mi        3.0Gi         9.0Mi        563Mi        3.5Gi
Swap:           2.0Gi          0B          2.0Gi
yacoub@new-yacoub-pi:~$ ps -o pid,comm,%cpu,%mem,rss -C python3
  PID COMMAND      %CPU %MEM  RSS
  1362 python3         0.0  0.5 21676
```

Then create `pi-validation-notes.md`

Now the notes can say:

```
Pi model observed
OS/kernel observed
Python version observed
IP observed
RAM command output observed
thermal command output observed
middleware process observed
high/low n8n calls succeeded
```

▼ pi ram

```
yacoub@new-yacoub-pi:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi        208Mi        3.0Gi         9.0Mi        563Mi        3.5Gi
Swap:           2.0Gi          0B          2.0Gi
```

For RAM, from `free -h`, do not overthink it. Put the result in the notes file. In the CSV `ram_mb`, you can use `not measured` unless you calculate exact middleware RAM.

▼ pi temp

```
yacoub@new-yacoub-pi:~$ vcgencmd measure_temp
temp=45.7'C
```

▼ gatehred all 4 evidenec

What you gathered means:

1. Full workflow latency
 - ✓ pi-workflow-run-final.csv
 - ✓ n8n on PC -> Pi middleware -> fan_on/fan_off
 - ✓ latency measured
2. Direct Pi middleware latency
 - ✓ fan_on direct endpoint latency screenshot
 - ✓ fan_off direct endpoint latency screenshot
3. CPU/RAM of middleware process
 - ✓ Python middleware process visible
 - ✓ CPU percent visible
 - ✓ memory RSS visible
 - ✓ middleware RAM MB calculated
4. Pi temperature

```
✓ vcgencmd measure_temp
✓ temp = 49.6°C
```

This is **more than enough** for Step 10. You have gone beyond the minimum “limited real-hardware measurements.”

▼ 1. full workflow latency measure

▼ final evidence

[pi-workflow-run-final.csv](#)

connect to hotspot

▼ setup pi terminal

Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\WINDOWS\system32> ssh yacoub@new-yacoub-pi.local
```

The authenticity of host 'new-yacoub-pi.local (fe80::dea6:32ff:fe07:e7b7%19)' can't be established.

ED25519 key fingerprint is SHA256:Qi/aHag17KNoSXJ2ubjBPByxAIDnGos8Nj9lhQTm3vk.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? y

Please type 'yes', 'no' or the fingerprint: yes

Warning: Permanently added 'new-yacoub-pi.local' (ED25519) to the list of known hosts.

yacoub@new-yacoub-pi.local's password:

Permission denied, please try again.

yacoub@new-yacoub-pi.local's password:

Linux new-yacoub-pi 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

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the exact distribution terms for each program are described in the

individual files in /usr/share/doc/*/copyright.

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```
yacoub@new-yacoub-pi:~$ uname -a
```

```
Linux new-yacoub-pi 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64
```

```
GNU/Linux
```

```
yacoub@new-yacoub-pi:~$ cat /proc/device-tree/model
```

```
Raspberry Pi 4 Model B Repython3 --version
```

```
Python 3.13.5
```

```
yacoub@new-yacoub-pi:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	3.7Gi	187Mi	3.2Gi	9.0Mi	446Mi	3.5Gi
Swap:	2.0Gi	0B	2.0Gi			

```
yacoub@new-yacoub-pi:~$ hostname -I
```

```
10.131.76.89
```

```
yacoub@new-yacoub-pi:~$ vcgencmd measure_temp
```

```
temp=40.9°C
```

```
yacoub@new-yacoub-pi:~$ git status
```

```
-bash: git: command not found
```

```
yacoub@new-yacoub-pi:~$ sudo apt update
```

```

[sudo] password for yacoub:
Hit:1 http://deb.debian.org/debian trixie InRelease
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://archive.raspberrypi.com/debian trixie InRelease [54.9 kB]
Get:4 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:5 http://deb.debian.org/debian-security trixie-security/main arm64 Packages [130 kB]
Get:6 http://archive.raspberrypi.com/debian trixie/main arm64 Packages [438 kB]
Get:7 http://deb.debian.org/debian-security trixie-security/main armhf Packages [123 kB]
Get:8 http://deb.debian.org/debian-security trixie-security/main Translation-en [82.8 kB]
Get:9 http://archive.raspberrypi.com/debian trixie/main armhf Packages [433 kB]
Fetched 1,262 kB in 1s (846 kB/s)
4 packages can be upgraded. Run 'apt list --upgradable' to see them.
yacoub@new-yacoub-pi:~ $ git status
-bash: git: command not found
yacoub@new-yacoub-pi:~ $ sudo apt install git -y
Installing:
git

Installing dependencies:
git-man liberror-perl

Suggested packages:
git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 4
Download size: 10.9 MB
Space needed: 53.1 MB / 55.5 GB available

Get:1 http://deb.debian.org/debian trixie/main arm64 liberror-perl all 0.17030-1 [26.9 kB]
Get:2 http://deb.debian.org/debian trixie/main arm64 git-man all 1:2.47.3-0+deb13u1 [2,205 kB]
Get:3 http://deb.debian.org/debian trixie/main arm64 git arm64 1:2.47.3-0+deb13u1 [8,666 kB]
Fetched 10.9 MB in 3s (3,683 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 66257 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17030-1_all.deb ...
Unpacking liberror-perl (0.17030-1) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.47.3-0+deb13u1_all.deb ...
Unpacking git-man (1:2.47.3-0+deb13u1) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.47.3-0+deb13u1_arm64.deb ...
Unpacking git (1:2.47.3-0+deb13u1) ...
Setting up liberror-perl (0.17030-1) ...
Setting up git-man (1:2.47.3-0+deb13u1) ...
Setting up git (1:2.47.3-0+deb13u1) ...
Processing triggers for man-db (2.13.1-1) ...
yacoub@new-yacoub-pi:~ $ git clone https://github.com/Rumple12/new-yacoub-thesis.git
Cloning into 'new-yacoub-thesis'...
remote: Enumerating objects: 264, done.
remote: Counting objects: 100% (264/264), done.
remote: Compressing objects: 100% (191/191), done.
remote: Total 264 (delta 93), reused 222 (delta 53), pack-reused 0 (from 0)
Receiving objects: 100% (264/264), 1.32 MiB | 2.61 MiB/s, done.
Resolving deltas: 100% (93/93), done.
yacoub@new-yacoub-pi:~ $ cd new-yacoub-thesis

```

```

yacoub@new-yacoub-pi:~/new-yacoub-thesis $ MIDDLEWARE_HOST=0.0.0.0 python3 -m
middleware.api.app
new-yacoub middleware running at http://0.0.0.0:8000
Press Ctrl+C to stop.
10.131.76.47 - "GET /status HTTP/1.1" 200 -
10.131.76.47 - "POST /fan/on HTTP/1.1" 200 -
10.131.76.47 - "POST /fan/off HTTP/1.1" 200 -
10.131.76.47 - "POST /fan/on HTTP/1.1" 200 -
10.131.76.47 - "POST /fan/off HTTP/1.1" 200 -
10.131.76.47 - "POST /fan/off HTTP/1.1" 200 -
client_loop: send disconnect: Connection reset
PS C:\WINDOWS\system32> cd ~/new-yacoub-thesis
cd : Cannot find path 'C:\Users\Jake_\new-yacoub-thesis' because it does not exist.
At line:1 char:1

• cd ~/new-yacoub-thesis

• + CategoryInfo          : ObjectNotFound: (C:\Users\Jake_\new-yacoub-thesis:String) [Set-Location], ItemNotFoundEx ception
  + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\WINDOWS\system32> ssh yacoub@10.131.76.89
ssh: connect to host 10.131.76.89 port 22: Connection timed out
PS C:\WINDOWS\system32> ssh yacoub@new-yacoub-pi.local
ssh: Could not resolve hostname new-yacoub-pi.local: Ingen s\345dan v\344rd \344r k\344nd.
PS C:\WINDOWS\system32> ssh yacoub@10.131.76.89
The authenticity of host '10.131.76.89 (10.131.76.89)' can't be established.
ED25519 key fingerprint is SHA256:Qi/aHag17KNoSXJ2ubjBPByxAIDnGos8Nj9lhQTm3vk.
This host key is known by the following other names/addresses:
C:\Users\Jake_\ssh\known_hosts:1: new-yacoub-pi.local
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.131.76.89' (ED25519) to the list of known hosts.
yacoub@10.131.76.89's password:
Linux new-yacoub-pi 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
Last login: Mon May  4 22:32:16 2026 from fe80::2e60:5577:12bc:886e%wlan0
yacoub@new-yacoub-pi:~ $ cd ~/new-yacoub-thesis
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ nohup env MIDDLEWARE_HOST=0.0.0.0 python3 -m
middleware.api.app > pi-middleware.log 2>&1 &
[1] 1842
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ ps aux | grep "middleware.api.app"
yacoub   1842  1.4  0.5 32280 21164 pts/0    S   00:23   0:00 python3 -m middleware.api.app
yacoub   1844  0.0  0.0 6392  2028 pts/0    S+  00:24   0:00 grep --color=auto middleware.api.app
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ tail -f pi-middleware.log
nohup: ignoring input
Invoke-RestMethod -Method Get -Uri "http://<PI_IP>:8000/status"^Z
[2]+  Stopped                  tail -f pi-middleware.log
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ ps aux | grep "middleware.api.app"
yacoub   1842  0.4  0.5 32280 21164 pts/0    S   00:23   0:00 python3 -m middleware.api.app
yacoub   1848  0.0  0.0 6392  2028 pts/0    S+  00:24   0:00 grep --color=auto middleware.api.app
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ Invoke-RestMethod -Method Get -Uri
"http://10.131.76.89:8000/status"

```

```

-bash: Invoke-RestMethod: command not found
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ client_loop: send disconnect: Connection reset
PS C:\WINDOWS\system32> ssh yacoub@10.131.76.89 yacoub@10.131.76.89's password:
Linux new-yacoub-pi 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue May 5 00:22:57 2026 from 10.131.76.47
yacoub@new-yacoub-pi:~ $ cd ~/new-yacoub-thesis
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ nohup env MIDDLEWARE_HOST=0.0.0.0 python3 -m
middleware.api.app > pi-middleware.log 2>&1 &
[1] 1918
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ ps aux | grep "middleware.api.app"
yacoub 1842 0.0 0.5 106072 21444 pts/0 S 00:23 0:00 python3 -m middleware.api.app
yacoub 1921 0.0 0.0 6392 2028 pts/1 S+ 01:02 0:00 grep --color=auto middleware.api.app
[1]+ Exit 1 nohup env MIDDLEWARE_HOST=0.0.0.0 python3 -m middleware.api.app > pi-
middleware.log 2>&1
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ tail -f pi-middleware.log
File "/usr/lib/python3.13/socketserver.py", line 457, in init
self.server_bind()
~~~~~^
File "/usr/lib/python3.13/http/server.py", line 136, in server_bind
socketserver.TCPServer.server_bind(self)
~~~~~^
File "/usr/lib/python3.13/socketserver.py", line 478, in server_bind
self.socket.bind(self.server_address)
~~~~~^
OSError: [Errno 98] Address already in use
^C
yacoub@new-yacoub-pi:~/new-yacoub-thesis $ Invoke-RestMethod -Method Get -Uri
"http://10.131.76.89:8000/status"
-bash: Invoke-RestMethod: command not found
yacoub@new-yacoub-pi:~/new-yacoub-thesis $

```

▼ run

1. Measure full workflow latency

Use your existing Step 9 script.

On your laptop, from repo root:

```
cd "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kaniddat\new-yacoub-thesis"
```

Then run:

```
pythonscripts/collect_metrics.py --modedeterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8" --runs1 --outputevaluation/results/pi-validation/pi-workflow-run-01.csv
```

But important: because this is a **test webhook**, you must click **Execute workflow** in n8n before running the command.

If it works, that CSV should contain latency values measured by Python.

That is your best “exact latency” for Step 10.

If the test webhook only works once and fails after, then either:

click Execute workflow again before each run

or activate the workflow and use the production URL:

<http://localhost:5678/webhook/e1812616-d784-4be8-a796-03dc8fb69eb8>

Production URL is easier for repeated measurement.

1. Measure full workflow latency

Use your existing Step 9 script.

On your laptop, from repo root:

```
</> PowerShell
cd "C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kandidat\new-yacoub-thesis"
```

Then run:

```
</> PowerShell
python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
```

But important: because this is a **test webhook**, you must click **Execute workflow** in n8n before running the command.

If it works, that CSV should contain latency values measured by Python.

That is your best “exact latency” for Step 10.

If the test webhook only works once and fails after, then either:

click Execute workflow again before each run

or activate the workflow and use the production URL:

<http://localhost:5678/webhook/e1812616-d784-4be8-a796-03dc8fb69eb8>

▼ 2. Measure direct Pi middleware latency

▼ evidence

```
14 C:\Users\Jake_OneDrive - Mittuniversitet\EDU\4. RED\1. kandidat\new-yacoub-thesis> python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
15 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
16 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
17 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
18 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
19 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
20 python scripts/collect_metrics.py --mode deterministic --webhook-url "http://localhost:5678/webhook-test/e1812616-d784-4be8-a796-03dc8fb69eb8"
```

2. Measure direct Pi middleware latency

This is simpler and useful as supporting evidence.

On your laptop, replace `<PI_IP>`:


```
$piUrl="http://<PI_IP>:8000/fan/on"
```

```
python-c"import time, urllib.request; url='$piUrl'; start=time.perf_counter(); urllib.request.urlopen(urllib.request.Reque  
st(url, method='POST')).read(); end=time.perf_counter(); print(round((end-start)*1000, 2), 'ms')"
```

For fan off:

```
$piUrl="http://<PI_IP>:8000/fan/off"
```

```
python-c"import time, urllib.request; url='$piUrl'; start=time.perf_counter(); urllib.request.urlopen(urllib.request.Reque  
st(url, method='POST')).read(); end=time.perf_counter(); print(round((end-start)*1000, 2), 'ms')"
```

This measures:

```
laptop -> Pi middleware endpoint -> laptop
```

Not the full n8n workflow.

▼ 3. Measure CPU/RAM of the middleware process

▼ evidence

```
yacoub@new-yacoub-pi:~$ pgrep -af "middleware.api.app"
1842 python3 -m middleware.api.app
yacoub@new-yacoub-pi:~$ PID=$(pgrep -f "middleware.api.app" | head -n 1)
yacoub@new-yacoub-pi:~$ ps -o "$PID" -o pid,comm,%cpu,%mem,rss,args
PID COMMAND %CPU %MEM RSS COMMAND
1842 python3 0.0 0.5 21448 python3 -m middleware.api.app
yacoub@new-yacoub-pi:~$ PID=$(pgrep -f "middleware.api.app" | head -n 1)
yacoub@new-yacoub-pi:~$ RSS_KB=$(ps -p "$PID" -o rss=)
yacoub@new-yacoub-pi:~$ echo "Middleware RAM MB: $(awk "BEGIN {printf \"%.2f\\n\", $RSS_KB/1024}")"
Middleware RAM MB: 20.95
yacoub@new-yacoub-pi:~$ PID=$(pgrep -f "middleware.api.app" | head -n 1)
yacoub@new-yacoub-pi:~$ ps -p "$PID" -o %cpu=
0.0
yacoub@new-yacoub-pi:~$
```

3. Measure CPU/RAM of the middleware process

Open your **second SSH session** into the Pi.

Run:

```
pgrep -af "middleware.api.app"
```

You should see something like:

```
1234 python3 -m middleware.api.app
```

The number is the process ID.

Now run:

```
PID=$(pgrep -f "middleware.api.app" | head -n 1)
ps -p "$PID" -o pid,comm,%cpu,%mem,rss,args
```

This gives:

```
%CPU = CPU usage
%MEM = memory percent
RSS = memory in KB
```

To get RAM in MB:

```
PID=$(pgrep -f"middleware.api.app" | head -n 1)
RSS_KB=$(ps -p"$PID" -o rss=)
echo"Middleware RAM MB:${awk"BEGIN {printf \"%.2f\\",$RSS_KB/1024}\"}"
```

For CPU only:

```
PID=$(pgrep -f"middleware.api.app" | head -n 1)
ps -p"$PID" -o%cpu=
```

Take a screenshot of this.

▼ 4. pi measurements

```
yacoub@new-yacoub-pi:~ $ vcgencmd measure_temp
temp=49.6'C
yacoub@new-yacoub-pi:~ $
```

vcgencmd measure_temp

-

▼ codex creating validation notes and csv -

▼ files

2 files changed +108 -0	Undo ↶	Review ↷	✱
evaluation/results/pi-validation/pi-run-01.csv +3 -0	•	▼	
evaluation/results/pi-validation/pi-validation-notes.md +105 -0	•	▼	

▼ q

Create/update the final Step 10 Raspberry Pi validation evidence files.

Use the real evidence collected from the Raspberry Pi validation.

Create or update:

- evaluation/results/pi-validation/pi-validation-notes.md
- evaluation/results/pi-validation/pi-run-01.csv

Use these existing evidence files:

- evaluation/results/pi-validation/pi-workflow-run-final.csv
- screenshots saved under evaluation/results/pi-validation/

Observed result:

- n8n ran locally on the PC.
- Raspberry Pi ran the Python middleware with:
MIDDLEWARE_HOST=0.0.0.0 python3 -m middleware.api.app
- n8n production webhook successfully called the Raspberry Pi middleware.
- High-temperature test:
 - input: 31.4 C
 - expected_action: fan_on
 - observed_action: fan_on
 - HTTP status: 200
 - success: true
 - full workflow latency: 437.003 ms

```

- Low-temperature test:
  - input: 24.5 C
  - expected_action: fan_off
  - observed_action: fan_off
  - HTTP status: 200
  - success: true
  - full workflow latency: 200.206 ms

Direct Pi middleware latency evidence:
- fan_on direct endpoint latency: 135.44 ms
- fan_off direct endpoint latency: 233.27 ms
- clarify that direct endpoint latency is supporting evidence, not the full n8n workflow latency.

CPU/RAM evidence:
- middleware process: python3 -m middleware.api.app
- CPU: 0.0 %
- MEM: 0.5 %
- RSS: 21048 KB
- Middleware RAM MB: 20.95

Thermal evidence:
- vcgencmd measure_temp returned:
  temp=49.6'C
- thermal_c: 49.6

Important:
- Do not invent any values.
- Clearly state this is Tier 1.5 Raspberry Pi validation.
- Clearly state n8n stayed on the PC.
- Clearly state the Raspberry Pi ran the middleware/action endpoint side.
- Mention that full system deployment to Pi was not attempted.
- Mention that direct endpoint latency is supporting evidence only.
- Mention that pi-workflow-run-final.csv is the main full-workflow measurement evidence.
- Do not modify middleware, workflows, or Step 11 files.

```

▼ a

Done. I created the final Step 10 Pi validation evidence files:

- pi-validation-notes.md
- pi-run-01.csv

What changed:

- Documented Step 10 as **Tier 1.5 Raspberry Pi validation**.
- Clearly stated that **n8n stayed on the PC**.
- Clearly stated that the **Raspberry Pi ran the middleware/action endpoint side**.
- Recorded the middleware command: `MIDDLEWARE_HOST=0.0.0.0 python3 -m middleware.api.app`.
- Recorded both full workflow runs:
 - `31.4 C -> fan_on`, HTTP `200`, success `true`, `437.003 ms`
 - `24.5 C -> fan_off`, HTTP `200`, success `true`, `200.206 ms`
- Recorded direct Pi endpoint latency as **supporting evidence only**:
 - `fan_on`: `135.44 ms`
 - `fan_off`: `233.27 ms`
- Recorded CPU/RAM/thermal evidence:
 - CPU `0.0 %`
 - MEM `0.5 %`
 - RSS `21048 KB`
 - middleware RAM `20.95 MB`

- thermal 49.6 C
- Marked pi-workflow-run-final.csv as the main full-workflow measurement evidence.
- Left middleware, workflows, and Step 11 untouched.

Validation passed: pi-run-01.csv parses cleanly with 2 rows and 23 columns.

-

▼ report scribe

▼ q

Create the report note for Step 10.

Create:

docs/report-notes/step-10-raspberry-pi-validation.md

Use only evidence already present in the repo.

Use these sources:

- infrastructure/os/raspberry-pi-notes.md
- infrastructure/docker/pi-deployment-notes.md
- evaluation/results/pi-validation/README.md
- evaluation/results/pi-validation/pi-validation-notes.md
- evaluation/results/pi-validation/pi-run-01.csv
- evaluation/results/pi-validation/pi-workflow-run-final.csv
- evaluation/results/pi-validation/
- docs/evaluation/evaluation-protocol.md
- docs/decisions.md
- README.md
- docs/plans/new-yacoub-14-step-process.md

Include:

- what was built / validated
- why this matters for the thesis
- evidence produced
- which report chapters it feeds
- limitations or assumptions
- screenshots/logs still needed

Important:

- do not invent results
- clearly state that this is Tier 1.5 Raspberry Pi validation
- clearly state that n8n stayed on the PC
- clearly state that the Raspberry Pi ran the middleware/action endpoint side
- clearly state that this was not a full Raspberry Pi deployment of the whole system
- mention that the production n8n webhook successfully called the Raspberry Pi middleware
- mention high-temperature 31.4 C produced fan_on

- mention low-temperature 24.5 C produced fan_off
- mention full workflow latency came from pi-workflow-run-final.csv
- mention the summarized results in pi-run-01.csv
- mention direct endpoint latency only as supporting evidence
- mention CPU/RAM/thermal evidence from [pi-validation-notes.md](#)
- include the observed thermal value only if present in the evidence: 49.6 C
- include middleware RAM only if present in the evidence: 20.95 MB
- mention that this strengthens the edge-computing angle without changing the main PC-first thesis scope
- do not expand into Tier 2
- do not start Step 11
- do not claim real GPIO/fan hardware was used; the action endpoint is still simulated middleware behavior

▼ a

Step 10 - Raspberry Pi Validation

What was built / validated

- Optional Tier 1.5 Raspberry Pi validation evidence was added under [evaluation/results/pi-validation/](#).
- The validation used a split setup:
 - n8n stayed on the PC.
 - The Raspberry Pi ran the Python middleware/action endpoint side.
- The Raspberry Pi middleware was started with:

```
MIDDLEWARE_HOST=0.0.0.0 python3 -m middleware.api.app
```

- The production n8n webhook on the PC successfully called the Raspberry Pi middleware.
- This was not a full Raspberry Pi deployment of the whole system.
- The fan action endpoint remained simulated middleware behavior; no real GPIO or physical fan hardware was used.

Why it matters for the thesis

This strengthens the edge-computing angle without changing the main PC-first thesis scope. It shows that the already working PC-hosted n8n workflow can call a Raspberry Pi-hosted middleware/action endpoint for the same temperature-to-fan scenario.

The evidence supports the thesis discussion around constrained hardware and deployment feasibility while keeping Raspberry Pi validation as Tier 1.5 support evidence, not the primary baseline.

Evidence produced

- [infrastructure/os/raspberry-pi-notes.md](#)
- [infrastructure/docker/pi-deployment-notes.md](#)
- [evaluation/results/pi-validation/README.md](#)

- `evaluation/results/pi-validation/pi-validation-notes.md`
- `evaluation/results/pi-validation/pi-run-01.csv`
- `evaluation/results/pi-validation/pi-workflow-run-final.csv`
- `evaluation/results/pi-validation/2. Measure direct Pi middleware latency.png`
- `evaluation/results/pi-validation/3. Measure CPU_RAM of the middleware process.png`
- `evaluation/results/pi-validation/pi-temp-measurement.png`
- `evaluation/results/pi-validation/pi-high-temp-fan-on-canvas.png`
- `evaluation/results/pi-validation/pi-high-temp-fan-on-output.png`
- `evaluation/results/pi-validation/pi-low-temp-fan-off-canvas.png`
- `evaluation/results/pi-validation/pi-low-temp-fan-off-output.png`
- `evaluation/results/pi-validation/pi-terminal-high-temp-fan-on.png`
- `evaluation/results/pi-validation/pi-terminal-low-temp-fan-off.png`

`pi-workflow-run-final.csv` records the final full-workflow latency evidence:

- high temperature `31.4 C` produced `fan_on`, HTTP `200`, success `true`, full workflow latency `437.003 ms`
- low temperature `24.5 C` produced `fan_off`, HTTP `200`, success `true`, full workflow latency `200.206 ms`

`pi-run-01.csv` records the same successful production webhook run in the Step 9-style result format, including the input temperatures, expected and observed actions, HTTP status, success flag, latency fields, and observed simulated middleware responses.

Direct Raspberry Pi middleware endpoint latency is supporting evidence only:

- `fan_on`: `135.44 ms`
- `fan_off`: `233.27 ms`

CPU/RAM/thermal evidence from `pi-validation-notes.md`:

- observed middleware process: `python3 -m middleware.api.app`
- CPU: `0.0 %`

`python3 -m`

- MEM: `0.5 %`
- RSS: `21048 KB`
- middleware RAM: `20.95 MB`
- thermal command: `vcgencmd measure_temp`
- observed thermal value: `49.6 C`

Report chapters it feeds

- Chapter 4 - Choice of approach: PC-first development with Raspberry Pi as later Tier 1.5 validation.
- Chapter 5 - Implementation: split deployment note where PC-hosted n8n calls Raspberry Pi-hosted middleware.
- Chapter 6 - Results: optional Pi validation results, full workflow latency, and limited Pi resource/thermal observations.
- Chapter 7 - Discussion: practical edge-computing limitations, split deployment tradeoffs, and why the PC baseline remains primary.

Limitations or assumptions

- This is Tier 1.5 Raspberry Pi validation only.
- n8n was not deployed on the Raspberry Pi.
- The whole system was not moved to the Raspberry Pi.
- The Raspberry Pi ran only the middleware/action endpoint side.
- The middleware action remained simulated; no real GPIO or physical fan was controlled.
- The full workflow latency values come from `pi-workflow-run-final.csv`.
- Direct endpoint latency is supporting evidence only and should not be treated as full workflow latency.
- `pi-run-01.csv` includes `thermal_c=not_available` in the Step 9-style row; Pi thermal evidence is separately documented in `pi-validation-notes.md`.
- The run count is limited and should be interpreted as validation evidence, not broad benchmarking.
- This step does not expand into Tier 2 deployment work.
- Step 11 freeze work was not started here.

Screenshots/logs still needed

- Clear terminal transcript showing the Raspberry Pi middleware startup command and IP/network setup.
- Optional repeated Pi runs if stronger final report tables are needed.
- A concise screenshot set selected for the thesis appendix, avoiding template and failed setup files.
- Any future evidence would need to stay within Tier 1.5 unless the scope is explicitly changed.

-

▼ broad multi agent not implemented

A **broad multi-agent system** would mean multiple specialized AI agents working together, not just one simple LLM decision node.

Example:

- one agent reads sensor data
- one agent plans the action
- one agent checks safety
- one agent executes tool calls
- one agent reviews/logs the result

They might communicate through shared memory, tools, or workflows.

For your thesis, this would be much bigger because you would need to design, test, and compare several agents instead of one minimal agent-enhanced path.

-

▼ Goal - Upgrade the thesis from Tier 1 to Tier 1.5 without redesigning the system.

Strengthen Yacoub's edge-computing angle by adding a thin Tier 1.5 hardware validation.

▼ Work to complete - Only after the local system works:

- deploy the same system or relevant parts to Raspberry Pi
- run the same or simplified scenario
- collect at least limited real-hardware measurements

- limited latency/resource measurement
- basic thermal observation if possible
- document differences between PC and Raspberry Pi behavior

```
infrastructure/os/raspberry-pi-notes.md
infrastructure/docker/pi-deployment-notes.md
evaluation/results/pi-validation/
```

- What this proves - This strengthens Yacoub's edge-computing angle

▼ Important constraint

Raspberry Pi validation must not block the local thesis system from being complete.

If time collapses, the thesis can still survive with the local Tier 1 proof of concept.

▼ Evidence produced

- Pi deployment notes
- Pi test logs
- Pi hardware measurement notes
- hardware limitations discussion

-

▼ chapter relevance

▼ Chapter 4 - Choice of approach

justification for PC-first then Pi validation

```
PC-first vs Pi-first decision
docs/decisions.md
```

▼ Chapter 5 - Implementation

edge deployment notes

```
infrastructure/os/
pi deployment notes
```

▼ Chapter 6 - Results

- optional Pi validation results

```
evaluation/results/pi-validation/
```

▼ Chapter 7 - Discussion

- limitations and practical edge discussion

```
Pi limitations
edge hardware constraints
local vs Pi comparison
```


▼ new full step 10

Step 10 — Add Raspberry Pi validation if stable

Work to complete

Only after local system works:

Run:

- same or simplified scenario on Raspberry Pi

Evidence produced

- Pi deployment notes
- Pi logs
- Pi measurements
- hardware limitation notes

Chapter feeds from

Chapter 4 — Choice of approach feeds from:

```
PC-first vs Pi-first decision
docs/decisions.md
```

Chapter 5 — Implementation feeds from:

```
infrastructure/os/
pi deployment notes
```

Chapter 6 — Results feeds from:

```
evaluation/results/pi-validation/
```

Chapter 7 — Discussion feeds from:

```
Pi limitations
edge hardware constraints
local vs Pi comparison
```

Chapter deliverables

This step helps deliver:

Chapter 4

- justification for PC-first then Pi validation

Chapter 5

- edge deployment notes

Chapter 6

- optional Pi validation results

Chapter 7

- limitations and practical edge discussion

▼ (Write the thesis)

▼ Step 11 — Freeze implementation

chapter 5 mplementation

chapter 7 Discussion

▼ new new step 11

```
Step 11 - Freeze implementation
-
Goal - Stop building. Protect writing time.
-
what to do
no new workflows
no new devices
no new architecture
only fix broken evidence names, missing notes, or obvious bugs
mark the final repo state
final screenshots and architecture diagram
-
outputs
docs/decisions.md final-state note
final workflow exports
final architecture diagram
final evidence inventory
Git commit/tag

***** PROMPT *****

Step 11 – Freeze implementation.

Freeze the current Tier 1.5 implementation state. Do not add new architecture, devices, workflows, MCP, real GPIO, or new agent features.

Tasks:
1. Add a final-state note to docs/decisions.md.
2. Create docs/report-notes/step-11-implementation-freeze.md.
3. Create a concise evidence inventory listing final workflow exports, result CSVs, screenshots, Raspberry Pi validation files, and report notes.
4. Confirm the frozen scope:
  - n8n runs locally on PC with Docker.
  - middleware/action endpoint can run locally and on Raspberry Pi.
  - deterministic and minimal agent-enhanced workflows exist.
  - safety/contract layers exist.
  - evaluation and Raspberry Pi validation evidence exist.
5. Clearly state that all further work is report writing, analysis, figure/table cleanup, and bug fixes only.

Do not modify runtime behavior.
Do not start Step 12 content writing yet.
```

▼ new step 11

Step 11 — Freeze implementation

Goal

Stop expanding and protect the writing phase.

Work to complete

- fix only bugs
- no new devices

- no new workflows
- no new architecture
- no MCP unless already stable and necessary
- tag or document final implementation state

Evidence produced

```
final Git commit/tag
final screenshots
docs/decisions.md final-state note
workflow exports
final architecture image
```

Chapter feeds from

Chapter 5 — Implementation feeds from:

```
final repo state
final architecture
final screenshots
```

Chapter 7 — Discussion feeds from:

```
final limitations
deferred work
frozen scope decisions
```

Chapter deliverables

This step helps deliver:

Chapter 5

- final implementation narrative

Chapter 7

- limitations
- work method reflection
- why scope was frozen

▼ Goal

Stop expanding and protect the report-writing phase.

▼ Work to complete

- fix bugs only
- no new devices
- no new workflows
- no new architecture
- no MCP unless already stable and necessary
- tag or mark the final implementation state

▼ Evidence produced

- final Git commit/tag
- final architecture screenshot
- final workflow screenshots
- final decision-log entry

▼ ~~Step 12 — Write the report from the artifacts~~

~~chapter 1 intro~~

~~chapter 2 theory~~

~~chapter 3 methodology~~

~~chapter 4 Choice of approach~~

~~chapter 5 implementation~~

▼ new new step 12

```
step 12
-
(write chapters 1-5)
Build the first half of the report from the artifacts.

chapters
-
1 Introduction
2 Theory / Related Work
3 Methodology
4 Choice of Approach / System Design
5 Implementation

key instructions
-
Write from evidence, not memory.

Use:
-
docs/report-notes/
docs/decisions.md
docs/evaluation/evaluation-protocol.md
middleware/
shared_interfaces/
cognitive_logic/
safety_layer/
evaluation/
infrastructure/
workflow screenshots

Important Ali points to handle here:
-
research questions early in Chapter 1
Background must explain why existing tools alone are not enough
Explain what the agent-enhanced workflow adds, even if small
Make the report read as an argument, not a diary

Important Ali points to handle here:
-
- research questions early in Chapter 1
- Background must explain why existing tools alone are not enough
- Explain what the agent-enhanced workflow adds, even if small
- Make the report read as an argument, not a diary
```

▼ new step 12

Step 12 — Write Chapters 1–5 from artifacts

Goal

Write the first half of the thesis using already-produced material.

Work to complete

Write:

- Chapter 1 — Introduction
- Chapter 2 — Theory
- Chapter 3 — Methodology
- Chapter 4 — Choice of approach
- Chapter 5 — Implementation

Evidence used

```
docs/ongoing/  
docs/plans/  
docs/decisions.md  
docs/architecture/  
infrastructure/  
middleware/  
shared_interfaces/  
cognitive_logic/  
safety_layer/  
workflow screenshots
```

Chapter feeds from

This step feeds directly from all previous planning and implementation artifacts.

Chapter deliverables

Chapter 1

- background
- aim
- research questions
- scope
- AI tools usage
- division of work

Chapter 2

- relevant theory
- related work

Chapter 3

- scientific method
- project method
- evaluation method

Chapter 4

- alternatives
- comparison
- chosen approach

Chapter 5

- architecture
- infrastructure
- middleware
- contracts
- workflows
- safety
- measurement setup

Goal

Write the thesis from evidence, not memory.

Write Chapters 1–5 first

Chapter 1 — Introduction

Use:

- research questions
- project overview
- Yacoub scope
- Obid minimum integration
- scope limitations

Chapter 2 — Theory

Use:

- agentic AI notes
- edge computing notes
- n8n notes
- middleware/API contract notes
- validation/HITL notes
- Docker/Raspberry Pi notes

Chapter 3 — Methodology

Use:

- implementation plan
- decisions
- evaluation protocol
- repeated-run strategy

- measurement method

Chapter 4 — Choice of approach

Use:

- deterministic vs agent-enhanced comparison
- PC-first vs Pi-first tradeoff
- direct action vs validation/approval
- full multi-agent vs minimal workflow
- mock sensor vs real sensor

Chapter 5 — Implementation

Use:

- repo structure
- architecture diagram
- infrastructure files
- middleware
- contracts
- workflows
- safety layer
- measurement setup

Evidence produced

- draft Chapters 1–5
- missing figure/table checklist

▼ ~~Step 13 — Write results and discussion~~

~~chapter 6 Results~~

~~chapter 7 Discussion~~

▼ new new step 13

```
step 13
-
Write Results(6) and Discussion(7)

Goal
-
Turn evidence into findings.

chapters
-
6 results
7 discussions

chapters 6 results
-
should be objective
what worked
what failed or was limited
measured values
```

```
tables
figures
screenshots
```

```
chapter 7 discussion
-
- what the results mean
- whether the research questions were answered
- why the narrowed scope is still valuable
- limitations near the start
- future work
```

▼ new step 13

Step 13 — Write Results and Discussion

Goal

Turn the evidence into findings and interpretation.

Work to complete

Write:

- Chapter 6 — Results
- Chapter 7 — Discussion

Evidence used

```
evaluation/results/
evaluation/metrics/
screenshots
workflow outputs
allowed/blocked cases
baseline-vs-agent results
Pi validation results if available
```

Chapter feeds from

Chapter 6 — Results feeds from:

```
summary CSVs
result tables
charts
screenshots
test logs
```

Chapter 7 — Discussion feeds from:

```
Chapter 6 results
limitations
scope decisions
method reflections
future work notes
```

Chapter deliverables

Chapter 6

- resulting system
- measurement results
- baseline-vs-agent comparison
- safety results
- optional Pi validation results

Chapter 7

- analysis of results
- project/work method discussion
- scientific discussion
- consequence analysis/recommendation
- ethical and societal discussion

▼ Goal

Turn measured data into report conclusions.

▼ Chapter 6 — Results

Use:

- summary CSVs
- screenshots
- charts
- workflow outputs
- safety cases
- baseline-vs-agent comparison

▼ Chapter 7 — Discussion

Answer:

- did the system work?
- where did latency appear?
- how stable was the runtime?
- what resource effects were observed?
- did the safety layer help?
- what did the narrowed scope exclude?
- what would full Obid expansion add?

▼ Evidence produced

- results chapter
- discussion chapter
- figures/tables

▼ Step 14 — Conclusions, polish, and submission package

chapter 8 conculsion

▼ new new step 14

```
step 14
-
Make the initial report submission-ready.

tasks
-
- Chapter 8 Conclusion
- Future work
- Abstract
- Sammanfattning
- Keywords
- Terminology
- References
- Appendix A: source-code link
- export final PDF
```

▼ new step 14

Step 14 — Conclusions, polish, and submission package

Goal

Make the thesis submission-ready.

Work to complete

Write:

- Chapter 8 — Conclusions
- future work
- abstracts
- terminology
- references
- appendix source-code note

Prepare:

```
thesis PDF
GitHub repo
workflow exports
result CSVs
figures
appendix/source-code link
```

Chapter feeds from

Chapter 8 — Conclusions feeds from:

```
research questions
Chapter 6 results
Chapter 7 discussion
future work notes
```

Front matter feeds from:

```
final abstract
Swedish sammanfattning
keywords
terminology list
```

Appendix A feeds from:

```
GitHub repo link
source-code structure
workflow exports
result files
```

Chapter deliverables

This step helps deliver:

Abstract / Sammanfattning

- final short summary of aim, method, results, conclusion

Chapter 8

- direct answers to research questions
- final conclusion
- future work

Appendix A

- source-code link
- repo reference
- implementation artifact location

Goal

Make the thesis submission-ready, not perfect.

Work to complete

Chapter 8 — Conclusions

Directly answer the research questions:

1. can the narrowed system run?
2. what resource/latency behavior was observed?
3. can the minimal safety layer mediate one physical action?
4. what is the difference between deterministic and agent-enhanced mode?

Future work

Move deferred work here:

- richer ReAct logic
- broader memory strategies
- more advanced guardrails
- more sensors/devices
- deeper Raspberry Pi optimization

- MCP integration

Final package

Prepare:

- thesis PDF
- GitHub source repo
- workflow exports
- result CSVs
- figures
- Appendix A source-code link

Final polished process in one line

The full process is:

Lock scope → build local runtime → build middleware → define contract → build baseline → add minimal agent path → add safety → measure → optionally validate on Pi → freeze → write report from artifacts.

That is the cleanest version of your draft. It keeps the same content, but removes the confusion and makes the step order usable.

▼ step 15 - important last

- sammanfatning/ abstract
- AI tools usage

▼ figures / tables

```
figures_tables

Figure 1: Final system architecture
Figure 2: Deterministic baseline workflow screenshot
Figure 3: Agent-enhanced workflow screenshot
Figure 4: Raspberry Pi validation setup
-
Optional chart:
Latency comparison bar chart
* Do not overdo charts. Tables are enough if time is tight.

Minimum tables:
Table 1: Research questions and evaluation method
Table 2: Middleware endpoints
Table 3: Test cases
Table 4: Deterministic/agent result summary
Table 5: Raspberry Pi validation results
Table 6: Limitations and future work
```

▼ appendix

- * Appendix for source code

▼ references

25 references

build 10 by 10